RESEARCH ARTICLE
Applied Mathematics

Nonlinear Dynamics of Pulsatile Blood Flow in Viscoelastic Vessels: A Dispersive Wave Approach

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Funding information
This publication is based upon work supported by the Khalifa University of Science and Technology under Award No. FSU-2023-014. AC is grateful to NSERC of Canada for support through a Discovery grant RGPIN-2024-04308.

In this study, we investigate the dynamics of pulsatile flow in viscoelastic vessels, beginning with the axisymmetric full Euler equations with a free surface. Though highly accurate, this comprehensive model presents considerable analytical and computational challenges. To address these difficulties, we derive the fully nonlinear cylindrical Serre–Green–Naghdi (SGN) equations, which describe the propagation of long-crested pulses in cylindrical vessels. However, due to the complexity of these equations, we proceed by applying the weak nonlinearity assumption, leading to the derivation of the cylindrical Boussinesq equations. These equations offer a more tractable model while still capturing essential dispersive and nonlinear effects. In the final step, we derive the classical unidirectional approximations, such as the Korteweg–de Vries (KdV) and Benjamin–Bona–Mahony (BBM) equations, further simplifying the wave propagation analysis in this setting. Additionally, we investigate the modulation of periodic waves and establish the existence of Modulational Instability (MI) within this context, providing new insights into the stability of wave trains in cylindrical geometries. The results presented in this work contribute to a deeper understanding of fluid-structure interactions

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in biomedical applications, particularly in modelling blood flow in large arteries.

KEYWORDS
blood flows; axisymmetric flows; compliant vessels; long wave model; nonlinear dispersive systems

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1 | INTRODUCTION

All models are approximations. Essentially, all models are wrong, but some are useful. However, the approximate nature of the model must always be borne in mind.

George E.P. Box

The study of pulsatile flow in viscoelastic vessels is a fundamental topic in fluid dynamics with significant applications in biomedical science, particularly in the modelling of blood flow in large arteries (Fung, 1997). Accurately describing fluid-structure interactions within vessels is critical to understanding hemodynamic processes, such as the propagation of pressure pulses generated by the heart. Large arteries exhibit both elastic and viscoelastic properties, storing elastic energy during the systolic phase and releasing it during diastole, which makes their modelling mathematically challenging (Chandran et al., 2012; Quarteroni and Formaggia, 2004; van de Vosse and Stergiopulos, 2011). The present manuscript builds upon the PhD thesis of the first author, Rim El Cheikh, and aims to develop a systematic framework to describe the dynamics of such flows using asymptotic modelling techniques.

The Euler equations, which describe the motion of an inviscid fluid, provide a starting point for analyzing blood flow, but they neglect the viscous effects that are present in real blood vessels. Although blood is not a perfect inviscid fluid, viscous effects are largely limited to the boundary layer, and therefore can be assumed to be small in large arteries, where the flow is fast and the diameter is large, compared to smaller vessels where viscous effects become more significant. Furthermore, the assumption of axisymmetry simplifies the problem by reducing the number of spatial dimensions to two, which is a common approach in fluid mechanics. However, it is important to note that real blood vessels may have complex geometries, especially at bifurcations and bends. The choice of a cylindrical geometry allows for a more manageable mathematical treatment, and it is a common approximation for long, straight vessels.

Previous research has proposed several models to tackle the complexities of fluid flow in elastic and viscoelastic vessels. In Mitsotakis et al. (2018), the authors derived new asymptotic one-dimensional equations of Boussinesq type, which are weakly nonlinear and weakly dispersive, for the study of inviscid and irrotational fluid flow in elastic vessels. These models serve as an alternative to the Euler equations and offer significant computational advantages

while maintaining accuracy in simulating blood flow in arteries. Additionally, the work extended these models to incorporate dissipative effects arising from fluid viscosity, providing a more comprehensive framework for arterial blood flow.

Building on the framework developed in Mitsotakis et al. (2018), the subsequent study (Mitsotakis et al., 2019) considered the inclusion of viscoelastic effects in the vessel walls, further refining the model to account for the realistic biomechanical behaviour of arteries. This study introduced asymptotic models describing the propagation of solitary and periodic waves in viscoelastic vessels, providing valuable insights into the wave dynamics in such systems. These two contributions laid the groundwork for investigating more complex interactions between the fluid and the vessel walls, including both dispersive and nonlinear effects.

In this manuscript, we advance this line of research by investigating the dynamics of pulsatile flow in viscoelastic vessels, beginning with the axisymmetric full Euler equations with a free surface. While this comprehensive model offers a high degree of accuracy, it also presents significant analytical and computational challenges. To mitigate these complexities, we derive the fully nonlinear cylindrical Serre–Green–Naghdi (SGN) equations, which provide a more tractable model for the propagation of long-crested pulses in cylindrical geometries. Subsequently, we simplify the system further by applying the weak nonlinearity assumption, leading to the derivation of the cylindrical Boussinesq equations, which capture essential dispersive and nonlinear effects while being more amenable to analysis and computation.

The SGN equations, despite being one-dimensional (1D) models, still retain the fully nonlinear character of the Euler equations, which can be important for capturing certain types of wave propagation effects. By retaining the nonlinear terms, the Serre equations allow for the modelling of phenomena such as wave steepening and the formation of shocks, which may arise in pulsatile blood flow. However, unlike the full Euler equations, the Serre equations do not require solving a system of partial differential equations in two spatial dimensions, but rather, they reduce the dimensionality of the problem, leading to a more efficient computational approach. It is important to note that the Serre equations are derived under the assumption of long waves, which means that they are most accurate when the wavelength of the disturbances is significantly larger than the vessel radius. They provide a framework for studying the propagation of pressure pulses in arteries, but their limitations are clear: the one-dimensional representation is a significant simplification of the actual complex geometry of the circulatory system, and they are less accurate when the wavelength of the pressure and flow disturbances becomes comparable to or smaller than the vessel radius.

The cylindrical Boussinesq equations are derived from the SGN equations under a weak nonlinearity assumption, and they maintain some dispersive effects that are essential for capturing certain aspects of wave propagation in blood vessels. The balance between nonlinear and dispersive effects is known to be important for understanding the formation of stable pulses, and these equations allow the study of this balance through various analytical and numerical techniques. The Boussinesq equations, like the Serre equations, are also 1D models. However, by applying this assumption, they will generally lose their capacity to model the steepening of waves or shocks as they are derived under the assumption that nonlinear terms are small. While they are mathematically simpler than the Serre equations, they may still require significant computational resources for numerical wave propagation and stability studies.

Finally, we derive classical unidirectional approximations, including the Korteweg–de Vries (KdV) and Benjamin–Bona–Mahony (BBM) equations, to further simplify the wave propagation analysis. These unidirectional models are particularly useful in studying wave modulation and stability. Notably, we explore the modulation of periodic waves and establish the existence of Modulational Instability (MI), which provides new insights into the stability of wave trains in cylindrical vessels. Our findings contribute to a deeper understanding of fluid-structure interactions in biomedical applications and offer new perspectives on the modelling of blood flow in large arteries. The results presented in this paper contribute to a deeper understanding of fluid-structure interactions in biomedical applications,

particularly in the modelling of blood flow in large arteries. By providing a hierarchy of models ranging from the full Euler equations to simplified unidirectional approximations, this work offers a comprehensive framework for analyzing pulsatile flow in viscoelastic vessels across various scales and levels of complexity.

The Korteweg–de Vries (KdV) and Benjamin–Bona–Mahony (BBM) equations are classical unidirectional models which describe wave propagation, and they are widely used in various fields of physics and engineering to study long-wave phenomena. Both of these equations are further simplifications of Boussinesq systems. While the KdV and BBM equations are both unidirectional models, they differ in their treatment of dispersive effects. Both equations capture the balance between nonlinearity and dispersion in a certain way, which leads to solitary wave solutions. They can also both be more amenable to wave modulation and stability studies due to their relatively simple form. The KdV and BBM equations are derived under strong assumptions, including that the wave propagation is primarily unidirectional and that nonlinear and dispersive effects are weak, so their applicability is limited to specific regimes of the flow. While these models may not fully capture the complexity of blood flow, they offer valuable insight into the basic mechanisms of wave propagation and can be useful for studying wave modulation and stability.

In the context of blood flow in large arteries, these models can help to explain the relationship between the pulsatile nature of blood flow, the elastic and viscoelastic properties of the vessel wall, and the propagation of pressure waves. By considering a range of models of different levels of complexity, we will be able to evaluate the importance of different physical effects and parameters, such as the viscosity of blood, the stiffness of the vessel walls, and the presence of geometric features such as bifurcations and bends, and assess their influence on wave propagation and energy transport. The MI analysis offers insights into the long-term stability of wave trains in arteries, which is important for understanding phenomena such as pulse wave reflections, the interaction of waves, and wave energy dissipation. This can contribute to a more complete understanding of blood flow dynamics in the cardiovascular system and the effects of factors such as ageing and disease on the mechanical properties of blood vessels.

This manuscript is organized as follows. First, in Section 2, we present the parent-level mathematical model and the underlying physical assumptions, including both dimensional and dimensionless formulations. We then derive a new fully nonlinear weakly dispersive model, detailing the asymptotic expansion of the velocity field, mass conservation, and intermediate results. This is followed by the classical cylindrical Boussinesq equations and unidirectional reductions, such as the Korteweg–de Vries (KdV) and Benjamin–Bona–Mahony (BBM) models. We proceed with an in-depth analysis of the dispersion relations of these models (Section 3), shedding light on their wave propagation characteristics. The manuscript then explores numerical simulations (Section 4), Stokes expansions (Section 5), and the conservation laws and symmetries inherent in the models (Section 6). A dedicated section addresses the MI analysis (Section 7). The paper concludes with a discussion of the main findings (Section 8) and perspectives for future research (Section 9). Several supporting sections address acknowledgements, conflict of interest, data availability, and supplementary information. Finally, Appendix A presents a Windkessel model for pulsatile flow in viscoelastic vessels based on Serre-type approximations, connecting our continuum models to classical hemodynamic analogies.

2 | MATHEMATICAL MODELLING

Mathematical models of blood flow range from complex three-dimensional CFD-type models (Roe, 2005) to very simplified zero-dimensional lumped models (Milišić and Quarteroni, 2004; Alastruey et al., 2008) (see also Appendix A for our Windkessel model approach). In the present study, we take an intermediate point of view, where we follow the system's evolution in time and along the longitudinal direction[1] (Sherwin et al., 2003; Alastruey et al., 2011) while

[1]The so-called unsteady one-dimensional models.

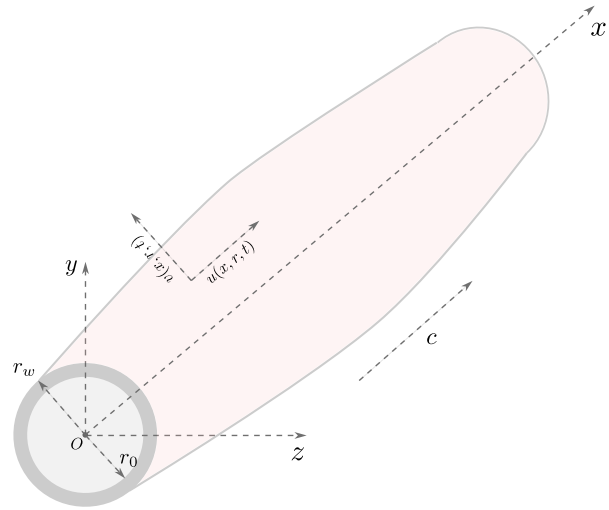


FIGURE 1 Sketch of a cylindrical elastic vessel subject to an axi-symmetric deformation filled with an ideal fluid.

the processes taking place in the transverse direction are modelled based on the long wave assumption (Lavrentiev, 1947; Serre, 1956; Bona et al., 2005; Khakimzyanov et al., 2020). This point of view has already been adopted in our previous studies (Mitsotakis et al., 2018, 2019) in the weakly nonlinear regime. In the present study, we relieve this assumption in our developments as far as possible.

We assume that we have an incompressible flow[2] of an ideal and homogeneous[3] fluid. This flow takes place in an elastic cylindrical vessel as schematically depicted in Figure 1. Moreover, the blood flow in large arteries is mostly laminar[4]. In order to simplify the problem and reduce its physical dimensionality, we assume the flow and, consequently, the deformation of the vessel's walls to be axisymmetric[5]. The difference between axisymmetric and general deformation is schematically depicted in Figure 2.

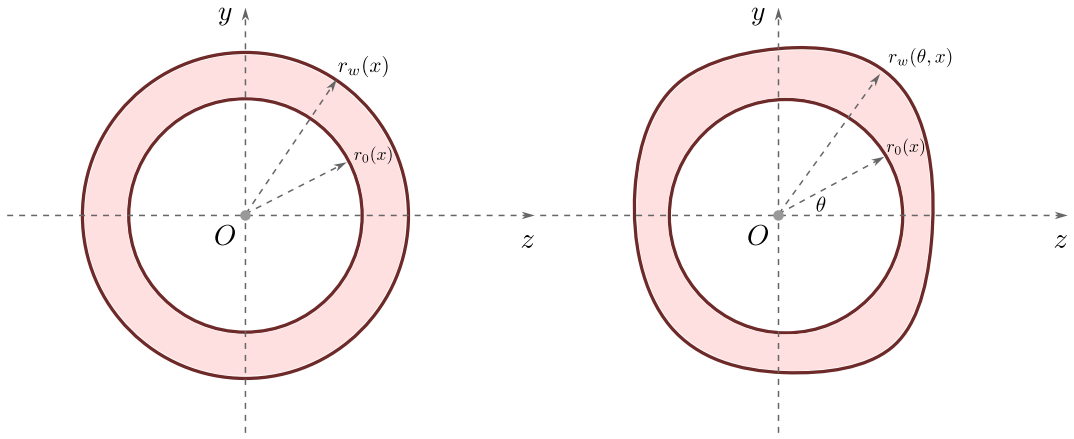
Finally, we apply the long wave approximation (Dutykh and Dias, 2007a; Khakimzyanov et al., 2020). This simplifying assumption can be justified in the following way. Indeed, the Moens–Korteweg celerity $c_{MK} = O\left(10^2\right)$ cm/s (Tijsseling and Anderson, 2012; Lagrée, 2000) and the hear pulsation period is $t_h = O\left(1\right)$ s. Consequently, the pulse wavelength can be estimated to be $\lambda = c_{MK} \cdot t_h = O\left(10^2\right)$ cm, which is much larger than a typical diameter or arteries approximately equal to $O\left(1\right)$ cm. Henceforth, the long wave assumption can be justified in this way. It is also well known that long waves are much less affected by viscosity (Boussinesq, 1895; Lamb, 1932; Dutykh and Dias, 2007a), which allows us to adopt the ideal fluid assumption. If necessary, the dissipative effects can be added later along the lines of our previous works on this topic (Dutykh and Dias, 2007b; Dutykh, 2009b,a; Mitsotakis et al., 2019). We

[2] Rough estimates show that the Mach number for blood flows is of the order of $O\left(10^{-3}\right)$ which justifies the incompressibility assumption.

[3] The fluid homogeneity assumption is justified by comparing the typical pressure variations with the bulk modulus of the blood. A simple computation shows that the relative variation of the density is of the order of $O\left(5 \times 10^{-6}\right)$, which justifies our homogeneity assumption.

[4] The laminarity assumption can be discussed further. Despite the relatively high velocities in large arteries, the blood viscosity and the vessels' small diameter keep the Reynolds number in a range where laminar flow predominates. In addition, blood flow in the arteries is pulsatile due to rhythmic pumping of the heart. This pulsatility can actually stabilize the flow and delay the transition to turbulence.

[5] This approximate symmetry applies particularly to major arteries, such as the aorta and carotid arteries, which are approximately cylindrical with a nearly circular cross-section under normal physiological conditions. The lack of curvature minimizes asymmetric flow disturbances in straight sections of the arteries, supporting the axisymmetric assumption.



(a) An axi-symmetric deformation.

(b) A generic deformation.

FIGURE 2 A schematic illustration of an axi-symmetric (left panel) and a generic (right panel) deformations of initially cylindrical vessel cross-section. In the sequel, we are going to employ the polar coordinates in the Oyz plane.

would like to mention that the long wave approximation is not new *per se*. However, historically, it has been used to derive hydrostatic models, *cf.* Lambert (1958); Formaggia et al. (2003); Quarteroni and Formaggia (2004); Delestre and Lagrée (2013). The models resulting from this approach are usually of hyperbolic type similar to Saint-Venant equations originating from hydraulics and water pipe flow modeling (Bourdarias and Gerbi, 2007, 2008; Bourdarias et al., 2012). The peculiarity of our approach is that we go one step further in the hierarchy of models, namely, we include frequency dispersion effects (Khakimzyanov et al., 2018).

2.1 | Base model: dimensional formulation

Taking into account the simplifying assumptions stated above, we can finally formulate our base model in scaled (physical and dimensional) variables by following Mitsotakis et al. (2018, 2019):

$$u_t + u u_x + v u_r + \frac{1}{\rho} p_x = 0, \tag{1}$$

$$v_t + u v_x + v v_r + \frac{1}{\rho} p_r = 0, \tag{2}$$

$$r u_x + (r v)_r = 0, \tag{3}$$

where ρ is the constant fluid density, u and v are horizontal and radial particle velocity vector components, respectively, and p is the fluid pressure. The subscripts with independent variables denote the partial derivatives with respect to those variables. We denote by $r^w(x,t)$ the distance between the cylinder axis and the elastic vessel's wall at a distance x and at time instance $t \geqslant 0$. The static equilibrium position of the elastic wall is denoted by $r_0(x)$. See the sketch of the fluid/solid domain in Figure 1. The elastic wall radial excursion will be denoted by $\eta(x,t)$ and the following relation holds:

$$r^w(x,t) = r_0(x) + \eta(x,t).$$

Equations (1) to (3) should be supplemented with appropriate initial and boundary conditions to obtain a well-posed[6] IBVP. In order to avoid some obvious singularities, we impose the following condition:

$$v\,(x,\,0,\,t)\;=\;0\,.$$

The elastic wall kinematic impermeability condition reads:

$$\eta_t\;+\;r_x^w\cdot u\;=\;v\qquad\text{at}\qquad r\;=\;r^w\,.$$

The dynamic boundary condition is nothing else but the expression of the force balance written on vessel walls. Modelling the wall as a Kelvin–Voigt viscoelastic solid, the elastic restoring stress is supplemented by a viscous stress proportional to the rate of deformation η_t, so that

$$\rho^w\,h\,\eta_{tt}\;=\;p^w\;-\;\frac{E\,h}{r_0^2\,(1\;-\;\nu^2)}\,\left(\eta\;+\;\gamma^*\,\eta_t\right)\qquad\text{at}\qquad r\;=\;r^w\,,$$

where ρ^w is the wall density, p^w is the pressure exerted by the fluid on the elastic wall, h is the wall thickness, E and ν characterize the elastic properties (Young modulus and Poisson ratio correspondingly) of the vessel, and $\gamma^*\;\geqslant\;0$ is the wall *stress-relaxation time* (with the dimension of time) characterising the wall viscoelasticity; the purely elastic wall is recovered in the limit $\gamma^*\;=\;0$. Together with the fluid Rayleigh damping introduced below, this term accounts for the dissipative (non-conservative) mechanisms of the model. In this study, we assume for simplicity that E, ν, ρ^w, h, and γ^* are constants.

2.2 | Potential model: dimensional formulation

We next recast the governing equations in terms of a velocity potential. Assuming the flow to be incompressible, irrotational, and axisymmetric, we introduce a smooth potential

$$\phi\;=\;\phi(x,r,t),\qquad(u,v)^\top\;=\;\nabla\phi\;=\;(\phi_x,\phi_r)^\top,$$

so that the Euler equations reduce to the Cauchy–Lagrange integral on the moving wall (Mitsotakis et al., 2019):

$$\phi_t+\tfrac{1}{2}\,\phi_x^2+\tfrac{1}{2}\,\phi_r^2+\frac{1}{\rho}\,p+\kappa\,\phi\;=\;0,\qquad r=r^w. \tag{4}$$

Here the instantaneous wall position is $r^w\;=\;r_0(x)+\eta(x,t)$, with r_0 the unperturbed radius and η the wall displacement. Under the potential-flow assumption the continuity equation becomes

$$r\,\phi_{xx}\;+\;\left(r\,\phi_r\right)_r\;=\;0,\qquad 0\;<\;r\;<\;r^w\,, \tag{5}$$

[6]The well-posedness proof for this Initial Boundary Value Problem (IBVP) is left to future works on this subject.

while the kinematic boundary conditions read

$$\phi_r = 0, \qquad r = 0, \tag{6}$$

$$\phi_r = \eta_t + r_x^w \phi_x, \qquad r = r^w(x,t). \tag{7}$$

Equations (4)–(7) furnish the starting point for the long-wave asymptotics carried out in the sequel of this study.

2.2.1 | Variational principles

Two complementary variational frameworks are commonly employed for irrotational surface-wave problems: Luke's Lagrangian formulation (Luke, 1967) and the Hamiltonian formulation of Petrov and Zakharov (Petrov, 1964; Zakharov, 1968). Surveys by Radder (1999) and Zakharov and Kuznetsov (1997) provide extensive historical background. Luke's principle incorporates the velocity potential directly and imposes irrotationality *a priori*, whereas Zakharov's Hamiltonian requires both incompressibility and bottom impermeability to be satisfied identically, thereby rendering it more restrictive. In either case the free-surface conditions arise from the stationarity of an action functional in which part of the bulk dynamics is built in exactly and the remainder is obtained by variation. The earliest use of a pressure-based Lagrangian is due to Clebsch (1859) and Hargreaves (1908), although the concomitant free-surface boundary conditions were not explicitly recognised at the time. The following derivation highlights this connection. Recently, the variational techniques have been successfully applied to model water waves in various asymptotic regimes (Clamond and Dutykh, 2012; Clamond et al., 2024).

Lagrangian formulation.

First, for the irrotational case, let $\phi(x, r, t)$ be the velocity potential of a fluid lying between $r = 0$ and $r = r^w(x, t)$. The variational principle (Luke, 1967) is given by

$$\delta J = \delta \int_{t_1}^{t_2} \int_{x_1}^{x_2} \mathcal{L}\, \mathrm{d}x\, \mathrm{d}t = 0, \quad \text{with} \quad \mathcal{L} = \int_0^{r^w} \left(\phi_t + \frac{1}{2}\phi_x^2 + \frac{1}{2}\phi_r^2\right) r\, \mathrm{d}r - \frac{1}{2}\alpha\eta_t^2 - \frac{1}{2}\beta\eta^2 + p^w\eta, \tag{8}$$

where $\phi(x, r, t)$ and $r^w(x, t)$ are allowed to vary subject to the restrictions $\delta\phi = 0$ and $\delta r^w = 0$ at x_1, x_2, t_1 and t_2. The only difference from classical formulations using pressure expressions is that $r^w(x, t)$ variations are permitted here. Following standard procedures from the calculus of variations, equation (8) becomes

$$\begin{aligned}\delta J = \int_{t_1}^{t_2} \int_{x_1}^{x_2} \Bigg[\left(\phi_t + \frac{1}{2}\phi_x^2 + \frac{1}{2}\phi_r^2\right) r \Big|_{r=r^w} \delta r^w \\ + \int_0^{r^w} (\phi_x\, \delta\phi_x + \phi_r\, \delta\phi_r + \delta\phi_t)\, r\, \mathrm{d}r \Bigg] \mathrm{d}x\, \mathrm{d}t \\ + \delta \int_{t_1}^{t_2} \int_{x_1}^{x_2} \left(\frac{1}{2}\alpha\eta_t^2 - \frac{1}{2}\beta\eta^2 - p^w\eta\right) \mathrm{d}x\, \mathrm{d}t = 0. \end{aligned} \tag{9}$$

Natural boundary conditions at $r = 0$ and $r = r^w$ follow from the careful integration of (8). We consider the following decomposition:

1. $\displaystyle\int_0^{r^w} r\,\phi_x\,\delta\phi_x\,\mathrm{d}r = \int_0^{r^w} \left[\, r\,\phi_x\,\delta\phi \,\right]_x \mathrm{d}r - \int_0^{r^w} r\,\phi_{xx}\,\delta\phi\,\mathrm{d}r = -\,r_x^w\,r^w\,\phi_x\,\delta\phi\Big|_{r=r^w} + \partial_x \int_0^{r^w} r\,\phi_x\,\delta\phi\,\mathrm{d}r\,,$

2. $\displaystyle\int_0^{r^w} r\,\phi_r\,\delta\phi_r\,\mathrm{d}r = \left[\, r\,\phi_r\,\delta\phi \,\right]_{r=0}^{r=r^w} - \int_0^{r^w} \left(\phi_r + r\,\phi_{rr}\right)\delta\phi\,\mathrm{d}r\,,$

3. $\displaystyle\int_0^{r^w} r\,\delta\phi_t\,\mathrm{d}r = \int_0^{r^w} \partial_t\left(r\,\delta\phi\right)\mathrm{d}r = -\,r^w\,r_t\,\delta\phi\Big|_{r=r^w} + \partial_t \int_0^{r^w} r\,\delta\phi\,\mathrm{d}r\,.$

Additionally, we compute

$$\delta \int_{t_1}^{t_2}\int_{x_1}^{x_2}\left(-\frac{1}{2}\,\alpha\,\eta_t^2 - \frac{1}{2}\,\beta\,\eta^2 + p^w\,\eta\right)\mathrm{d}x\,\mathrm{d}t = \int_{t_1}^{t_2}\int_{x_1}^{x_2}\left(-\,\alpha\,\eta_t\,\delta\eta_t - \beta\,\eta\,\delta\eta + p^w\,\delta\eta\right)\mathrm{d}x\,\mathrm{d}t\,.$$

Substituting all these contributions into (8), we obtain

$$\begin{aligned}\delta J = \int_{t_1}^{t_2}\int_{x_1}^{x_2}\Bigg[&\left(\phi_t + \frac{1}{2}\,\phi_x^2 + \frac{1}{2}\,\phi_r^2\right) r\Big|_{r=r^w}\delta r^w \\ &- \left[\, r\,\phi_r\,\delta\phi\,\right]_{r=0} - \int_0^{r^w}\left(r\,\phi_{xx} + \phi_r + r\,\phi_{rr}\right)\delta\phi\,\mathrm{d}r\Bigg]\mathrm{d}x\,\mathrm{d}t \\ &+ \int_{t_1}^{t_2}\int_{x_1}^{x_2}\left(-\,\alpha\,\eta_{tt} - \beta\,\eta + p^w\right)\delta\eta\,\mathrm{d}x\,\mathrm{d}t = 0\,.\end{aligned}$$

Taking $\delta r^w = 0$ and assuming $\delta\phi\big|_{r=r^w} = \delta\phi\big|_{r=0} = 0$, and noting that $\delta\phi$ is otherwise arbitrary, we deduce the following system:

$$\phi_t + \frac{1}{2}\,\phi_x^2 + \frac{1}{2}\,\phi_r^2 - \alpha\,\eta_{tt} - \beta\,\eta + p^w = 0\,, \qquad r = r^w(x,t)\,, \tag{10}$$

$$r\,\phi_{xx} + \left(r\,\phi_r\right)_r = 0\,, \qquad 0 < r < r^w(x,t)\,, \tag{11}$$

$$\phi_r = 0\,, \qquad r = 0\,, \tag{12}$$

$$\phi_r = \eta_t + r_x^w\,\phi_x\,, \qquad r = r^w(x,t)\,, \tag{13}$$

$$p^w = \alpha\,\eta_{tt} + \beta\,\eta\,, \qquad r = r^w(x,t)\,. \tag{14}$$

Thus, we observe that the variational principle (8) elegantly recovers the complete potential flow formulation for our problem. Equation (11) represents the Laplace equation in cylindrical coordinates, which governs the irrotational flow in the fluid domain. The boundary conditions are given by the impermeability condition at the axis of symmetry (12) and the kinematic condition at the vessel wall (13). The dynamic condition (10) corresponds to the Cauchy–Lagrange integral at the vessel wall, while (14) describes the pressure law for the viscoelastic wall. This demonstrates the power of variational methods in deriving the governing equations for fluid-structure interaction problems in a systematic and physically consistent manner.

Hamiltonian formulation.

This section belongs to the relatively recent trend of applying concepts from Hamiltonian mechanics to problems in fluid dynamics. By 'Hamiltonian mechanics', we refer to the entirety of classical mechanics in the spirit of foundational texts such as Lanczos (1970); Goldstein et al. (2001); Arnold (1997), encompassing both particle mechanics and field theory. Hamiltonian approaches have proven indispensable in classical and quantum theories of particles and fields. If

their only contribution were a reformulation of known results, they would be of limited value. However, substantial evidence indicates that classical mechanics methods offer a powerful and unifying framework in fluid dynamics as well. Several factors account for the effectiveness of Hamiltonian approaches. Firstly, Hamilton's formalism provides an elegant and economical representation of dynamics. Secondly, the well-established connection between symmetries of the Hamiltonian and conservation laws facilitates the construction of approximate models that preserve analogues of exact invariants. Thirdly, Hamiltonian formulations are coordinate-invariant and thus adaptable to diverse geometrical settings. In Hamiltonian perturbation theory, for instance, dynamical approximations are naturally associated with transformations of the dependent variables. This flexibility enables one to simultaneously adjust the physical description and its mathematical representation to obtain models in their simplest possible form.

We now consider the motion governed by equations (10)–(14). The total energy of the system comprises the kinetic energy, potential energy, and surface energy, and is normalised such that the total energy vanishes in the quiescent state. The evolution equations take the abstract Hamiltonian form

$$\begin{pmatrix} \eta_t \\ \psi_t \end{pmatrix} = \mathcal{J} \begin{pmatrix} \dfrac{\delta \mathcal{H}}{\delta \eta} \\ \dfrac{\delta \mathcal{H}}{\delta \psi} \end{pmatrix}, \qquad \mathcal{J} = \begin{pmatrix} 0 & \mathbb{I} \\ -\mathbb{I} & 0 \end{pmatrix}, \tag{15}$$

where $\delta \mathcal{H}$ denotes the Gâteaux (variational) derivative, $\psi \equiv \phi(x, r, t)|_{r = r^w(x,t)}$ is the surface potential, and the Hamiltonian density and total energy are given by

$$H = \frac{1}{2} \int_0^{\eta} |\nabla \phi|^2 \, \mathrm{d}r + \left(-\frac{1}{2} \alpha \eta_t^2 - \frac{1}{2} \beta \eta^2 + p^w \eta \right), \qquad \mathcal{H} = \int_{\mathbb{R}} H \, \mathrm{d}x \,. \tag{16}$$

We now compute the variation of the Hamiltonian functional under the perturbation $\eta \mapsto \eta + \delta\eta$:

$$\begin{aligned} \mathcal{H}[\eta + \delta\eta, \phi] - \mathcal{H}[\eta, \phi] = {} & \frac{1}{2} \int_{\mathbb{R}} \int_0^{\eta + \delta\eta} |\nabla \phi|^2 \, \mathrm{d}r \, \mathrm{d}x \\ & - \frac{1}{2} \alpha \int_{\mathbb{R}} (\eta + \delta\eta)_t^2 \, \mathrm{d}x - \frac{1}{2} \beta \int_{\mathbb{R}} (\eta + \delta\eta)^2 \, \mathrm{d}x + \int_{\mathbb{R}} p^w (\eta + \delta\eta) \, \mathrm{d}x \\ & - \frac{1}{2} \int_{\mathbb{R}} \int_0^{\eta} |\nabla \phi|^2 \, \mathrm{d}r \, \mathrm{d}x + \frac{1}{2} \alpha \int_{\mathbb{R}} \eta_t^2 \, \mathrm{d}x + \frac{1}{2} \beta \int_{\mathbb{R}} \eta^2 \, \mathrm{d}x - \int_{\mathbb{R}} p^w \eta \, \mathrm{d}x \,. \end{aligned}$$

After simplification, we obtain

$$\mathcal{H}[\eta + \delta\eta, \phi] - \mathcal{H}[\eta, \phi] = \frac{1}{2} \int_{\mathbb{R}} \int_{\eta}^{\eta + \delta\eta} |\nabla \phi|^2 \, \mathrm{d}r \, \mathrm{d}x - \alpha \int_{\mathbb{R}} \eta_t \, \delta\eta_t \, \mathrm{d}x - \beta \int_{\mathbb{R}} \eta \, \delta\eta \, \mathrm{d}x + \int_{\mathbb{R}} p^w \, \delta\eta \, \mathrm{d}x \,.$$

Applying the mean value theorem to the integral over the interval $[\eta, \eta + \delta\eta]$, we finally obtain

$$\mathcal{H}[\eta + \delta\eta, \phi] - \mathcal{H}[\eta, \phi] = \frac{1}{2} \int_{\mathbb{R}} \left[|\nabla \phi|^2 \right]_{r = r^w} \delta\eta \, \mathrm{d}x - \alpha \int_{\mathbb{R}} \eta_{tt} \, \delta\eta \, \mathrm{d}x - \beta \int_{\mathbb{R}} \eta \, \delta\eta \, \mathrm{d}x + \int_{\mathbb{R}} p^w \, \delta\eta \, \mathrm{d}x \,.$$

Therefore, the non-canonical Hamiltonian evolution equations take the form

$$\begin{pmatrix} \eta_t \\ \psi_t \end{pmatrix} = \begin{pmatrix} \phi_r - r_x^w \phi_x \\ -\frac{1}{2} |\nabla \phi|^2 + \alpha \eta_{tt} + \beta \eta - p^w \end{pmatrix}, \tag{17}$$

which recover the dynamic and kinematic boundary conditions (10) and (13), respectively. It is important to emphasise that the operator $\mathcal{J}$ is skew-symmetric and satisfies the Jacobi identity (cf. Nutku (1987)), thus qualifying as a legitimate Hamiltonian operator.

In conclusion, the derivation and analysis of the non-canonical Hamiltonian structure provide a robust and systematic framework for investigating complex dynamical systems governed by nonlinear wave equations. The geometric properties of the Hamiltonian operator, including skew-symmetry and the Jacobi identity, ensure consistency with the underlying physics. This insight not only strengthens the theoretical foundations but also enhances the practical modelling of nonlinear dispersive systems in mathematical physics.

2.3 | Potential model: dimensionless formulation and the first derivation of the long wave model

To model blood flow in a vessel, we derive simplified equations from the Euler equations using non-dimensional variables and asymptotic techniques. This approach, relevant to arterial pulse propagation, accounts for nonlinearity and dispersion while simplifying the complex fluid-structure interaction at the vessel wall (Fung, 1993).

We introduce non-dimensional and scaled variables to normalize the Euler equations:

$$\eta^* = \frac{\eta}{a}, \quad x^* = \frac{x}{\lambda}, \quad r^* = \frac{r}{R}, \quad t^* = \frac{t}{T}, \quad \phi^* = \frac{\phi}{\lambda\varepsilon\tilde{c}}, \quad p^* = \frac{p}{\varepsilon\rho\tilde{c}^2}, \tag{18}$$

where η is the vessel wall displacement, a is the typical wall deviation, λ is the typical wavelength, R is the vessel radius, $T = \lambda/\tilde{c}$ is the time scale, $\tilde{c} = \sqrt{Eh/2\rho R}$ is the Moens-Korteweg speed, ϕ is the velocity potential, p is the pressure, ρ is the fluid density, E is the Young's modulus, and h is the wall thickness (Fung, 1993). The non-dimensional parameters are:

$$\varepsilon = \frac{a}{R}, \qquad \delta = \frac{R}{\lambda}, \tag{19}$$

where $\varepsilon = O(1)$ characterizes nonlinearity, and $\delta^2 \ll 1$ indicates weak dispersion. The external pressure is assumed to be zero, typical for blood flow models.

In non-dimensional form (dropping asterisks for simplicity), the governing equations and boundary conditions are:

$$\partial_t\phi + \frac{\varepsilon}{2}(\partial_x\phi)^2 + \frac{\varepsilon}{2\delta^2}(\partial_r\phi)^2 + p + \varepsilon\kappa\phi = 0, \quad \text{at } r = r^w, \tag{20}$$

$$\delta^2 r\partial_x^2\phi + \partial_r(r\partial_r\phi) = 0, \quad 0 < r < r^w, \tag{21}$$

$$\partial_r\phi = 0, \quad \text{at } r = 0, \tag{22}$$

$$\partial_r\phi = \delta^2\partial_t\eta + \delta^2[r_0(x) + \varepsilon\eta]\partial_x\phi, \quad \text{at } r = r^w, \tag{23}$$

$$p = \alpha\delta^2\partial_t^2\eta + \beta(\eta + \delta^2\gamma\partial_t\eta), \quad \text{at } r = r^w, \tag{24}$$

where $r^w = r_0(x) + \varepsilon\eta$ is the non-dimensional vessel radius, $\kappa = \kappa^*\tilde{c}\varepsilon/\lambda$ is a damping coefficient, $\alpha = \rho_w h/\rho R$, $\beta = 2R^2/r_0^2(x)$, $\gamma = \gamma^*\delta^2 T$, and ρ_w is the wall density (*cf.* Mitsotakis et al. (2019)). Here γ^* is the dimensional wall stress-relaxation time of the Kelvin–Voigt constitutive law introduced above, and γ is its dimensionless counterpart, scaled by $\delta^2 T$: it is the (non-negative) viscoelastic damping parameter that measures the strength of energy dissipation by the vessel wall. The viscoelastic contribution $\delta^2\beta\gamma\,\partial_t\eta$ in the pressure law (24) is therefore a weakly dispersive correction, of order δ^2, to the elastic wall response $\beta\eta$, and the elastic wall is recovered as $\gamma \to 0$. Following standard

asymptotic techniques (Bona et al., 2002), we expand the velocity potential ϕ as:

$$\phi(x,r,t) = \sum_{m=0}^{\infty} r^m \phi_m(x,t). \tag{25}$$

Substituting into (21), we obtain the recurrence relation:

$$\delta^2 \partial_x^2 \phi_{2m} + (2m+2)^2 \phi_{2m+2} = 0, \quad \phi_{2m+1} = 0, \tag{26}$$

for $m = 0, 1, 2, \ldots$. This yields

$$\phi_2 = -\frac{\delta^2}{4} \partial_x^2 \phi_0, \tag{27}$$

and for $m = 1, 2, \ldots$:

$$\phi_{2m+2} = \frac{\delta^4}{(2m+2)^2 (2m)^2} \partial_x^4 \phi_{2m-2} = O(\delta^4). \tag{28}$$

Thus, the velocity potential up to $O(\delta^6)$ is given by

$$\phi(x,r,t) = \phi_0(x,t) - \delta^2 \frac{r^2}{4} \partial_x^2 \phi_0(x,t) + \delta^4 \frac{r^4}{64} \partial_x^4 \phi_0(x,t) + O(\delta^6). \tag{29}$$

Using (29) in the boundary condition (23) at $r = r^w$, we approximate the mass conservation equation as follows:

$$\partial_t \eta + r_x^w \partial_x \phi_0 + \frac{r^w}{2} \partial_x^2 \phi_0 - \delta^2 \frac{(r^w)^2 r_x^w}{4} \partial_x^3 \phi_0 - \delta^2 \frac{(r^w)^3}{16} \partial_x^4 \phi_0 = O(\delta^4). \tag{30}$$

Substituting (29) and (24) into the Bernoulli equation (20) at $r = r^w$, we eliminate pressure to obtain the momentum equation:

$$\begin{aligned} &\partial_t \phi_0 - \delta^2 \frac{(r^w)^2}{4} \partial_x^2 \partial_t \phi_0 + \frac{\varepsilon}{2} \left[(\partial_x \phi_0)^2 - \delta^2 \frac{(r^w)^2}{2} \partial_x \phi_0 \partial_x^3 \phi_0 \right] + \frac{\varepsilon \delta^2}{8} (r^w)^2 (\partial_x^2 \phi_0)^2 \\ &+ \varepsilon \kappa \left[\phi_0 - \delta^2 \frac{(r^w)^2}{4} \partial_x^2 \phi_0 \right] + \alpha \delta^2 \partial_t^2 \eta + \beta (\eta + \delta^2 \gamma \partial_t \eta) = O(\delta^4). \end{aligned} \tag{31}$$

Defining the horizontal velocity at the vessel center as $w(x,t) = \partial_x \phi_0(x,t)$, we rewrite (30) and (31) as

$$\partial_t \eta + r_x^w w + \frac{r^w}{2} \partial_x w - \delta^2 \frac{(r^w)^2 r_x^w}{4} \partial_x^2 w - \delta^2 \frac{(r^w)^3}{16} \partial_x^3 w = O(\delta^4), \tag{32}$$

$$\partial_t w + \partial_x (\beta \eta) + \varepsilon w \partial_x w - \frac{\delta^2}{4} \partial_x \left[(r^w)^2 \Gamma \right] + \varepsilon \kappa \left[w - \frac{\delta^2}{4} \partial_x \left((r^w)^2 \partial_x w \right) \right] + \alpha \delta^2 \partial_x \partial_t^2 \eta + \delta^2 \gamma \partial_x (\beta \partial_t \eta) = O(\delta^4), \tag{33}$$

where

$$\Gamma = \partial_x \partial_t w + \varepsilon w \partial_x^2 w - \frac{\varepsilon}{2} (\partial_x w)^2. \tag{34}$$

The system (32)–(33) resembles a Serre-type system, capturing the leading-order dynamics of blood flow in a compliant vessel. The equations account for nonlinearity (ε), dispersion (δ^2), and vessel wall effects (α, β, γ). Higher-order approximations of the velocity potential could further refine this model, enhancing its accuracy for physiological applications (Mitsotakis et al., 2019).

2.3.1 | Enhanced Serre-type system

In this section, we refine the Serre-type system derived in (32)–(33) by evaluating the horizontal velocity at any radius within the vessel, improving the model's applicability to blood flow dynamics. We also briefly introduce a second approach using incompressible Euler equations, which will be explored further in subsequent analyses.

To enhance the system (32)–(33), we consider the horizontal velocity at any radius $r = \theta r^w$, where $0 \leqslant \theta \leqslant 1$ and $r^w = r_0(x) + \varepsilon\eta$ is the non-dimensional vessel radius. From the velocity potential expansion (29), the horizontal velocity is:

$$u(x,r,t) = \partial_x\phi = \partial_x\phi_0 - \delta^2\frac{r^2}{4}\partial_x^3\phi_0 + O(\delta^4).$$

Defining $w(x,t) = \partial_x\phi_0(x,t)$ and evaluating at $r = \theta r^w$, we obtain

$$u^\theta(x,t) \doteq u(x,\theta r^w,t) = w(x,t) - \delta^2\frac{\theta^2(r^w)^2}{4}\partial_x^2 w(x,t) + O(\delta^4). \tag{35}$$

Inverting this, we express w in terms of u^θ:

$$w(x,t) = u^\theta(x,t) + \delta^2\frac{\theta^2(r^w)^2}{4}\partial_x^2 u^\theta(x,t) + O(\delta^4). \tag{36}$$

Substituting (36) into the mass equation (32), we obtain

$$\partial_t\eta + r_x^w u^\theta + \frac{r^w}{2}\partial_x u^\theta + \delta^2\frac{(2\theta^2-1)(r^w)^2 r_x^w}{4}\partial_x^2 u^\theta + \delta^2\frac{(2\theta^2-1)(r^w)^3}{16}\partial_x^3 u^\theta = O(\delta^4). \tag{37}$$

Using the vessel wall velocity $r_t^w = \varepsilon\partial_t\eta$ and approximating from (37):

$$r_t^w = \varepsilon\partial_t\eta = -\varepsilon r_x^w u^\theta - \varepsilon\frac{r^w}{2}\partial_x u^\theta + O(\varepsilon\delta^2), \tag{38}$$

we substitute (36) into the momentum equation (33) to obtain

$$\partial_t u^\theta + \partial_x(\beta\eta) + \varepsilon u^\theta\partial_x u^\theta + \frac{\delta^2}{4}\left\{\theta^2(r^w)^2\partial_x\Gamma - \partial_x[(r^w)^2\Gamma]\right\} + \varepsilon\kappa Q + R = O(\delta^4), \tag{39}$$

where

$$\begin{aligned}
\Gamma &= \partial_x\partial_t u^\theta + \varepsilon u^\theta\partial_x^2 u^\theta - \frac{\varepsilon}{2}(\partial_x u^\theta)^2,\\
Q &= u^\theta + \frac{\delta^2}{4}(\theta^2-1)(r^w)^2\partial_x^2 u^\theta - \delta^2 r^w r_x^w\partial_x u^\theta,\\
R &= \alpha\delta^2\partial_x\partial_t^2\eta + \delta^2\gamma\partial_x(\beta\partial_t\eta).
\end{aligned} \tag{40}$$

Note, that from (37) $\partial_t \eta = -\left(r_x^w u^\theta + \frac{r^w}{2}\partial_x u^\theta\right) + O(\delta^2)$. Using this approximation, we can replace the quantity R by the most convenient for numerical studies quantity

$$\tilde{R} = -\delta^2 \partial_x \left[\alpha \partial_t \left(r_x^w u^\theta + \frac{r^w}{2}\partial_x u^\theta\right) + \gamma\beta \left(r_x^w u^\theta + \frac{r^w}{2}\partial_x u^\theta\right)\right] = R + O(\delta^4)\,, \tag{41}$$

where r_t^w can be substituted by (38) whenever it appears in (41).

Choosing $\theta^2 = 1/2$ simplifies (37) and (39), yielding a familiar Serre-type formulation that balances dispersion and nonlinearity effectively for blood flow modeling. The refined system (37)–(39) generalizes the velocity to any radial position, enhancing the model's flexibility for capturing blood flow variations across the vessel cross-section. The parameter θ allows tuning to specific radial locations, with $\theta^2 = \frac{1}{2}$ recovering a standard Serre-type system. The terms involving Γ, Q, and R account for higher-order dispersive, nonlinear, and wall interaction effects, critical for accurate modeling of arterial waves (Mitsotakis et al., 2019).

As an alternative approach, the mathematical modeling of blood flow suggests using the incompressible, ideal, and radially symmetric Euler equations. This framework, which assumes inviscid flow, will be developed further to complement the asymptotic system derived above, offering additional insights into arterial pulse dynamics.

2.4 | Base model: dimensionless formulation

In this Section, we elaborate on the scaled formulation for the Euler equations in cylindrical coordinates presented above. The first step consists of introducing non-dimensional independent variables, which are defined as follows:

$$\eta^\star = \frac{\eta}{a}\,, \quad x^\star = \frac{x}{\lambda}\,, \quad r^\star = \frac{r}{R}\,, \quad t^\star = \frac{t}{T}\,, \quad u^\star = \frac{1}{\varepsilon\,\tilde{c}}\,u, \quad v^\star = \frac{1}{\varepsilon\,\delta\,\tilde{c}}\,v, \quad p^\star = \frac{1}{\varepsilon\,\rho\,\tilde{c}^2}\,p\,,$$

where a is a typical amplitude of the vessel wall displacement, λ a typical wavelength of a pulse, R is a vessel's typical radius, $t = \lambda / \tilde{c}$ the characteristic timescale, while $\tilde{c} = \sqrt{E\,h/2\rho R}$ is the Moens-Korteweg characteristic speed (Fung, 1997). The parameters ε and δ characterize the non-linearity and the non-hydrostatic effects in the system:

$$\varepsilon \coloneqq \frac{a}{R}\,, \quad \delta \coloneqq \frac{R}{\lambda}\,.$$

In the present work, we aim to keep $\varepsilon = O(1)$ as long as possible, while assuming from the outset that $\delta^2 \ll 1$ is a small parameter.

Omitting the $\star$ symbol from the notation below for the sake of simplicity, the non-dimensional Euler equations take the form (Mitsotakis et al., 2018)

$$u_t + \varepsilon\,u\,u_x + \varepsilon\,v\,u_r + p_x = 0\,, \tag{42}$$

$$\delta^2\,(v_t + \varepsilon\,u\,v_x + \varepsilon\,v\,v_r) + p_r = 0\,, \tag{43}$$

$$r\,u_x + (r\,v)_r = 0\,, \tag{44}$$

where $0 < r < r^w \coloneqq r_0 + \varepsilon\,\eta$. We also add the dimensionless flow irrotationality condition that will be needed in

our developments below (Landau and Lifshitz, 1987):

$$\delta^2 v_x = u_r . \tag{45}$$

The scaled version of the kinematic boundary condition reads

$$v(x, r^w, t) = \eta_t(x, t) + r_x^w u(x, r^w, t), \tag{46}$$

while the dynamic boundary condition reads

$$p^w(x, t) = p(x, r^w, t) = \alpha\delta^2 \eta_{tt}(x, t) + \beta(x)\eta(x, t) + \delta^2\beta\gamma\eta_t . \tag{47}$$

The non-singularity condition along the symmetry axis reads:

$$v(x, 0, t) = 0. \tag{48}$$

Above, we introduced the following notation:

$$\tilde{\alpha} := \frac{\rho^w h}{\rho} \quad \text{and} \quad \tilde{\beta}(x) := \frac{E h}{\rho r_0^2(x)} .$$

Then,

$$\alpha := \frac{\tilde{\alpha}}{R}, \qquad \beta(x) := \frac{2R^2\rho}{E h}\tilde{\beta}(x) \quad \text{and} \quad \gamma := \frac{\tilde{\gamma}}{\delta^2 T} ,$$

where ρ^w is the wall density, h is the thickness of the vessel wall, E is the young modulus of elasticity, R is a vessel's radius, $T := \lambda/c$ is the characteristic timescale. Below we use the scaled governing equations in order to propose a simplified model.

2.5 | Derivation of a new fully nonlinear weakly dispersive model

In this Section, we perform the dimensionality reduction of the axisymmetric Euler system presented above. In order to achieve this goal, we use essentially two mathematical tools: depth averaging and asymptotic expansions. In the derivation below, we follow the footsteps of our previous works on this topic (Dutykh et al., 2013b; Clamond et al., 2017; Khakimzyanov et al., 2020; Clamond et al., 2024).

The horizontal velocity is approximated by its depth-averaged value

$$\bar{u}(x, t) := \frac{1}{r_0 + \varepsilon\eta}\int_0^{r^w} u(x, r, t)\,\mathrm{d}r . \tag{49}$$

We start the derivation by integrating the continuity Equation (44):

$$\int_0^{r^w} r u_x \,\mathrm{d}r + \int_0^{r^w} (r v)_r \,\mathrm{d}r = 0,$$

where the second term can be readily integrated:

$$\int_0^{r^w} r u_x \, \mathrm{d}r + r^w v(x, r^w, t) - r^w v(x, 0, t) = 0.$$

After taking into account boundary conditions given in Equations (46) and (48), we obtain

$$\int_0^{r^w} r u_x \, \mathrm{d}r + r^w \eta_t(x, t) + r^w r_x^w u(x, r^w, t) = 0. \tag{50}$$

In a similar way, we integrate the axial momentum balance Equation (42):

$$\int_0^{r^w} u_t \, \mathrm{d}r + \varepsilon \int_0^{r^w} u u_x \, \mathrm{d}r + \varepsilon \int_0^{r^w} v u_r \, \mathrm{d}r = -\int_0^{r^w} p_x \, \mathrm{d}r. \tag{51}$$

The integral of the pressure on the right-hand side can be transformed using the classical Leibniz rule and the dynamic boundary condition expressed in Equation (47) to yield

$$\int_0^{r^w} p_x \, \mathrm{d}r = \frac{\partial}{\partial x} \int_0^{r^w} p \, \mathrm{d}r - p(x, r^w, t) r_x^w = \left[r^w \bar{p}\right]_x - \left(\alpha \delta^2 \eta_{tt} + \eta\beta + \delta^2 \beta\gamma \eta_t\right) r_x^w, \tag{52}$$

where $\bar{p}$ denotes the depth-averaged value of the pressure defined similarly to Equation (49). In order to calculate $\bar{p}$, we use the radial momentum balance Equation (43) recast as

$$p_r(x, r, t) = -\delta^2 \Gamma(x, r, t), \qquad \text{with} \qquad \Gamma \coloneqq v_t + \varepsilon u v_x + \varepsilon v v_r.$$

Integrating the derivative p_r from r to r^w, we obtain:

$$p(x, r^w, t) - p(x, r, t) = -\delta^2 \int_r^{r^w} \Gamma(x, z, t) \, \mathrm{d}z$$

and after a slight rewrite

$$-p(x, r, t) = -p(x, r^w, t) - \delta^2 \int_r^{r^w} \Gamma(x, z, t) \, \mathrm{d}z.$$

Integrating the last equation one more time over r from 0 to r^w and by substituting the dynamic boundary condition (47) for $p(x, r^w, t)$, we obtain:

$$-\int_0^{r^w} p(x, r, t) \, \mathrm{d}r = -\int_0^{r^w} \left(\alpha \delta^2 \eta_{tt} + \beta\eta + \delta^2 \beta\gamma \eta_t\right) \mathrm{d}r - \delta^2 \int_0^{r^w} \int_r^{r^w} \Gamma(x, z, t) \, \mathrm{d}z \, \mathrm{d}r.$$

After introducing the depth-averaged pressure $\bar{p}$ and computing the first integral on the right-hand side, we obtain:

$$r^w \bar{p} = \left(\alpha \delta^2 \eta_{tt} + \beta\eta + \delta^2 \beta\gamma \eta_t\right) r^w + \delta^2 \int_0^{r^w} \int_r^{r^w} \Gamma(x, z, t) \, \mathrm{d}z \, \mathrm{d}r.$$

With this result in hands, we can return to Equation (52) to substitute $r^w \bar{p}$ there to find:

$$\left(\alpha\delta^2\eta_{tt} + \beta\eta + \delta^2\beta\gamma\eta_t\right) r_x^w + \left(\alpha\delta^2\eta_{tt} + \beta\eta + \delta^2\beta\gamma\eta_t\right)_x r^w + \delta^2 \frac{\partial}{\partial x}\int_0^{r^w}\int_r^{r^w} \Gamma(x,z,t)\,\mathrm{d}z\,\mathrm{d}r - \left(\alpha\delta^2\eta_{tt} + \beta\eta + \delta^2\beta\gamma\eta_t\right) r_x^w .$$

Finally, we can now determine the integral of p_x:

$$\int_0^{r^w} p_x\,\mathrm{d}r = \left(\alpha\delta^2\eta_{tt} + \beta\eta + \delta^2\beta\gamma\eta_t\right)_x r^w + \delta^2\frac{\partial}{\partial x}\int_0^{r^w}\int_r^{r^w}\Gamma(x,z,t)\,\mathrm{d}z\,\mathrm{d}r .$$

After substituting the last equation into Equation (51), we obtain

$$\int_0^{r^w} u_t\,\mathrm{d}r + \varepsilon\int_0^{r^w} u u_x\,\mathrm{d}r + \left(\alpha\delta^2\eta_{tt} + \beta\eta + \delta^2\beta\gamma\eta_t\right)_x r^w + \delta^2\frac{\partial}{\partial x}\int_0^{r^w}\int_r^{r^w}\Gamma(x,z,t)\,\mathrm{d}z\,\mathrm{d}r = -\varepsilon\int_0^{r^w} v u_r\,\mathrm{d}r .$$

Further progress can be made after we approximate the horizontal velocity variable dependence on the flow depth.

2.5.1 | Velocity field asymptotic expansion

The horizontal velocity variable will be approximated with its second-order Taylor expansion around the cylinder axis $r = 0$:

$$u(x,r,t) = u_0(x,t) - \frac{1}{4}\delta^2 r^2\frac{\partial^2 u_0}{\partial x^2} + O(\delta^4), \tag{53}$$

Notice that the incompressibility condition (44) on the velocity fields implies the following approximation of the second (vertical) velocity component:

$$v(x,r,t) = -\frac{r}{2}\frac{\partial u_0}{\partial x} + O(\delta^2). \tag{54}$$

Averaging of Equation (53) yields the following relation:

$$u_0 = \bar{u} + \frac{1}{12}\delta^2 (r^w)^2\frac{\partial^2\bar{u}}{\partial x^2} + O(\delta^4).$$

Thus, the variable u_0 can be eliminated from Equation (53) for the profit of the depth-averaged horizontal velocity variable:

$$u(x,r,t) = \bar{u} + \frac{1}{12}\delta^2 (r^w)^2\frac{\partial^2\bar{u}}{\partial x^2} - \frac{1}{4}\delta^2 r^2\frac{\partial^2\bar{u}}{\partial x^2} + O(\delta^4). \tag{55}$$

The same substitution can be made in Equation (54) as well:

$$v(x,r,t) = -\frac{r}{2}\frac{\partial\bar{u}}{\partial x} + O(\delta^2). \tag{56}$$

Equation (55) allows us also to approximate the derivative u_x :

$$u_x = \bar{u}_x + \frac{1}{12}\delta^2 2 r_x^w r^w \bar{u}_{xx} + \frac{1}{12}\delta^2 (r^w)^2 \bar{u}_{xxx} - \frac{1}{4}\delta^2 r^2 \bar{u}_{xxx} + O(\delta^4)$$

along with the nonlinear product $u\,u_x$:

$$u\,u_x = \bar{u}\bar{u}_x + \frac{1}{6}\delta^2 r_x^w r^w \bar{u}\bar{u}_{xx} + \frac{1}{12}\delta^2 (r^w)^2 \bar{u}\bar{u}_{xxx} - \frac{1}{4}\delta^2 r^2 \bar{u}\bar{u}_{xxx} + \frac{1}{12}\delta^2 (r^w)^2 \bar{u}_{xx}\bar{u}_x - \frac{1}{4}\delta^2 r^2 \bar{u}_{xx}\bar{u}_x + O(\delta^4).$$

These asymptotic representations allow us to evaluate also the integrals over fluid depth of $u\,u_x$:

$$\int_0^{r^w} u\,u_x \,\mathrm{d}r = r^w \bar{u}\bar{u}_x + \frac{1}{6}\delta^2 r_x^w (r^w)^2 \bar{u}\bar{u}_{xx} + O(\delta^4)$$

and of u_t :

$$\int_0^{r^w} u_t \,\mathrm{d}r = r^w \bar{u}_t + \frac{1}{6}\delta^2 r_t^w (r^w)^2 \bar{u}_{xx} + O(\delta^4).$$

The asymptotic evaluation of the following integral requires the representation (56) along with the irrotationality condition (45):

$$\int_0^{r^w} v\,u_r \,\mathrm{d}r = \delta^2 \int_0^{r^w} v\,v_x \,\mathrm{d}r = \delta^2 \int_0^{r^w} \left(\frac{-r}{2}\frac{\partial \bar{u}}{\partial x}\right)\left(\frac{-r}{2}\frac{\partial^2 \bar{u}}{\partial x^2}\right)\mathrm{d}r + O(\delta^4)$$
$$\equiv \delta^2 \int_0^{r^w} \frac{r^2}{4}\bar{u}_x \bar{u}_{xx}\,\mathrm{d}r + O(\delta^4) = \frac{1}{12}\delta^2 \bar{u}_x \bar{u}_{xx} (r^w)^3 + O(\delta^4).$$

Equation (56) can be substituted into the fluid particle vertical acceleration $\Gamma(x, r, t)$ to obtain the following asymptotic approximation:

$$\Gamma(x, r, t) = v_t + \varepsilon u v_x + \varepsilon v v_r = -\frac{r}{2}\bar{u}_{xt} - \frac{r}{2}\varepsilon \bar{u}\bar{u}_{xx} + \varepsilon \frac{r}{4}(\bar{u}_x)^2 + O(\delta^2) = -\frac{r}{2}\left[\bar{u}_{xt} + \varepsilon \bar{u}\bar{u}_{xx} - \frac{\varepsilon}{2}(\bar{u}_x)^2\right] + O(\delta^2).$$

Combining the previous developments, we can state asymptotically one of the governing equations of the cylindrical Serre–Green–Naghdi (SGN) system:

$$r^w \bar{u}_t + \frac{\delta^2}{6} r_t^w (r^w)^2 \bar{u}_{xx} + \varepsilon r^w \bar{u}\bar{u}_x + \frac{\varepsilon\delta^2}{6} r_x^w (r^w)^2 \bar{u}\bar{u}_{xx} + \left(\alpha\delta^2 \eta_{tt} + \beta\eta + \delta^2 \beta\gamma\eta_t\right)_x r^w + \frac{\varepsilon\delta^2}{12}(r^w)^3 \bar{u}_x \bar{u}_{xx} - \frac{\delta^2}{6}\frac{\partial}{\partial x}\left[(r^w)^3 \left(\bar{u}_{xt} + \varepsilon \bar{u}\bar{u}_{xx} - \frac{\varepsilon}{2}(\bar{u}_x)^2\right]\right] = O(\delta^4).$$

2.5.2 | The mass conservation

In this Section, we are going to derive the second equation of what will become the cylindrical SGN system. Namely, this equation will play the rôle of the mass conservation equation in more classical shallow water-type systems (Dutykh and Clamond, 2016, 2011; Dutykh et al., 2015; Dutykh and Goubet, 2016; Clamond et al., 2019). For this, we return to the integrated version of the continuity Equation (50), and we substitute the approximation (55) for the horizontal velocity:

$$\int_0^{r^w} r\,u_x\,\mathrm{d}r = \int_0^{r^w} r\left(\bar{u} + \left(\frac{1}{12}\,\delta^2\,(r^w)^2 - \frac{1}{4}\,\delta^2 r^2\right)\bar{u}_{xx}\right)_x \mathrm{d}r + O(\delta^4) =$$
$$\bar{u}_x \int_0^{r^w} r\,\mathrm{d}r + \frac{\delta^2}{6}\,r_x^w r^w\,\bar{u}_{xx} \int_0^{r^w} r\,\mathrm{d}r + \frac{\delta^2}{12}\,\bar{u}_{xxx} \int_0^{r^w} r\,(r^w)^2\,\mathrm{d}r - \frac{\delta^2}{4}\,\bar{u}_{xxx} \int_0^{r^w} r^3\,\mathrm{d}r =$$
$$\bar{u}_x\,\frac{(r^w)^2}{2} + \frac{1}{12}\,\delta^2\,(r^w)^3\,r_x^w\,\bar{u}_{xx} + \frac{1}{12}\,\delta^2\,\bar{u}_{xxx}\,\frac{(r^w)^4}{2} - \frac{1}{4}\,\delta^2\,\bar{u}_{xxx}\,\frac{(r^w)^4}{4} + O(\delta^4).$$

After a slight rewriting of the last integral, Equation (50) can be represented asymptotically as

$$\frac{(r^w)^2}{2}\,\bar{u}_x + \frac{\delta^2}{12}\,r_x^w\,(r^w)^3\,\bar{u}_{xx} - \frac{\delta^2}{48}\,(r^w)^4\,\bar{u}_{xxx} + r^w\,\eta_t + r^w\,r_x^w\,u^w(x,t) = O(\delta^4).$$

The last equation after division by r^w constitutes the desired mass conservation equation that will be explicitly written in the following Section.

2.5.3 | Intermediate conclusions

In this Section, we would like to summarize the developments made so far. We also remind that no assumption on the nonlinearity parameter has been made so far, *i.e.* $\varepsilon = O(1)$. On the other hand, all this progress has been made thanks to the long wave assumption that can be expressed as $\delta \ll 1$. We summarize below the two governing equations, which will be the prototype of the future cylindrical SGN system to be presented below after some little additional efforts[7]:

$$\frac{r^w}{2}\,\bar{u}_x + \frac{\delta^2}{12}\,r_x^w\,(r^w)^2\,\bar{u}_{xx} - \frac{\delta^2}{48}\,(r^w)^3\,\bar{u}_{xxx} + \eta_t + r_x^w\,u^w(x,t) = O(\delta^4), \tag{57}$$

$$\bar{u}_t + \frac{\delta^2}{6}\,r_t^w\,r^w\,\bar{u}_{xx} + \varepsilon\,r^w\,\bar{u}\,\bar{u}_x + \frac{\varepsilon\,\delta^2}{6}\,r_x^w\,r^w\,\bar{u}\,\bar{u}_{xx} + \left(\alpha\,\delta^2\,\eta_{tt} + \beta\,\eta + \delta^2\,\beta\,\gamma\,\eta_t\right)_x +$$
$$\frac{\varepsilon\,\delta^2}{12}\,(r^w)^2\,\bar{u}_x\,\bar{u}_{xx} - \frac{\delta^2}{6\,r^w}\,\frac{\partial}{\partial x}\left[(r^w)^3\left(\bar{u}_{xt} + \varepsilon\,\bar{u}\,\bar{u}_{xx} - \frac{\varepsilon}{2}\,(\bar{u}_x)^2\right)\right] = O(\delta^4). \tag{58}$$

In order to achieve our goals, we have to determine the trace of the horizontal velocity u^w at the vessel wall. The following Section is entirely devoted to this task.

[7] In reality, we just divided both equations by $r^w > 0$ which does not vanish in normal (healthy) conditions.

2.5.4 | Horizontal velocity trace at the vessel wall

The integration of the irrotationality condition (45) over r from r to r^w yields

$$u(x,r,t) = u(x,r^w,t) - \delta^2 \int_r^{r^w} v_x(x,s,t)\,\mathrm{d}s. \tag{59}$$

We introduce the following function[8] by following the steps of (Mitsotakis et al., 2019):

$$Q(x,r,t) = \frac{1}{r}\int_0^r s\,u(x,s,t)\,\mathrm{d}s.$$

An asymptotic expression of this function Q can be readily obtained by using the just obtained horizontal velocity exact representation (59):

$$Q(x,r,t) = \frac{1}{r}\int_0^r s\,u^w(x,t)\,\mathrm{d}s + O(\delta^2) = \frac{r}{2}u^w(x,t) + O(\delta^2). \tag{60}$$

The integration of the continuity Equation (44) this time from $r=0$ to $r=r^w$, we obtain

$$\int_0^r (s\,v)_s\,\mathrm{d}s = -\int_0^r s\,u_x\,\mathrm{d}s,$$

and after an elementary evaluation of the integral on the left-hand side by using the condition (48), we come to the following *exact* representation of the vertical velocity variable:

$$v(x,r,t) = -\frac{1}{r}\int_0^r s\,u_x\,\mathrm{d}s \equiv -Q_x(x,r,t).$$

Combining the last identity with the asymptotic formula (60), we get

$$v(x,r,t) = -\frac{r}{2}u^w_x(x,t) + O(\delta^2). \tag{61}$$

The last asymptotic formula can be used to eliminate the vertical velocity from the Equation (59):

$$u(x,r,t) = u^w(x,t) + \delta^2\int_r^{r^w}\frac{s}{2}u^w_{xx}(x,t)\,\mathrm{d}s + O(\delta^4).$$

After evaluating the integral and flipping the equality sides of variables u and u^w, we obtain

$$u^w(x,t) = u(x,r,t) - \delta^2 u^w_{xx}(x,t)\,\frac{(r^w)^2 - r^2}{4} + O(\delta^4).$$

Using the last asymptotic formula recursively, one has

$$u^w(x,t) = u - \delta^2\left(u(x,r,t) - \delta^2 u^w_{xx}\,\frac{(r^w)^2 - r^2}{4}\right)_{xx}\cdot\frac{(r^w)^2 - r^2}{4} + O(\delta^4)$$

[8] In the literature this quantity is usually referred to as the average volume flux.

and after some simplification, we arrive at the following simple formula:

$$u^w(x,t) = u - \delta^2 u_{xx}(x,r,t)\,\frac{(r^w)^2 - r^2}{4} + O(\delta^4).$$

In the last equation, we may use the asymptotic representation (55) for $u(x,r,t)$ in terms of $\bar{u}(x,t)$:

$$u^w(x,t) = \bar{u}(x,t) + \delta^2\left[\frac{1}{12}(r^w)^2 - \frac{1}{4}r^2\right]\bar{u}_{xx}(x,t) - \delta^2\left(\bar{u}(x,t) + \frac{1}{12}\delta^2(r^w)^2\bar{u}_{xx}(x,t) - \frac{1}{4}\delta^2 r^2\bar{u}_{xx}(x,t)\right)_{xx}\cdot\frac{(r^w)^2 - r^2}{4} + O(\delta^4)$$

and after a series of asymptotic simplifications, we obtain the final result of this Section:

$$u^w(x,t) = \bar{u} - \frac{1}{6}\delta^2(r^w)^2\bar{u}_{xx} + O(\delta^4).$$

2.5.5 | Fully nonlinear weakly dispersive equations

The asymptotic formula obtained in the previous Section allows us to rewrite the intermediate system (57), (58) in terms of the depth-averaged velocity variable:

$$\frac{(r^w)^2}{2}\bar{u}_x - \frac{\delta^2}{12}r_x^w(r^w)^3\bar{u}_{xx} - \frac{\delta^2}{48}(r^w)^4\bar{u}_{xxx} + r^w\eta_t + r^w r_x^w\bar{u} = O(\delta^4),$$

$$r^w\bar{u}_t + \frac{\delta^2}{6}r_t^w(r^w)^2\bar{u}_{xx} + \varepsilon r^w\bar{u}\bar{u}_x + \frac{\varepsilon\delta^2}{6}r_x^w(r^w)^2\bar{u}\bar{u}_{xx} + \left(\alpha\delta^2\eta_{tt} + \beta\eta + \delta^2\beta\gamma\eta_t\right)_x r^w + \frac{\varepsilon\delta^2}{12}(r^w)^3\bar{u}_x\bar{u}_{xx} - \frac{\delta^2}{6}\frac{\partial}{\partial x}\left[(r^w)^3\left(\bar{u}_{xt} + \varepsilon\bar{u}\bar{u}_{xx} - \frac{\varepsilon}{2}(\bar{u}_x)^2\right)\right] = O(\delta^4).$$

In order to close the system above, we can remember that r^w can be expressed through a known function and the unknown η as $r^w \equiv r_0(x) + \varepsilon\eta(x,t)$. Then, we obtain the following system of PDEs with variable coefficients, which constitute the celebrated fully nonlinear weakly dispersive Serre–Green–Naghdi (SGN) equations in the cylindrical axi-symmetric setting:

$$\frac{r_0}{2}\bar{u}_x + \varepsilon\frac{\eta}{2}\bar{u}_x - \frac{\varepsilon\delta^2}{12}\eta_x(r_0+\varepsilon\eta)^2\bar{u}_{xx} - \frac{\delta^2}{48}(r_0+\varepsilon\eta)^3\bar{u}_{xxx} + \eta_t + \varepsilon\eta_x\bar{u} = 0, \tag{62}$$

$$\bar{u}_t + \varepsilon\bar{u}\bar{u}_x + \frac{\varepsilon\delta^2}{6}\eta_t(r_0+\varepsilon\eta)\bar{u}_{xx} + \frac{\varepsilon^2\delta^2}{6}\eta_x(r_0+\varepsilon\eta)\bar{u}\bar{u}_{xx} + \left(\alpha\delta^2\eta_{tt} + \beta\eta + \delta^2\beta\gamma\eta_t\right)_x + \frac{\varepsilon\delta^2}{12}(r_0+\varepsilon\eta)^2\bar{u}_x\bar{u}_{xx} - \frac{\delta^2}{6(r_0+\varepsilon\eta)}\frac{\partial}{\partial x}\left[(r_0+\varepsilon\eta)^3\left(\bar{u}_{xt} + \varepsilon\bar{u}\bar{u}_{xx} - \frac{\varepsilon}{2}(\bar{u}_x)^2\right)\right] = 0. \tag{63}$$

The system (62), (63) is written in scaled variables, in which ε and δ play the rôle of bookkeeping parameters tracking the orders of nonlinearity and dispersion. Passing to dimensional (physical) variables amounts to setting $\varepsilon = \delta = 1$ and replacing the scaled wall coefficients α, β by their dimensional counterparts $\tilde{\alpha} = \rho^w h/\rho$ and

$\tilde{\beta} = E h / (\rho r_0^2)$ (*cf.* (69)). The fully nonlinear cylindrical SGN system then reads

$$\frac{r_0}{2}\bar{u}_x + \frac{\eta}{2}\bar{u}_x - \frac{1}{12}\eta_x (r_0+\eta)^2 \bar{u}_{xx} - \frac{1}{48}(r_0+\eta)^3 \bar{u}_{xxx} + \eta_t + \eta_x \bar{u} = 0, \tag{64}$$

$$\bar{u}_t + \bar{u}\bar{u}_x + \frac{1}{6}\eta_t (r_0+\eta)\bar{u}_{xx} + \frac{1}{6}\eta_x (r_0+\eta)\bar{u}\bar{u}_{xx} + \left(\tilde{\alpha}\eta_{tt} + \tilde{\beta}\eta + \tilde{\beta}\gamma\eta_t\right)_x + \frac{1}{12}(r_0+\eta)^2 \bar{u}_x \bar{u}_{xx} - \frac{1}{6(r_0+\eta)}\frac{\partial}{\partial x}\left[(r_0+\eta)^3\left(\bar{u}_{xt} + \bar{u}\bar{u}_{xx} - \frac{1}{2}(\bar{u}_x)^2\right)\right] = 0. \tag{65}$$

For a constant unperturbed radius r_0, multiplying the mass equation (64) by $(r_0+\eta)$ recovers exactly the divergence form (147), (148) used in the conservation-law analysis of Section 6.

2.6 | Classical cylindrical Boussinesq equations

The cylindrical SGN system (62), (63) presented above currently seems to be intractable to our analytical and numerical techniques. In order to be able to continue our investigation, we would like to adopt the simplifying assumption of weak nonlinearity while staying in the so-called Boussinesq regime (Dougalis and Mitsotakis, 2008; Dutykh and Goubet, 2016; Dutykh and Dias, 2007a; Dougalis et al., 2007), *i.e.* $\varepsilon \ll 1$ while $\frac{\varepsilon}{\delta^2} = O(1)$. For the sake of simplicity, we shall also assume that the vessel radius $r_0 = \text{const}$. Under these assumptions, the SGN system (62), (63) can be simplified to the following classical Boussinesq-type equations:

$$\frac{r^w}{2}\bar{u}_x - \frac{\delta^2}{48}(r_0)^3 \bar{u}_{xxx} + \eta_t + r_x^w \bar{u} = O(\delta^4 + \varepsilon\delta^2), \tag{66}$$

$$\bar{u}_t + \varepsilon\bar{u}\bar{u}_x + \left(\alpha\delta^2\eta_{tt} + \beta\eta + \delta^2\beta\gamma\eta_t\right)_x - \frac{\delta^2}{6} r_0^2 \bar{u}_{xxt} = O(\delta^4 + \varepsilon\delta^2). \tag{67}$$

Substituting $r^w \equiv r_0 + \varepsilon\eta(x,t)$ in the last Boussinesq system, we obtain

$$\frac{r_0}{2}\bar{u}_x + \frac{\varepsilon}{2}\eta\bar{u}_x - \frac{\delta^2}{48} r_0^3 \bar{u}_{xxx} + \eta_t + \varepsilon\eta_x\bar{u} = O(\delta^4 + \varepsilon + \delta^2),$$

$$\bar{u}_t + \varepsilon\bar{u}\bar{u}_x + \alpha\delta^2\eta_{xtt} + \beta\eta_x + \delta^2\beta\gamma\eta_{xt} - \frac{\delta^2}{6} r_0^2 \bar{u}_{xxt} = O(\delta^4 + \varepsilon + \delta^2).$$

The last system can be easily recast in physical variables after neglecting the asymptotically small terms on the right-hand side:

$$\frac{r_0}{2}\bar{u}_x + \frac{\eta}{2}\bar{u}_x + \eta_t - \frac{r_0^3}{48}\bar{u}_{xxx} + \eta_x\bar{u} = 0, \tag{68}$$

$$\bar{u}_t + \bar{u}\bar{u}_x + \tilde{\alpha}\eta_{xtt} + \tilde{\beta}\eta_x + \tilde{\beta}\gamma\eta_{xt} - \frac{r_0^2}{6}\bar{u}_{xxt} = 0, \tag{69}$$

where $r_0 > 0$ is the constant vessel radius, $\tilde{\alpha} = \rho^w h/\rho > 0$ and $\tilde{\beta} = E h/(\rho r_0^2) \geqslant 0$ are, respectively, the (dimensional) wall-inertia and wall-stiffness coefficients, and $\gamma \geqslant 0$ is the dimensionless wall viscoelastic damping parameter introduced in (24).

We note in passing that in the non-dispersive limit $\delta \to 0$ the dispersive terms drop out and the system (68), (69) (and likewise the parent SGN system) collapses to a first-order hyperbolic system of nonlinear shallow-water (Saint-Venant) type. For a uniform vessel ($r_0 = \text{const}$) this hydrostatic limit is C-integrable in the sense of Calogero, that is, it can be linearised by a hodograph transformation exactly as the classical one-dimensional shallow-water equations. The dispersive ($O(\delta^2)$) terms retained above are precisely what set the present weakly dispersive models apart from this hydrostatic limit. The following statement makes the C-integrability explicit.

Lemma 1 (Hodograph linearisation of the dispersionless limit) *For $r_0 = \text{const}$ and $\gamma = 0$, the dispersionless ($\delta \to 0$) limit of (68)–(69),*

$$\eta_t + \bar{u}\,\eta_x + \tfrac{1}{2}(r_0+\eta)\,\bar{u}_x = 0, \qquad \bar{u}_t + \bar{u}\,\bar{u}_x + \tilde{\beta}\,\eta_x = 0, \tag{70}$$

is a strictly hyperbolic 2×2 system with characteristic speeds $\lambda_\pm = \bar{u} \pm \sqrt{\tilde{\beta}\,(r_0+\eta)/2}$. It therefore possesses two Riemann invariants and, away from simple waves, the hodograph transformation $(x,t) \mapsto (\eta, \bar{u})$ maps it to a linear *second-order equation of Euler–Poisson–Darboux type; the system is C-integrable in the sense of* Calogero.

Proof Writing (70) as $\mathbf{U}_t + \mathrm{A}(\mathbf{U})\,\mathbf{U}_x = 0$ with $\mathbf{U} = (\eta, \bar{u})^\top$ gives $\mathrm{A} = \left(\begin{smallmatrix} \bar{u} & (r_0+\eta)/2 \\ \tilde{\beta} & \bar{u} \end{smallmatrix}\right)$, whose eigenvalues are $\lambda_\pm = \bar{u} \pm \sqrt{\tilde{\beta}(r_0+\eta)/2}$ (computed in `scripts/verify_hodograph.py`); since $\tilde{\beta} > 0$ and $r_0 + \eta > 0$ these are real and distinct, so the system is strictly hyperbolic and genuinely nonlinear, hence endowed with Riemann invariants $R_\pm = \bar{u} \mp 2\sqrt{2\tilde{\beta}(r_0+\eta)}$ (up to normalisation). Because the coefficients of (70) are independent of (x,t), interchanging dependent and independent variables (the hodograph transformation) renders the equations linear in the unknowns $x(\eta,\bar{u})$, $t(\eta,\bar{u})$; eliminating one unknown yields a single linear second-order equation of Euler–Poisson–Darboux type, exactly as for the classical shallow-water system (Whitham, 1999). Linearisability by a point/hodograph change of variables is precisely C-integrability (Calogero, 1991).

To compare the present model with that proposed in Mitsotakis et al. (2019), we recall their dimensional cylindrical Boussinesq system (*cf.* Eqs. (3.6)–(3.7) therein), written here in the notation of the present paper:

$$\eta_t + \frac{1}{2}(r_0+\eta)\,u_x + (r_0+\eta)_x\,u = 0, \tag{71}$$

$$(1 - \tilde{\alpha}\,r_{0,xx})\,u_t + (\tilde{\beta}\,\eta)_x + u\,u_x - \frac{(4\tilde{\alpha}+r_0)\,r_0}{8}\,u_{xxt} + \frac{(3\tilde{\alpha}+r_0)\,r_{0,x}}{2}\,(\tilde{\beta}\,\eta)_{xx} + \kappa\,u - \gamma\left[\tilde{\beta}\left(r_{0,x}\,u + \tfrac{1}{2}\,r_0\,u_x\right)\right]_x = 0, \tag{72}$$

where $\tilde{\alpha}$ and $\tilde{\beta}$ are as defined above, κ is the viscous-frequency (Rayleigh) damping coefficient, and γ the wall viscoelastic relaxation parameter. Comparing (71), (72) with our system (68), (69), one observes that the latter was obtained from a potential-flow formulation, whereas the current study rests on a velocity–pressure formulation; consequently the dispersion is carried by different terms (a $\bar{u}_{xxx}$ term in our mass equation against a u_{xxt} term in their momentum equation). Furthermore, their model retains the fluid Rayleigh damping $\kappa\,u$, which is set to zero in our analysis. Accordingly, these distinctions underscore that the two systems are not equivalent representations of the Boussinesq model.

2.7 | Unidirectional model equations

In this Section, we continue the derivation of simplified weakly dispersive models for axi-symmetric flows. Namely, we are about to apply the unidirectional[9] wave propagation approximation. For this, we introduce the following non-dimensional variables:

$$\eta^\star = \frac{\eta}{a}, \qquad x^\star = \frac{x}{\lambda}, \qquad r^\star = \frac{r}{r_0}, \qquad t^\star = \frac{t}{T}, \qquad u^\star = \frac{u}{c_0},$$

where here $c_0 = \dfrac{a}{r_0}\sqrt{\dfrac{2Eh}{\rho r_0}}$ is a modified Moens-Korteweg characteristic speed and and $T = 2\dfrac{a\lambda}{r_0 c_0}$. System (68), (69) in these dimensionless variables reads

$$\eta^\star_{t^\star} + u^\star_{x^\star} + \varepsilon\eta^\star u^\star_{x^\star} + 2\varepsilon\eta^\star_{x^\star}u^\star - \frac{\delta^2}{24}u^\star_{x^\star x^\star x^\star} = 0, \tag{73}$$

$$u^\star_{t^\star} + \eta^\star_{x^\star} + 2\varepsilon u^\star u^\star_{x^\star} + \frac{1}{2}\delta^2\alpha\eta^\star_{x^\star t^\star t^\star} + \gamma^\star\delta^2\eta^\star_{x^\star t^\star} - \frac{1}{6}\delta^2 u^\star_{x^\star x^\star t^\star} = 0, \tag{74}$$

where $\gamma^\star := \dfrac{1}{2}\dfrac{\sigma}{\varepsilon\delta^2}\gamma$, $\sigma := \frac{c_0}{\lambda}$, and $\varepsilon := \frac{a}{r_0}$, $\delta := \frac{r_0}{\lambda}$. As it is well known, all previous systems of equations describe the two-way propagation of nonlinear waves. In this Section, we will derive equations that describe nonlinear waves travelling essentially in one chosen direction. The two most celebrated models of this class are the KdV and BBM equations (Korteweg and de Vries, 1895; Benjamin et al., 1972; Bona and Smith, 1975; Zabusky and Galvin, 1971; Dutykh et al., 2013a; Dutykh and Pelinovsky, 2014; Dutykh and Tobisch, 2015). The BBM equation, in particular, serves as the foundation for our zero-dimensional Windkessel model presented in Appendix A.

In order to derive such models, we observe first that Equation (73) can be read to the first order as

$$\eta^\star_{t^\star} = -u^\star_{x^\star} + O(\varepsilon + \delta^2),$$

and thus

$$\eta^\star_{x^\star t^\star} = -u^\star_{x^\star x^\star} + O(\varepsilon + \delta^2),$$

and also

$$\eta^\star_{x^\star t^\star t^\star} = -u^\star_{x^\star x^\star t^\star} + O(\varepsilon + \delta^2).$$

Following Whitham's low-order approximation for predominantly right-running waves (Whitham, 1999; Dutykh and Dias, 2007a), we postulate the leading-order kinematic relation between the dimensionless horizontal velocity and the free-surface elevation,

$$u^\star = \eta^\star + \varepsilon A + \delta^2 B + O(\varepsilon^2, \delta^4, \varepsilon\delta^2), \tag{75}$$

[9] In some works, mostly in the field of optics, *e.g.* Agrawal and Pattanayak (1979), it is also referred to as the paraxial approximation.

where the yet-unknown corrections $A(x^\star,\, t^\star)$ and $B(x^\star,\, t^\star)$ are smooth functions to be determined. Substituting this relation into Equations (73) and (74) we obtain

$$\eta^\star_{x^\star} + \eta^\star_{t^\star} + \varepsilon\big(A_{x^\star} + 3\,\eta^\star\,\eta^\star_{x^\star}\big) + \delta^2\big(B_{x^\star} - \frac{1}{24}\,\eta^\star_{x^\star x^\star x^\star}\big) = O\left(\varepsilon^2,\, \delta^4,\, \varepsilon\,\delta^2\right), \tag{76}$$

$$\eta^\star_{x^\star} + \eta^\star_{t^\star} + \varepsilon\big(A_{t^\star} + 2\eta^\star\,\eta^\star_{x^\star}\big) + \delta^2\big(B_{t^\star} - \frac{\alpha}{2}\,\eta^\star_{x^\star x^\star t^\star} - \gamma^\star\,\eta^\star_{x^\star x^\star} - \frac{1}{6}\,\eta^\star_{x^\star x^\star t^\star}\big) = O\left(\varepsilon^2,\, \delta^4,\, \varepsilon\,\delta^2\right). \tag{77}$$

Adding Equation (76) and Equation (77) and equating the coefficients of ε and δ^2 to zero yields

$$\varepsilon: \quad A_{x^\star} + 3\,\eta^\star\,\eta^\star_{x^\star} - A_{t^\star} - 2\,\eta^\star\,\eta^\star_{x^\star} = 0, \tag{78}$$

$$\delta^2: \quad B_{x^\star} - B_{t^\star} - \frac{1}{24}\,\eta^\star_{x^\star x^\star x^\star} + \frac{\alpha}{2}\,\eta^\star_{x^\star x^\star t^\star} + \gamma^\star\,\eta^\star_{x^\star x^\star} + \frac{1}{6}\,\eta^\star_{x^\star x^\star t^\star} = 0. \tag{79}$$

Assuming $\partial_{x^\star} = -\,\partial_{t^\star} + O\left(\varepsilon^2, \delta^4, \varepsilon\delta^2\right)$, Equations (78) and (79) give

$$A_{x^\star} = -\,\frac{1}{2}\,\eta^\star\,\eta^\star_{x^\star}, \qquad B_{x^\star} = \Big(\frac{5+12\,\alpha}{48}\Big)\,\eta^\star_{x^\star x^\star x^\star} - \frac{1}{2}\,\gamma^\star\,\eta^\star_{x^\star x^\star}.$$

Integrating the last relations in $x^\star$ we obtain

$$A = -\,\frac{1}{4}\,\eta^{\star 2}, \qquad B = \Big(\frac{5+12\,\alpha}{48}\Big)\,\eta^\star_{x^\star x^\star} - \frac{1}{2}\,\gamma^\star\,\eta^\star_{x^\star},$$

and the resulting KdV–Burgers equation reads

$$\eta^\star_{t^\star} + \eta^\star_{x^\star} + \frac{5}{2}\,\varepsilon\,\eta^\star\,\eta^\star_{x^\star} + \delta^2\Big(\frac{12\,\alpha+3}{48}\Big)\,\eta^\star_{x^\star x^\star x^\star} - \frac{1}{2}\,\delta^2\,\gamma^\star\,\eta^\star_{x^\star x^\star} = 0. \tag{80}$$

Dimensional form.

Restoring physical variables as in Section 2.7, Equation (80) becomes

$$\eta_t + \tilde{c}\,\eta_x + \frac{5}{2\,r_0}\,\tilde{c}\,\eta\,\eta_x + \frac{\tilde{c}\,r_0(12\,\tilde{\alpha}+3\,r_0)}{48}\,\eta_{xxx} - \frac{r_0}{4}\,\tilde{\beta}\,\gamma\,\eta_{xx} = 0, \tag{81}$$

where $\tilde{c} = \sqrt{E\,h/(2\,\rho\,r_0)}$ is the classical Moens–Korteweg velocity Fung (1997). The associated dispersion relation is

$$\omega = \tilde{c}\,k - \frac{r_0\,\tilde{c}(12\,\tilde{\alpha}+3\,r_0)}{48}\,k^3 - \frac{\mathrm{i}}{4}\,r_0\,\gamma\,\tilde{\beta}\,k^2.$$

BBM–Burgers limit.

Replacing the third-order spatial derivative by its BBM proxy yields (this regularization approach is also employed in our Windkessel model derivation in Appendix A)

$$\eta^\star_{t^\star} + \eta^\star_{x^\star} + \frac{5}{2}\,\varepsilon\,\eta^\star\,\eta^\star_{x^\star} - \delta^2\Big(\frac{12\,\alpha+3}{48}\Big)\,\eta^\star_{x^\star x^\star t^\star} - \frac{1}{2}\,\delta^2\,\gamma^\star\,\eta^\star_{x^\star x^\star} = 0. \tag{82}$$

In dimensional variables, the last equation becomes

$$\eta_t + \tilde{c}\,\eta_x + \frac{5}{2\,r_0}\,\tilde{c}\,\eta\,\eta_x - \frac{r_0(12\,\tilde{\alpha}+3\,r_0)}{48}\,\eta_{xxt} - \frac{r_0}{4}\,\gamma\,\tilde{\beta}\,\eta_{xx} = 0\,. \tag{83}$$

The linear dispersion relation of this model can be easily computed:

$$\omega = \frac{48\,\tilde{c}\,k}{48 + r_0(12\,\tilde{\alpha}+3\,r_0)k^2} - \mathrm{i}\,\frac{12\,\gamma\,r_0\,\tilde{\beta}\,k^2}{48 + r_0(12\,\tilde{\alpha}+3\,r_0)k^2}. \tag{84}$$

Classical solitary waves (dissipation-free).
If $\gamma = 0$, Equation (81) supports solitary waves propagating with $c_s = \tilde{c}(1 + 5A/(6\,r_0))$:

$$\eta(x,t) = \frac{6(c_s-\tilde{c})r_0}{5\tilde{c}}\ \operatorname{sech}^2\left(\sqrt{\frac{12(c_s-\tilde{c})}{\tilde{c}(12\,\tilde{\alpha}+3\,r_0)r_0}}\,(x-c_s t)\right).$$

A parallel computation for Equation (83) furnishes

$$\eta(x,t) = \frac{6(c_s-\tilde{c})r_0}{5\tilde{c}}\ \operatorname{sech}^2\left(\sqrt{\frac{12(c_s-\tilde{c})}{c_s(12\,\tilde{\alpha}+3\,r_0)r_0}}\,(x-c_s t)\right).$$

Indeed, setting $\eta = \eta(\zeta)$ with $\zeta = x - c_s t$, substituting into Equation (81), integrating twice delivers

$$\frac{\tilde{c}(12\,\tilde{\alpha}+3\,r_0)r_0}{96}\,(\eta')^2 + \frac{(-c_s+\tilde{c})}{2}\,\eta^2 + \frac{5}{12}\,\frac{\tilde{c}}{r_0}\,\eta^3 = 0,$$

whose closed-form solution matches the profiles given above.

We summarise these classical facts, together with the influence of the wall viscoelasticity, in the next two statements.

Proposition 1 (Solitary waves: existence, uniqueness and amplitude–speed law) *Let $\gamma = 0$ and $r_0 = \text{const}$. For every speed $c_s > \tilde{c}$ the cylindrical KdV reduction (81) admits a solitary wave*

$$\eta(x,t) = A\,\operatorname{sech}^2\big(K\,(x-c_s t)\big), \qquad A = \frac{6\,(c_s-\tilde{c})\,r_0}{5\,\tilde{c}}\,, \qquad K^2 = \frac{10\,A}{r_0^2\,(12\,\tilde{\alpha}+3\,r_0)}\,, \tag{85}$$

which is even, positive and exponentially localised. It is the unique (up to translation) solitary-wave solution, and the amplitude–speed relation $c_s = \tilde{c}\,(1 + 5A/6r_0)$ is a strictly increasing bijection of $(0,\infty)$ onto $(\tilde{c},\infty)$: taller pulses travel faster.

Proof With $\eta = \eta(\zeta)$, $\zeta = x - c_s t$, two integrations of (81) subject to $\eta, \eta', \eta'' \to 0$ as $|\zeta| \to \infty$ give the first integral displayed above, which we rewrite as

$$(\eta')^2 = \frac{48}{\tilde{c}\,(12\,\tilde{\alpha}+3\,r_0)\,r_0}\,\eta^2\left[(c_s-\tilde{c}) - \tfrac{5\tilde{c}}{6r_0}\,\eta\right].$$

The right-hand side is a cubic in η with a double root at $\eta = 0$ and a simple root at $\eta = A = 6(c_s-\tilde{c})r_0/5\tilde{c}$. It is positive on $(0, A)$ precisely when $c_s > \tilde{c}$; the phase plane then has a unique orbit homoclinic to the saddle at the origin, symmetric about its maximum $\eta = A$. Separating variables and integrating yields the sech^2 profile (85) with K^2

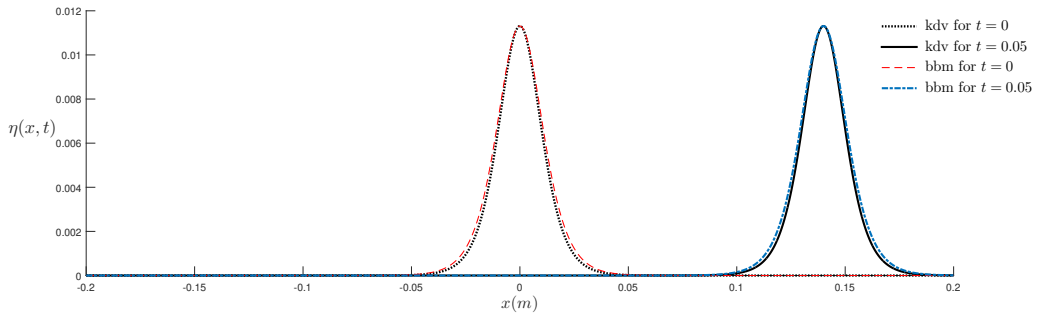


FIGURE 3 *Comparison of travelling-wave profiles generated by the KdV and BBM equations at several successive instants t.*

obtained by matching the quadratic behaviour near the crest; uniqueness up to translation follows from uniqueness of the homoclinic orbit. Strict monotonicity of $A \mapsto c_s = \tilde{c}(1 + 5A/6r_0)$ is immediate. The same argument applied to (83) gives the companion profile quoted above.

Lemma 2 (Adiabatic viscoelastic attenuation) *Let $0 < \gamma \ll 1$. To leading order in the slowly-varying (adiabatic) approximation, the amplitude of the KdV solitary wave (85) decays according to*

$$\frac{\mathrm{d}A}{\mathrm{d}t} = -\,C\,\gamma\,A^2, \qquad C = \frac{8\,\tilde{\beta}}{3\,r_0\,(12\,\tilde{\alpha} + 3\,r_0)}, \qquad \textit{so} \qquad A(t) = \frac{A_0}{1 + C\,\gamma\,A_0\,t}. \tag{86}$$

The pulse therefore attenuates algebraically, $A \sim (C\gamma t)^{-1}$, over the characteristic time $t_\star = (C\gamma A_0)^{-1}$.

Proof For $\gamma > 0$ the viscoelastic contribution to (81) is the Burgers-type diffusion $-\frac{r_0\tilde{\beta}\gamma}{4}\,\eta_{xx}$. Multiplying (81) by η and integrating over $\mathbb{R}$, the advective, nonlinear and dispersive terms are exact x-derivatives and drop out, leaving the exact balance

$$\frac{\mathrm{d}}{\mathrm{d}t}\int_{\mathbb{R}} \tfrac{1}{2}\,\eta^2\,\mathrm{d}x = -\,\nu\int_{\mathbb{R}} \eta_x^2\,\mathrm{d}x \le 0, \qquad \nu := \frac{r_0\,\tilde{\beta}\,\gamma}{4} \ge 0, \tag{87}$$

the divergence identity behind which is verified symbolically in `scripts/verify_energy.py`. Substituting the slowly-varying profile $\eta = A(t)\,\mathrm{sech}^2(K\zeta)$ with the width–amplitude relation $K^2 = 10A/[r_0^2(12\tilde{\alpha} + 3r_0)]$ and the integrals $\int \eta^2 = \frac{4A^2}{3K}$, $\int \eta_x^2 = \frac{16A^2K}{15}$ into (87) and simplifying gives (86) (the elementary algebra is checked in `scripts/verify_soliton_deca`). The excess mass is radiated into a small trailing shelf, consistent with the exact conservation of $\int \eta\,\mathrm{d}x$.

The initial disturbance is a solitary wave of amplitude $A = 0.0035$. The near-perfect overlap of the two families of curves at all reported times confirms that the BBM regularisation reproduces, to leading order, the dispersive and nonlinear balance captured by the KdV equation, thereby validating the asymptotic model for the haemodynamic parameters considered here. Figure 3 juxtaposes the solitary-wave solutions obtained from the KdV and BBM models for a vessel of undisturbed radius $r_0 = 0.01$ m, wall thickness $h = 3\times10^{-4}$ m, Young's modulus $E = 4.1\times10^5$ kg m^{-1} s^{-2}, wall density $\rho^{\omega} = 10^3$ kg m^{-3}, and computational length $L = 0.4$ m.

2.7.1 | Travelling-wave reduction of the cylindrical model

In this Section, we derive exact solitary-wave solutions for the variable-coefficient cylindrical Boussinesq-type model (68), (69). The derivation follows the computer-algebra strategy of Dutykh and Dias (2007a), implemented symbolically in Maple. We look for solutions that depend on a single travelling coordinate

$$\zeta \coloneqq x - c\,t,$$

so that

$$\bar{u}(x,t) = \bar{u}(\zeta), \qquad \eta(x,t) = \eta(\zeta), \qquad r^w = r_0 + \eta, \qquad r_0 = \text{const}.$$

Reduction of the governing equations.
Using the low-order approximation of the continuity Equation (61) together with the boundary condition (46), one finds[10]

$$\eta_{xtt} = -\left(\frac{1}{2} r^w \bar{u}_x + r_x^w \bar{u}\right)_{xt} + O(\delta^2).$$

Substituting this expression into Equation (67) yields the cylindrical Boussinesq system

$$\eta_t + \frac{r_0}{2}\bar{u}_x + \frac{\varepsilon}{2}\eta\bar{u}_x + \varepsilon\eta_x\bar{u} - \frac{\delta^2}{48} r_0^3 \bar{u}_{xxx} = O(\delta^4, \varepsilon\delta^2), \tag{88}$$

$$\bar{u}_t + \varepsilon\bar{u}\bar{u}_x - \frac{\alpha r_0}{2}\delta^2\bar{u}_{xxt} + \beta\eta_x + \delta^2\beta\gamma\eta_{xt} - \frac{\delta^2}{6} r_0^2 \bar{u}_{xxt} = O(\delta^4, \varepsilon\delta^2). \tag{89}$$

Henceforth, we set $\gamma = 0$ (the purely elastic wall case). Introducing the convenience coefficients

$$a = \frac{r_0}{2}, \qquad b = \frac{r_0^3}{48}, \qquad d = \frac{\alpha r_0}{2}, \qquad e = \beta, \qquad f = \frac{r_0^2}{6},$$

and rewriting Equations (88) and (89) in the travelling frame leads, after one integration with respect to ζ, to the coupled third-order ODE system

$$c\eta' + a\bar{u}' + \frac{\varepsilon}{2}\eta\bar{u}' + \varepsilon\eta'\bar{u} - \delta^2 b\bar{u}''' = 0, \tag{90}$$

$$c\bar{u}' + \varepsilon\bar{u}\bar{u}' - \delta^2 d\,c\bar{u}''' + e\eta' - \delta^2 f\,c\bar{u}''' = 0, \tag{91}$$

where primes denote differentiation with respect to ζ. Because solitary waves are spatially localised, all dependent variables and their derivatives vanish as $\zeta \to \pm\infty$.

Proportional flow ansatz.
Guided by extensive numerical experimentation, we set

$$\bar{u}(\zeta) = A\eta(\zeta), \tag{92}$$

[10] Terms of order $O(\delta^4)$ and higher are consistently neglected throughout.

where the proportionality constant A will be determined self-consistently. Substituting Equation (92) into Equations (90) and (91), integrating once under the zero-boundary conditions, and collecting like terms gives the pair

$$(c + aA)\,\eta \;-\; \delta^2\, b\, A\,\eta'' \;=\; -\frac{3}{4}\,\varepsilon\, A\,\eta^2, \tag{93}$$

$$(cA + e)\,\eta \;-\; \delta^2\, cA\,(d+f)\,\eta'' \;=\; -\frac{1}{2}\,\varepsilon\, A^2\,\eta^2. \tag{94}$$

Demanding that the algebraic factors multiplying η and η'' coincide in Equations (93) and (94) (otherwise only the trivial solution $\eta \equiv 0$ would be admissible) yields the linear system

$$2Ac \;+\; 2aA^2 \;=\; 3Ac \;+\; 3e, \qquad -Ac \;+\; 2aA^2 \;=\; 3e, \tag{95}$$

whose unique positive solution is[11]

$$A^2 \;=\; \frac{9\,e\,(d+f)}{6\,a\,(d+f) \;-\; 2\,b}, \qquad c \;=\; \frac{2\,b}{3\,(d+f)}\,A.$$

Restoring the original parameters,

$$A \;=\; 6\sqrt{\frac{\beta\,(3\alpha+1)}{r_0\,(36\alpha+11)}}, \qquad c \;=\; \frac{r_0}{6\alpha+2}\sqrt{\frac{\beta\,(3\alpha+1)}{r_0\,(36\alpha+11)}}\,.$$

Scalar solitary wave equation.

With the compatibility conditions (95) enforced, either of the relations (93), (94) becomes

$$A_1\,\eta' \;-\; B_1\,\eta''' \;=\; \eta\,\eta', \qquad A_1 := \frac{4be + 6ae(d+f)}{9e(d+f)\,\varepsilon}, \quad B_1 := \frac{2b\,\delta^2}{3\,\varepsilon}, \tag{96}$$

which is a well-known integrable third-order ODE of the KdV type (Newell, 1977). Provided $A_1 B_1 > 0$, equation (96) admits the classical sech^2 solitary-wave solution:

$$\eta(\zeta) \;=\; 3\,A_1\,\operatorname{sech}^2\Big[\frac{1}{2}\sqrt{A_1/B_1}\;(\zeta - \zeta_0)\Big], \qquad \zeta_0 \;\in\; \mathbb{R}.$$

Re-expressed through the physical parameters,

$$\eta(\zeta) \;=\; -\,\frac{3\,r_0\,(7+18\alpha)}{54\alpha+18}\,\operatorname{sech}^2\Big[\frac{\zeta}{2\delta}\sqrt{\frac{28+72\alpha}{r_0^2(3\alpha+1)}}\Big], \tag{97}$$

and, by virtue of (82),

$$\bar{u}(\zeta) \;=\; \frac{r_0\,(7+18\alpha)}{3\alpha+1}\sqrt{\frac{\beta\,(3\alpha+1)}{r_0\,(36\alpha+11)}}\,\operatorname{sech}^2\Big[\frac{\zeta}{2\delta}\sqrt{\frac{28+72\alpha}{r_0^2(3\alpha+1)}}\Big]. \tag{98}$$

The entire derivation has been cross-checked symbolically in Maple; the annotated worksheet is provided in the accompanying codes repository.

[11] The positive root is selected to guarantee a right-propagating pulse ($c > 0$).

2.8 | Consistency of the asymptotic model hierarchy

The systems derived above form a hierarchy of successive long-wave approximations. We now make precise, in the sense of asymptotic consistency (Lannes, 2013; Bona et al., 2005; Israwi, 2011), the order to which each model reproduces the parent axisymmetric dynamics. Here *consistency* refers to the size of the residual produced when a (uniformly bounded) smooth solution of the finer system is substituted into the coarser one; in particular, *no well-posedness of either system is assumed*—only an a priori bounded solution.

Definition A reduced system $\mathcal{S}$ is *consistent at order* $O(\mu)$ with the axisymmetric potential-flow problem (20)–(24) on $[0, T]$ if, for every solution (ϕ, η) of the latter whose relevant norms are bounded uniformly in $\delta, \varepsilon \in (0, 1]$ on $[0, T]$, the pair $(\eta, \bar{u})$—with $\bar{u}$ the depth-averaged axial velocity—satisfies $\mathcal{S}$ up to a residual $\mathbf{R}$ obeying $\|\mathbf{R}(t)\|_{H^s} \le C\,\mu$ for $t \in [0, T]$, the constant C depending only on those bounds.

Theorem 1 (Consistency of the cylindrical SGN system) *The cylindrical SGN system (62)–(63) is consistent at order $O(\delta^4)$ with the axisymmetric potential Euler problem.*

Proof By the axis condition (22) the velocity potential admits the even expansion $\phi = \sum_{m\ge 0} r^{2m}\phi_{2m}(x,t)$; substitution into the cylindrical Laplace equation (21) produces the recurrence (26), solved by $\phi_2 = -\frac{\delta^2}{4}\,\phi_{0,xx}$, $\phi_4 = \frac{\delta^4}{64}\,\phi_{0,xxxx}$ and $\phi_{2m} = O(\delta^{2m})$, i.e. the truncated potential (29). A direct computation shows that (29) solves (21) *identically through* $O(\delta^4)$: the coefficients of r^1 and r^3 in $\delta^2 r\,\phi_{xx} + \partial_r(r\,\phi_r)$ vanish, leaving the bulk residual $\frac{\delta^6}{64}\,r^5\,\partial_x^6\phi_0 = O(\delta^6)$, while the axis condition (22) holds exactly; this elementary but error-prone algebra is verified symbolically in `scripts/verify_consistency.py`. Inserting (29) into the wall conditions (23)–(24) and depth-averaging the continuity equation by the procedure of Section 2.3.1 reproduces (62)–(63), each discarded term carrying a factor δ^4. Under the assumed uniform bounds the residual is therefore $O(\delta^4)$ in H^s.

Corollary 1 (Consistency of the Boussinesq system) *In the Boussinesq regime $\varepsilon \ll 1$, $\varepsilon/\delta^2 = O(1)$, the cylindrical Boussinesq system (68)–(69) is consistent with the SGN system—and hence with the axisymmetric Euler problem—at order $O(\varepsilon\,\delta^2 + \delta^4)$.*

Proof Discarding from (62)–(63) the terms quadratic in the nonlinearity parameter retains all $O(1)$ and $O(\delta^2)$ contributions and removes only terms of order $O(\varepsilon\,\delta^2)$; together with the $O(\delta^4)$ truncation of Theorem 1 this yields the stated residual, precisely the order $O(\delta^4 + \varepsilon\,\delta^2)$ already recorded on the right-hand sides of (66)–(67).

Corollary 2 (Consistency of the unidirectional models) *For right-going initial data the KdV (81) and BBM (83) reductions are consistent with the cylindrical Boussinesq system at order $O(\varepsilon^2 + \varepsilon\,\delta^2)$ on the time scale $O(\varepsilon^{-1})$.*

Proof Factoring the bidirectional Boussinesq operator into counter-propagating parts and retaining only the right-going characteristic (Olver, 1988) introduces an $O(\varepsilon^2 + \varepsilon\,\delta^2)$ error from the neglected reflected field and from the leading-order substitution $\partial_t \simeq -\tilde{c}\,\partial_x + O(\varepsilon + \delta^2)$ used to recast the dispersive term in either the KdV or the BBM form; on the diffusive–dispersive time scale $O(\varepsilon^{-1})$ this residual remains controlled.

3 | DISPERSION RELATIONS

We first examine the linear dispersion characteristics of the full three-dimensional model and compare them with those obtained from the linearised Euler equations by following (Mitsotakis et al., 2018):

$$u_t + \frac{1}{\rho}\,p_x = 0, \tag{99}$$

$$v_t + \frac{1}{\rho} p_r = 0, \tag{100}$$

$$u_x + v_r + \frac{1}{r} v = 0, \tag{101}$$

subject to the boundary conditions

$$v\left(x, r^w, t\right) = \eta_t\left(x, t\right), \tag{102}$$

$$p^w\left(x, t\right) = \rho^w h \eta_{tt}\left(x, t\right) + \frac{E h}{\rho_0^2} \eta\left(x, t\right), \tag{103}$$

$$v\left(x, 0, t\right) = 0. \tag{104}$$

Normal-mode ansatz.

We consider normal modes of the form

$$u\left(x, r, t\right) = u_0\left(r\right) \exp\left(\mathrm{i}\left(k x - \omega t\right)\right), \qquad v\left(x, r, t\right) = v_0\left(r\right) \exp\left(\mathrm{i}\left(k x - \omega t\right)\right),$$

$$\eta\left(x, t\right) = \eta_0 \exp\left(\mathrm{i}\left(k x - \omega t\right)\right), \qquad p\left(x, r, t\right) = p_0\left(r\right) \exp\left(\mathrm{i}\left(k x - \omega t\right)\right),$$

and substitute these expressions into the linearised Euler system. The axial velocity amplitude $u_0\left(r\right)$ then satisfies the modified Bessel equation

$$r u_0''\left(r\right) + u_0'\left(r\right) - r k^2 u_0\left(r\right) = 0, \tag{105}$$

together with the regularity condition

$$u_0'\left(0\right) = 0 \tag{106}$$

and the wall conditions

$$u_0'\left(r_0\right) = \omega k \eta_0, \tag{107}$$

$$\rho^w h \omega^2 \eta_0 + \frac{\rho \omega}{k} u_0\left(r_0\right) - \frac{E h}{r_0^2} \eta_0 = 0. \tag{108}$$

The Bessel problem (105)–(107) admits the solution

$$u_0\left(r\right) = \eta_0 \omega \frac{I_0\left(k r\right)}{I_1\left(k r_0\right)},$$

where I_0 and I_1 are the modified Bessel functions of the first kind. Inserting this expression into (108) yields the *exact* linear dispersion relation

$$\omega_\varepsilon^2(k) = \frac{E\,h}{\rho\, r_0^3}\,\frac{r_0 k\, I_1(k r_0)}{\dfrac{\rho^w h}{\rho r_0}\, r_0 k\, I_1(k r_0) + I_0(k r_0)}. \tag{109}$$

Reduced Boussinesq system.
We next linearise the depth-averaged system (66)–(67) and again insert the normal-mode ansatz. This procedure leads to

$$\frac{r_0}{2}\,\mathrm{i}k\,\bar{u}_0 - \mathrm{i}\omega\,\eta_0 - \frac{\delta^2}{48}\,r_0^3\,(\mathrm{i}k)^3\bar{u}_0 = 0, \tag{110}$$

$$-\mathrm{i}\omega\,\bar{u}_0 - \alpha\delta^2 k\omega^2\eta_0 + \mathrm{i}k\,\beta\,\eta_0 + \delta^2\beta\gamma\,(\mathrm{i}k)\omega\,\eta_0 + \frac{\delta^2}{6}\,r_0^2 k^2\omega\,\bar{u}_0 = 0. \tag{111}$$

Elimination of $\bar{u}_0$ gives the quadratic dispersion relation

$$\left(24\,r_0 k + r_0^3 k^3\right)\left(\alpha k + \frac{48 + 8\delta^2 r_0^2 k^2}{24\,r_0 k + r_0^3 k^3}\right)\omega^2 + \mathrm{i}\delta^2\beta\gamma\, k\,\omega - \beta\, k = 0. \tag{112}$$

Modified Boussinesq system.
For the refined system (88)–(89) we proceed analogously and obtain the alternate quadratic

$$\left(24\,r_0 k + \delta^2 r_0^3 k^3\right)\left(-48 - 24\,\alpha\delta^2 r_0^2 k^2 - 8\,\delta^2 r_0^2 k^2\right)\omega^2 - \mathrm{i}\,\delta^2\beta\gamma\, k\,\omega + \beta\, k = 0. \tag{113}$$

Comparison of phase speeds.
The phase speeds furnished by (109), (112) and (113), together with those of the classical KdV and BBM reductions, are depicted in Figure 4 for

$$r_0 = 0.01\ \mathrm{m}, \qquad h = 3\times 10^{-4}\ \mathrm{m}, \qquad E = 4.1\times 10^5\ \mathrm{kg\,m^{-1}\,s^{-2}}, \qquad \rho^\omega = 10^3\ \mathrm{kg\,m^{-3}}, \qquad \rho = 1060\ \mathrm{kg\,m^{-3}}.$$

The graphical agreement of Figure 4 can be made precise: the reduced models reproduce the long-wave expansion of the exact dispersion relation to the order at which they were derived.

Lemma 3 (Asymptotic consistency of the dispersion relation) *Let $\tilde{\alpha} = \rho^w h/\rho$ and $\tilde{\beta} = E\,h/(\rho\, r_0^2)$. As $k\,r_0 \to 0$, the exact linearised-Euler dispersion relation (109) satisfies*

$$\omega_\varepsilon^2(k) = \tfrac{1}{2}\,\tilde{\beta}\, r_0\, k^2\left[1 - \left(\tfrac{1}{2}\,\tilde{\alpha}\, r_0 + \tfrac{1}{8}\, r_0^2\right) k^2 + O\big((k r_0)^4\big)\right]. \tag{114}$$

The leading coefficient equals the squared Moens–Korteweg speed, $\frac{1}{2}\tilde{\beta} r_0 = \tilde{c}^2$, and the cylindrical KdV reduction (81) reproduces both this coefficient and the $O(k^2)$ dispersive coefficient exactly; consequently

$$\omega_\varepsilon(k) - \omega_{\mathrm{KdV}}(k) = O\big((k r_0)^5\big),$$

i.e. the reduction is consistent with the full linear problem to relative order $O(\delta^4)$. The classical and modified Boussinesq

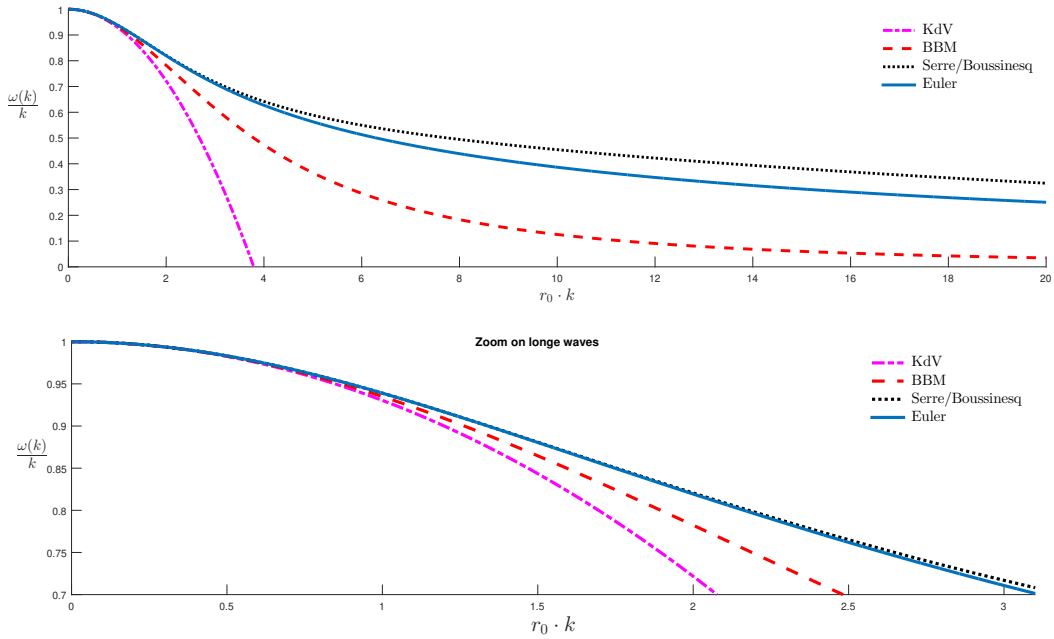


FIGURE 4 *Comparison of linear phase speeds predicted by various Boussinesq-type approximations with the reference Euler result.*

relations (112), (113) *share the same* $O(k^2)$ *branch.*

Proof Insert the Maclaurin series $I_0(z) = 1 + z^2/4 + z^4/64 + O(z^6)$ and $I_1(z) = z/2 + z^3/16 + O(z^5)$, $z = kr_0$, into (109) and expand the resulting rational function in k; collecting powers gives (114). The cylindrical KdV relation $\omega_{\mathrm{KdV}} = \tilde{c}\,k - \frac{\tilde{c}(12\tilde{\alpha}+3r_0)r_0}{48}k^3$ yields $\omega^2_{\mathrm{KdV}} = \tilde{c}^2k^2\big[1 - \frac{(12\tilde{\alpha}+3r_0)r_0}{24}k^2 + \dots\big]$, and $\frac{(12\tilde{\alpha}+3r_0)r_0}{24} = \frac{1}{2}\tilde{\alpha}r_0 + \frac{1}{8}r_0^2$ coincides with the bracket in (114); since $\tilde{c}^2 = \frac{1}{2}\tilde{\beta}r_0$ the two expansions agree through $O(k^4)$ in ω^2, whence the stated error. The symbolic computation is recorded in `scripts/verify_dispersion.py`.

Proposition 2 (Linear well-posedness and stability) *For the linearised constant-radius reductions:* (i) *if* $\gamma = 0$ *then* $\omega(k) \in \mathbb{R}$ *for every* k*, so every Fourier mode is bounded and the linear initial-value problem is well-posed in each* H^s*;* (ii) *if* $\gamma > 0$ *then* $\operatorname{Im}\omega(k) \le 0$ *for every* k*, so every mode decays and the linear evolution is an* L^2*-contraction. In particular none of the reductions exhibits a Hadamard (ill-posed) instability.*

Proof For the KdV reduction $\operatorname{Im}\omega = -\frac{1}{4}r_0\gamma\tilde{\beta}\,k^2$ and for the BBM reduction $\operatorname{Im}\omega = -12\gamma r_0\tilde{\beta}k^2/\big(48 + r_0(12\tilde{\alpha}+3r_0)k^2\big)$, both ≤ 0 for $\gamma \ge 0$ and vanishing at $\gamma = 0$. For the Boussinesq relation (112) with $\gamma = 0$ one solves $\omega^2 = \beta k/\big[\alpha k(24r_0k + r_0^3k^3) + 48 + 8\delta^2r_0^2k^2\big] > 0$, so ω is real. The sign computations are confirmed in `scripts/verify_dispersion.py`.

This settles, at the linear level, the well-posedness question deferred in Section 2; the viscoelastic parameter γ acts as a stabilising (dissipative) mechanism, while the modified Boussinesq formulation reduces the high-frequency error of the classical truncation, reinforcing its suitability for arterial pulse waves over physiologically relevant wavelengths.

4 | NUMERICAL SIMULATIONS

To study solitary wave dynamics in the Boussinesq system, we implement a pseudo-spectral solver with a filtering method (Chen and Bona, 2002). This approach leverages spectral methods for high-precision solutions and incorporates filtering techniques to stabilize solitary wave simulations, ensuring accurate capture of their propagation characteristics (Trefethen, 2000; Boyd, 2000).

The Boussinesq system, modeling nonlinear dispersive waves, is given by:

$$\partial_t \eta + \frac{r_0}{2}\partial_x \bar{u} + \frac{\varepsilon}{2}\eta\partial_x \bar{u} + \varepsilon\partial_x \eta \bar{u} - \frac{\delta^2 r_0^3}{48}\partial_x^3 \bar{u} = 0, \tag{115}$$

$$\partial_t \bar{u} + \varepsilon \bar{u}\partial_x \bar{u} - \frac{\alpha r_0 \delta^2}{2}\partial_x^2 \partial_t \bar{u} + \beta \partial_x \eta + \delta^2 \beta\gamma \partial_x \partial_t \eta - \frac{\delta^2 r_0^2}{6}\partial_x^2 \partial_t \bar{u} = 0, \tag{116}$$

where $\eta(x,t)$ is the free surface elevation, $\bar{u}(x,t)$ is the velocity, r_0 is a reference radius, ε and δ are nonlinearity and dispersion parameters, and α, β, γ are system parameters. For this study, we set $\gamma = 0$.

The pseudo-spectral solver transforms the variables $\eta(x,t)$ and $\bar{u}(x,t)$ into Fourier space using the Fourier transform, where $\hat{\eta}(k,t) = \mathcal{F}\{\eta\}$ and $\hat{u}(k,t) = \mathcal{F}\{\bar{u}\}$. The Fourier transform relationships are:

$$\mathcal{F}\{\partial_x \eta\} = \mathrm{i}k\hat{\eta}, \quad \mathcal{F}\{\partial_x^2 \bar{u}\} = -k^2\hat{u}, \quad \mathcal{F}\{\partial_x^3 \bar{u}\} = -\mathrm{i}k^3\hat{u}.$$

Applying the Fourier transform to (115) and (116), we obtain:

$$\partial_t \hat{\eta} = -\frac{r_0}{2}(\mathrm{i}k\hat{u}) - \frac{\varepsilon}{2}\mathcal{F}\{\eta\partial_x \bar{u}\} - \varepsilon\mathcal{F}\{(\partial_x \eta)\bar{u}\} + \frac{\delta^2 r_0^3}{48}(-\mathrm{i}k^3\hat{u}), \tag{117}$$

$$\partial_t \hat{u} = -\frac{\varepsilon \mathrm{i}k\mathcal{F}\{\bar{u}\bar{u}\} + \beta \mathrm{i}k\hat{\eta}}{1 + \frac{r_0\delta^2 k^2(3\alpha + r_0)}{6}}, \tag{118}$$

where the denominator in (118) accounts for the dispersive terms involving $\partial_x^2 \partial_t \bar{u}$.

4.1 | Filtering Method and Time Integration

To stabilize solitary wave solutions and reduce numerical noise, we employ a filtering method, which effectively dampens high-frequency oscillations while preserving the solitary wave's key features (Chen and Bona, 2002; Trefethen, 2000). The computational grid is defined with a refined spatial discretization to capture the wave profile accurately, and the time-stepping is adjusted to ensure stability. The solver initializes $\eta(x,0)$ and $\bar{u}(x,0)$ with given solitary wave profiles, transforming them into Fourier space to obtain $\hat{\eta}(k,0)$ and $\hat{u}(k,0)$. Temporal integration is performed using MATLAB's ODE113 solver, a variable-step, variable-order (VSVO) Adams–Bashforth–Moulton PECE solver with orders 1 to 13 (typically using order 12 with a 13th-order error estimate for extrapolation). This solver is well-suited for both stiff and non-stiff systems, providing efficient and accurate time evolution (Boyd, 2000). Initial tests of the pseudo-spectral solver with filtering demonstrate reliable convergence and stable solitary wave propagation, consistent with the expected behavior of the Boussinesq system. The filtering method effectively mitigates numerical

instabilities, yielding precise solutions. Future work will involve thorough validation against analytical solutions and experimental data to ensure accuracy and robustness (Chen and Bona, 2002; Trefethen, 2000).

To simulate solitary wave dynamics in the Boussinesq system, accurate numerical approximations of solitary waves are essential. While generating clean solitary waves in a laboratory is challenging, we demonstrate that numerical generation is straightforward using a filtering procedure, even without exact solutions Bona et al. (2008). This method produces reliable initial conditions for studying wave collisions and interactions.

The filtering procedure generates clean solitary waves by evolving an approximate wave profile and isolating the leading solitary wave from dispersive tails. The process is as follows:

Initial Wave Approximation: Start with an initial wave profile $\eta(x,0)$ and $\bar{u}(x,0)$ resembling a solitary wave, such as a low-order approximation. The profile need not be exact but should approximate the expected shape and amplitude.

Numerical Evolution: Evolve the initial profile in a long computational domain using the Boussinesq system (115)–(116), allowing the solitary wave to separate from dispersive tails and smaller waves over time.

Isolation of Solitary Wave: Once the leading solitary wave is distinct, isolate it by setting the remaining disturbances (dispersive tails) to zero. Shift the solitary wave to the left of the spatial domain for further refinement.

Iterative Refinement: Repeat the evolution and isolation process as needed. Typically, one iteration suffices to produce a clean solitary wave.

This filtering method effectively removes numerical noise and dispersive components, yielding high-precision solitary wave solutions (Bona et al., 2008).

The Boussinesq system is solved using the pseudo-spectral solver described previously, with spatial derivatives computed in Fourier space and temporal integration performed using MATLAB's ODE113 solver. The computational domain has a length $L = 400$ and grid spacing $\Delta X = \frac{2}{N}$ with $N = 32\,768$. Parameters are set as $\alpha = \beta = r_0 = 1$, and the characteristic speed is derived as:

$$\tilde{c} = \sqrt{\frac{\beta r_0}{2}}.$$

This value is obtained by analyzing the linearized Boussinesq system up to $O(\varepsilon + \delta^2)$, as shown in the following proof:

Proof Consider the linearized Boussinesq system:

$$\begin{cases} \partial_t \eta + \frac{r_0}{2} \partial_x \bar{u} = 0, \\ \partial_t \bar{u} + \beta \partial_x \eta = 0. \end{cases} \tag{119}$$

Assume solutions of the form:

$$\eta(x,t) = \eta_0 \exp(\mathrm{i}k(x - \tilde{c}t)), \quad \bar{u}(x,t) = \bar{u}_0 \exp(\mathrm{i}k(x - \tilde{c}t)).$$

Substituting into (119), we obtain:

$$\begin{cases} -\mathrm{i}k\tilde{c}\eta_0 + \frac{r_0}{2}\mathrm{i}k\bar{u}_0 = 0, \\ -\mathrm{i}k\tilde{c}\bar{u}_0 + \beta\mathrm{i}k\eta_0 = 0. \end{cases}$$

For non-trivial solutions, the determinant of the coefficient matrix must vanish, yielding:

$$\tilde{c}^2 = \frac{\beta r_0}{2} \implies \tilde{c} = \sqrt{\frac{\beta r_0}{2}}.$$

4.1.1 | Initial Conditions and Solitary Wave Solutions

To generate initial conditions, we use solitary wave solutions of the Korteweg–de Vries (KdV) equation with $\gamma = 0$:

$$\partial_t \eta + \tilde{c}\partial_x \eta + \frac{5}{2}\frac{1}{r_0}\tilde{c}\eta\partial_x \eta + \frac{\tilde{c}(12\tilde{\alpha} + 3r_0)r_0}{48}\partial_x^3 \eta = 0. \tag{120}$$

The first-order solitary wave solution is:

$$\eta(x,t) = A\,\mathrm{sech}^2\left(\frac{1}{2}\kappa(x - ct)\right), \tag{121}$$

$$\bar{u}(x,t) = \frac{c_0}{a}A\,\mathrm{sech}^2\left(\frac{1}{2}\kappa(x - ct)\right) - \frac{c_0}{4ar_0}\left[A\,\mathrm{sech}^2\left(\frac{1}{2}\kappa(x - ct)\right)\right]^2 + \frac{c_0 r_0(5 + 12\alpha)}{48a}\partial_x^2\left[A\,\mathrm{sech}^2\left(\frac{1}{2}\kappa(x - ct)\right)\right], \tag{122}$$

where $c_0 = \frac{a}{r_0}\sqrt{\frac{2Eh}{\rho r_0}}$ is a modified Moens-Korteweg speed, and the wave parameters are:

$$\kappa = \sqrt{\frac{40\tilde{c}A}{3r_0^2(4\tilde{\alpha}\tilde{c} + r_0)}}, \quad c = \tilde{c}\left(1 + \frac{5A}{6r_0}\right).$$

These solutions are used as initial conditions, evolved, and filtered to produce clean solitary waves.

4.1.2 | Numerical Results

Solitary waves of varying amplitudes A from 0.05 to 1.0 were generated, producing heights A_s from 0.0431 to 1.1603, as detailed in Table 1. The table compares simulated speeds c_s with theoretical speeds c.

Figure 5 illustrates the space-time evolution of a solitary wave with initial amplitude $A = 0.1$ from $t = 0$ to $t = 1050$, showing stable propagation. Figure 6 depicts the time evolution for $A = 0.4$ (height $A_s = 0.3760$) at times $t = 1, 80, 100, 150$, demonstrating the development of a solitary wave followed by a dispersive tail. At $t = 150$, the solitary wave is isolated by shifting it left by 130.4 units and setting the tail to zero, yielding a clean solution. Figure 7 compares the amplitude-speed relationship of the Boussinesq system with the KdV model, showing good agreement for small amplitudes.

The filtering procedure generates clean solitary wave solutions by iteratively refining approximate profiles, effectively isolating the solitary wave from dispersive tails. This method is robust for the Boussinesq system, where exact solutions may not exist due to nonlinear term complexities (Bona et al., 2008). The numerical results, validated through Table 1 and Figures 5–7, confirm stable solitary wave propagation and good agreement with KdV predictions for small amplitudes. Large-amplitude solutions, while less physically relevant due to model assumptions, test the

TABLE 1 Estimated solitary wave speed and amplitude for the cylindrical Boussinesq system (88)–(89).

A	A_s	c_s	c	A	A_s	c_s	c
0.05	0.0431	0.7324	0.7325	0.55	0.5460	0.9521	1.0288
0.1	0.0853	0.7551	0.7573	0.6	0.6066	0.9736	1.0645
0.15	0.1297	0.7785	0.7835	0.65	0.6693	0.9938	1.1015
0.2	0.1757	0.8015	0.8106	0.7	0.7338	1.0130	1.1395
0.25	0.2231	0.8242	0.8385	0.75	0.8003	1.0317	1.1787
0.3	0.2723	0.8466	0.8675	0.8	0.8687	1.0498	1.2190
0.35	0.3232	0.8687	0.8975	0.85	0.9389	1.0679	1.2603
0.4	0.3760	0.8910	0.9428	0.9	1.0109	1.0853	1.13028
0.45	0.4307	0.9116	0.9609	0.95	1.0847	1.1022	1.3462
0.5	0.4874	0.9327	0.9943	1	1.1603	1.1189	1.3908

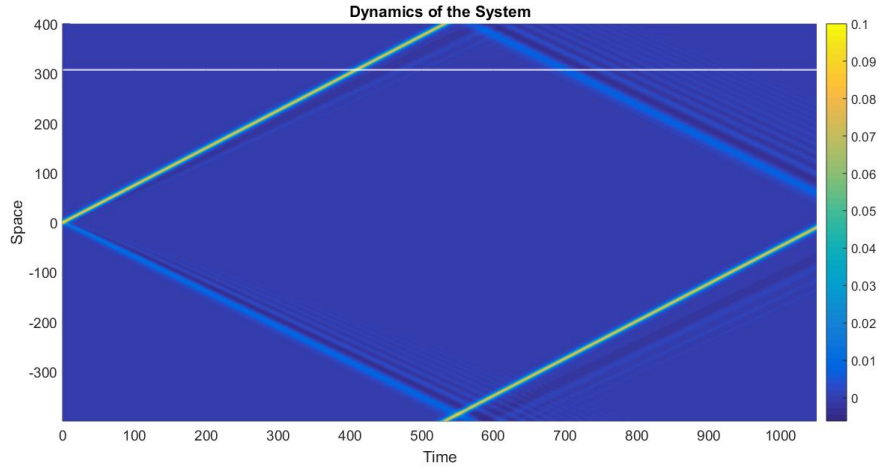


FIGURE 5 Space-time plot of the KdV soliton (120) evolution under the Boussinesq system (115)–(116) dynamics for initial amplitude $A = 0.1$ from $t = 0$ to $t = 1050$. Parameters are reported in Table 1.

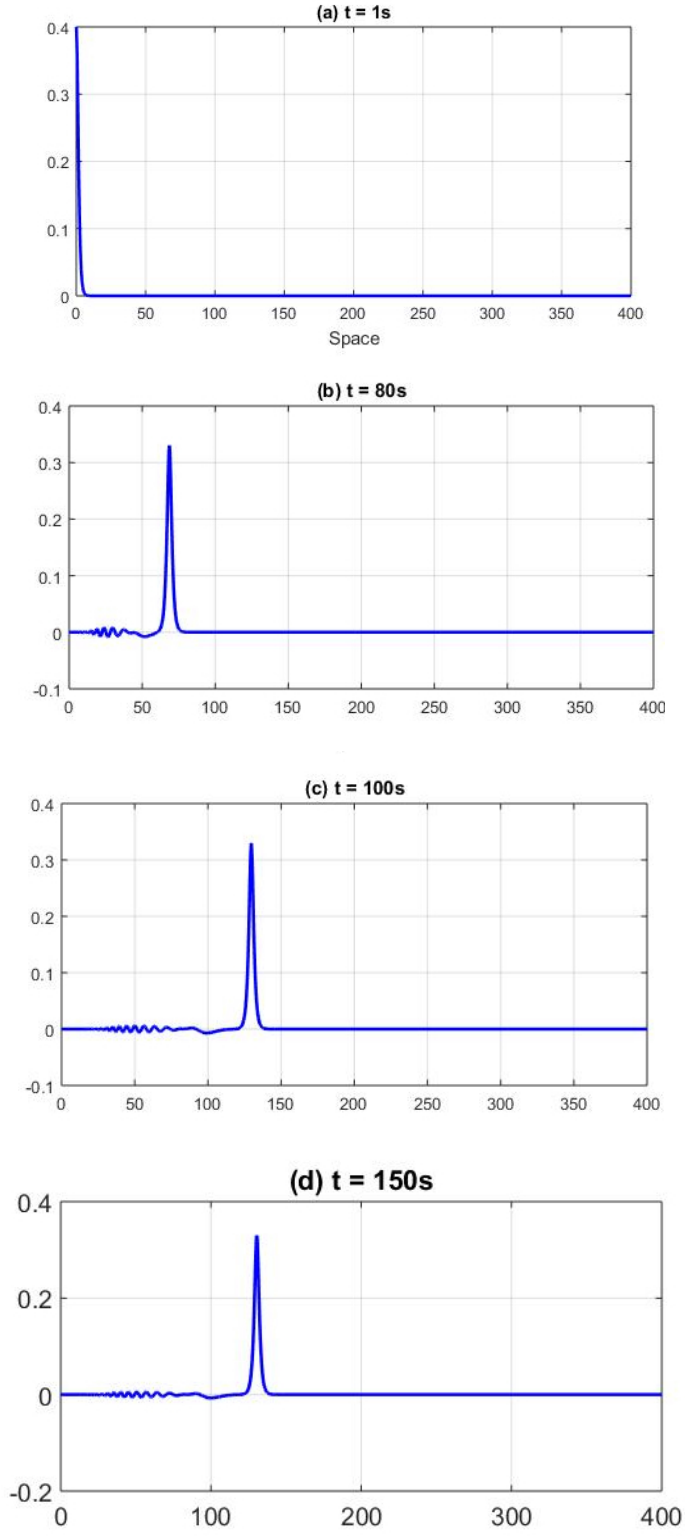


FIGURE 6 Time evolution of a solitary wave with initial amplitude $A = 0.4$ at $t = 1, 80, 100, 150$, showing the solitary wave followed by a dispersive tail.

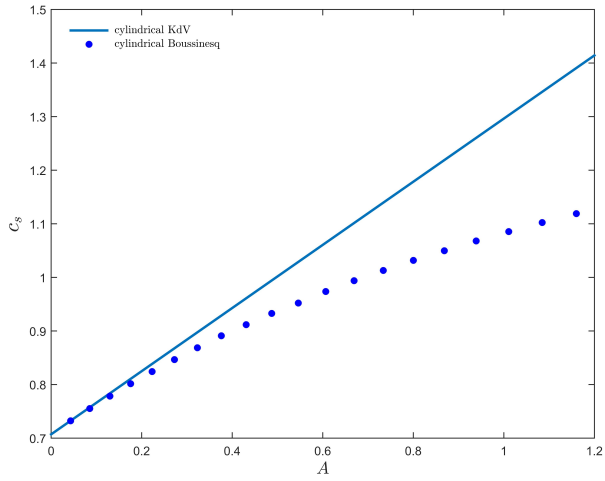


FIGURE 7 Comparison of amplitude-speed relationship between the Boussinesq system and the KdV model.

algorithm's robustness. This approach provides reliable initial conditions for studying wave interactions, enhancing our understanding of nonlinear wave dynamics in shallow water systems.

5 | STOKES EXPANSIONS

In this section, we employ Stokes expansion techniques to analyze the nonlinear wave solutions of our derived models. Stokes expansions provide a systematic approach to construct periodic wave solutions in weakly nonlinear regimes, revealing how nonlinearity affects the dispersion relation and wave profiles. We will focus particularly on applying this methodology to the KdV and BBM reductions of our cylindrical Boussinesq system, demonstrating how the interplay between nonlinearity and dispersion shapes the propagation characteristics of arterial pulse waves. This analysis will further elucidate the amplitude-dependent modifications to wave speed and structure that are essential for accurately modeling blood flow dynamics in elastic vessels.

5.1 | Weakly nonlinear Stokes expansion for the KdV reduction

Stokes' pioneering investigations of periodic gravity waves, first published in 1847, revealed two fundamental facts that remain central to non-linear dispersive-wave theory: *(i)* coherent periodic wave-trains are admissible even when the governing equations are non-linear, and *(ii)* the dispersion relation acquires an explicit dependence on the wave amplitude. The latter feature gives rise to qualitatively new phenomena rather than mere numerical corrections.

We apply Stokes' method to the KdV reduction

$$\eta_t + \tilde{c}\,\eta_x + \frac{5}{2}\frac{\tilde{c}}{r_0}\,\eta\,\eta_x + \frac{\tilde{c}\,(12\tilde{\alpha} + 3r_0)\,r_0}{48}\,\eta_{xxx} = 0, \tag{123}$$

obtained by setting $\gamma = 0$ in (81). Introducing the small parameter ε and the travelling phase $\theta = kx - \omega t$, we seek an

expansion

$$\frac{\eta}{r_0} = \varepsilon\,\xi_1(\theta) + \varepsilon^2\,\xi_2(\theta) + \varepsilon^3\,\xi_3(\theta) + \dots. \tag{124}$$

With the abbreviations

$$a = \tilde{c}, \qquad b = \frac{5\tilde{c}}{2r_0}, \qquad c = \frac{\tilde{c}\,(12\tilde{\alpha} + 3r_0)\,r_0}{48},$$

equation (123) yields the hierarchy

$$(\omega - ak)\,\xi_1' - c\,k^3\,\xi_1''' = 0, \tag{125}$$

$$(\omega - ak)\,\xi_2' - c\,k^3\,\xi_2''' = -b\,k\,\xi_1\,\xi_1', \tag{126}$$

$$(\omega - ak)\,\xi_3' - c\,k^3\,\xi_3''' = -b\,k\,(\xi_1^2)' - \omega_2\,\xi_1', \tag{127}$$

where $\omega = \omega_0(k) + \varepsilon^2\omega_2(k) + \dots$ and $\omega_0(k) = ak - ck^3$. Equation (125) admits the normalised solution $\xi_1(\theta) = \cos\theta$. Solving (126) with this choice gives

$$\xi_2(\theta) = \frac{b}{12\,c\,k^2}\cos(2\theta).$$

The right-hand side of (127) contains a resonant term proportional to $\sin\theta$; eliminating it fixes $\omega_2(k) = \dfrac{b^2}{24\,c\,k}$. A particular solution of (127) is then $\xi_3(\theta) = \dfrac{b^2}{192\,c^2\,k^4}\cos(3\theta)$. The above sequence of calculations can be summarized in the form of the following theorem, which provides a closed-form expression for the periodic wave solution and the corresponding nonlinear dispersion relation up to third order in the perturbation parameter.

Theorem 2 *Let $\varepsilon \ll 1$ and let $k > 0$ be fixed. To third order in ε the KdV equation (123) possesses a $2\pi/k$-periodic Stokes wave of the form*

$$\frac{\eta}{r_0} = \varepsilon\cos\theta + \varepsilon^2\frac{b}{12\,c\,k}\cos(2\theta) + \varepsilon^3\frac{b^2}{192\,c^2\,k^4}\cos(3\theta) + O(\varepsilon^4), \qquad \theta = kx - \omega t, \tag{128}$$

where the non-linear dispersion relation is

$$\omega(k,\varepsilon) = ak - ck^3 + \varepsilon^2\frac{b^2}{24\,c\,k} + O(\varepsilon^4). \tag{129}$$

The phase speed thus increases quadratically with amplitude, demonstrating the amplitude-dependent dispersion characteristic of Stokes waves.

Theorem 2 recovers the classical Stokes expansion for the KdV equation, with coefficients determined by the underlying physical parameters $\tilde{c}$, $\tilde{\alpha}$, and r_0. The result confirms that even weak non-linearity modifies the linear dispersion relation by an $O(\varepsilon^2)$ term proportional to the square of the wave amplitude.

5.2 | Weakly nonlinear Stokes expansion for the BBM reduction

Consider the Benjamin–Bona–Mahony (BBM) reduction obtained from the long-wave model,

$$\eta_t + \tilde{c}\,\eta_x + \frac{5\tilde{c}}{2r_0}\,\eta\,\eta_x - \frac{(12\tilde{\alpha} + 3r_0)\,r_0}{48}\,\eta_{xxt} = 0, \tag{130}$$

where, for simplicity, $\gamma = 0$ has been set. Denoting

$$a^* = \tilde{c}, \qquad b^* = \frac{5\tilde{c}}{2r_0}, \qquad c^* = \frac{\tilde{c}(12\tilde{\alpha} + 3r_0)\,r_0}{48},$$

we expand the solution in a Stokes series,

$$\frac{\eta}{r_0} = \varepsilon\,\xi_1(\theta) + \varepsilon^2\,\xi_2(\theta) + \varepsilon^3\,\xi_3(\theta) + \ldots, \qquad \theta = kx - \omega t, \tag{131}$$

with $\varepsilon \ll 1$.

Hierarchy of amplitude equations.

Insertion of (131) into (130) gives, at successive orders in ε,

$$(\omega - a^*k)\,\xi_1' - c^*\,\omega\,k^2\,\xi_1''' = 0, \tag{132}$$

$$(\omega - a^*k)\,\xi_2' - c^*\,\omega\,k^2\,\xi_2''' = b^*k\,\xi_1\xi_1', \tag{133}$$

$$(\omega - a^*k)\,\xi_3' - c^*\,\omega\,k^2\,\xi_3''' = b^*k\,(\xi_1^2)' - \omega_2\,\xi_1' + c^*\omega_2 k^2\,\xi_1'''. \tag{134}$$

The dispersion relation is again expanded as

$$\omega(k,\varepsilon) = \omega_0(k) + \varepsilon^2\,\omega_2(k) + \ldots, \qquad \omega_0(k) = \frac{a^*k}{1 + c^*k^2}.$$

Taking $\omega_1 = 0$ eliminates secular terms at $O(\varepsilon)$.

First- and second-order solutions.

Equation (132) admits the normalised solution

$$\xi_1(\theta) = \cos\theta.$$

Solving (133) for ξ_2 gives

$$\xi_2(\theta) = \frac{b^*(1 + c^*k^2)}{12\,a^*c^*k^2}\cos(2\theta). \tag{135}$$

Elimination of resonances at third order.

The right-hand side of (134) contains the resonant term $\propto \sin\theta$. Choosing

$$\omega_2(k) = \frac{b^{*2}}{24\,c^*k}, \tag{136}$$

cancels this term. Integrating once then yields

$$\xi_3(\theta) = \frac{b^{*2}(1 + c^* k^2)}{192\, c^{*2} k^4} \cos(3\theta). \tag{137}$$

The above sequence of calculations can be summarized in the form of the following theorem, which provides a closed-form expression for the periodic wave solution and the corresponding nonlinear dispersion relation up to third order in the perturbation parameter.

Theorem 3 *Let $\varepsilon \ll 1$ and $k > 0$. The BBM equation (130) possesses a $2\pi/k$-periodic Stokes wave of the form*

$$\frac{\eta}{r_0} = \varepsilon \cos\theta + \varepsilon^2 \frac{b^*(1 + c^* k^2)}{12\, a^* c^* k^2} \cos(2\theta) + \varepsilon^3 \frac{b^{*2}(1 + c^* k^2)}{192\, c^{*2} k^4} \cos(3\theta) + O(\varepsilon^4), \qquad \theta = kx - \omega t, \tag{138}$$

where the non-linear dispersion relation is

$$\omega(k, \varepsilon) = \frac{a^* k}{1 + c^* k^2} + \varepsilon^2 \frac{b^{*2}}{24\, c^* k} + O(\varepsilon^4). \tag{139}$$

Consequently, the phase speed obeys $c_{\text{ph}} = \omega/k$ and increases quadratically with the wave amplitude, reflecting the weakly non-linear character of the BBM model.

Theorem 3 mirrors the result obtained earlier for the KdV equation, but with the BBM-specific linear dispersion $\omega_0(k) = a^* k/(1 + c^* k^2)$ and modified amplitude coefficients. Together with Theorem 2, it emphasizes that both the KdV and BBM reductions support periodic Stokes waves whose speeds are augmented by nonlinearity, thereby enriching the spectrum of observable dynamics in pulsatile blood flow models.

6 | CONSERVATION LAWS AND SYMMETRIES

Every transformation that maps the solution manifold of a system of differential equations (DEs) into itself is termed a *symmetry*. The classical Lie method furnishes a systematic procedure for constructing single- and multi-parameter Lie groups of point symmetries and has been successfully employed for algebraic, ordinary, and partial differential equations Cheviakov (2007). Analogous computational frameworks exist for the determination of contact and higher-order symmetries, as well as for non-local symmetries. Closely related methodologies have been developed for approximate symmetries, symmetries of difference schemes, and those of integro-differential equations; see, for instance, the extensive compendium in Ibragimov (1995).

Information about a PDE system's conservation laws is indispensable for a profound understanding of its symmetry structure. Conservation laws encode the fundamental physical invariants of the underlying process and underpin analytical results on existence, uniqueness, and stability Olver (1993); Evans (2010). They likewise play a pivotal rôle in geometric analyses of solution manifolds Bluman and Kumei (1987) and in the design of robust numerical schemes that preserve key physical quantities LeVeque (2002).

In this section we introduce the recently developed Maple package `GeM` Cheviakov (2007). The package provides fully automated routines for the local symmetry analysis of ODEs and PDEs and for the systematic search of conservation laws. All determining equations are generated symbolically in a form amenable to efficient automatic reduction and solution.

6.1 | Conservation laws

Divergence expressions that vanish on the solution set of a PDE system represent its *local conservation laws*. Such laws arise ubiquitously: classical examples include the conservation of mass, momentum, charge, and energy. For time-dependent systems on bounded domains they yield first integrals, while for ODEs they provide conserved quantities Cheviakov (2010). Conservation laws play a decisive rôle in the analytical theory of nonlinear PDEs, notably in the study of stability and in the construction of numerical schemes Lax (1968); Knops and Stuart (1984). Moreover, a local conservation law often serves to introduce non-local (potential) variables, thereby generating auxiliary PDE systems with identical solution sets Bluman and Kumei (1987); Sjöberg and Mahomed (2004).

The four standard flux-construction techniques, together with the direct method for conservation-law derivation, have all been implemented symbolically in `GeM` Cheviakov (2007). Package routines return the determining equations in a form suitable for automatic reduction and solution. They are capable of treating high-order systems with numerous dependent and independent variables and can typically be executed on a standard desktop computer.

Consider an N-equation PDE system of order K for

$$x = (x^1, \dots, x^n), \qquad u(x) = (u^1(x), \dots, u^m(x)),$$

written compactly as

$$L^\sigma[u] \;=\; L^\sigma(x, u, \partial u, \dots, \partial^K u) \;=\; 0, \qquad \sigma \in \{1, \dots, N\}. \tag{140}$$

We denote first-order derivatives by

$$u^\nu_i \;=\; \frac{\partial u^\nu}{\partial x^i},$$

and collect all derivatives of order p in

$$\partial^p u \;=\; \left\{ \partial_{x^{i_1} \dots x^{i_p}} u^\nu \;\middle|\; \nu \in \{1, \dots, m\},\ i_1, \dots, i_p \in \{1, \dots, n\} \right\}. \tag{141}$$

A *solved form* of (140) is obtained when

$$L^\sigma[u] \;=\; u^{j_\sigma}_{i_{\sigma,1} \dots i_{\sigma,s}} \;-\; M^\sigma(x, u, \partial u, \dots, \partial^K u) \;=\; 0, \qquad \sigma = 1, \dots, N, \tag{142}$$

where $1 \leqslant j_\sigma \leqslant m$, $s \leqslant K$, and $1 \leqslant i_{\sigma,k} \leqslant n$. The set $\{u^{j_\sigma}_{i_{\sigma,1} \dots i_{\sigma,s}}\}_{\sigma=1}^N$ contains N linearly independent leading derivatives that do not appear in the right-hand sides M^σ.

Total and Euler operators.

The total derivative with respect to x^i is

$$D_i \;=\; \frac{\partial}{\partial x^i} \;+\; u^\nu_i \frac{\partial}{\partial u^\nu} \;+\; u^\nu_{i i_1} \frac{\partial}{\partial u^\nu_{i_1}} \;+\; u^\nu_{i i_1 i_2} \frac{\partial}{\partial u^\nu_{i_1 i_2}} \;+\; \dots, \qquad i \in \{1, \dots, n\}.$$

The Euler operator acting on a differential function $U(x)$ is

$$E_U \;=\; \frac{\partial}{\partial U} \;-\; D_i \frac{\partial}{\partial U_i} \;+\; \ldots \;+\; (-1)^s\, D_{i_1} \ldots D_{i_s} \frac{\partial}{\partial U_{i_1 \ldots i_s}} \;+\; \ldots \tag{143}$$

Local conservation laws.

A divergence expression

$$\operatorname{div} \Phi \;\equiv\; D_i\, \phi^i[u] \;=\; D_1 \phi^1 \;+\; \ldots \;+\; D_n \phi^n \;=\; 0 \tag{144}$$

valid on the solution set of (140) is called a local conservation law. The fluxes ϕ^i may depend on x, u, and finitely many derivatives of u.

A conservation law (144) is *trivial* if

$$\phi^i \;=\; A^i[u] \;+\; D_j H^{ij}[u],$$

where A^i vanishes on solutions of (140) and $\operatorname{div} H \equiv 0$ identically.

Direct construction by multipliers.

Non-trivial conservation laws arise from multipliers $\{\Lambda_\sigma(x, U, \partial U, \ldots, \partial^\ell U)\}_{\sigma=1}^N$ that satisfy

$$\Lambda_\sigma[U]\, L^\sigma[U] \;\equiv\; D_i\, \phi^i[U], \tag{145}$$

identically in the jet space. On any solution $U = u$ of (140) the right-hand side of (145) vanishes, yielding a local conservation law. Completeness of the direct method has been established for broad classes of systems, including those in (generalised) Kovalevskaya form Olver (1993):

Theorem 4 *For every local conservation law $D_i \phi^i[u] = 0$ of a system in solved form (142), there exists an equivalent conservation law whose fluxes contain no leading derivatives, viz.*

$$D_i\, \tilde{\phi}^i[U] \;=\; \widetilde{\Lambda}_\sigma[U]\, (U^{j\sigma}_{i_{\sigma,1} \ldots i_{\sigma,s}} - M^\sigma[U]),$$

for some non-singular multipliers $\{\widetilde{\Lambda}_\sigma\}_{\sigma=1}^N$. The correspondence between conservation laws and multiplier sets is, however, not always one-to-one.

The `GeM` workflow.

A typical sequence for conservation-law computation in `GeM` proceeds as follows:

(i) Initialisation:

```
restart;
read("d:/gem32_12.mpl"):
with(GeM): with(linalg): with(DEtools): with(PDEtools):
```

(ii) Declaration of variables:

```
gem_decl_vars(indeps=[...], deps=[...], freefunc=[...], freeconst=[...]);
```

(iii) Input of the PDE system:

```
gem_decl_eqs([...], solve_for=[...]);
```

(iv) Generation of determining equations:

```
det_eqs := gem_conslaw_det_eqs([...]);
```

(v) Selection of unknown multipliers:

```
CL_multipliers := gem_conslaw_multipliers();
```

(vi) Reduction of the over-determined system:

```
simplified_eqs := DEtools[rifsimp](det_eqs, CL_multipliers, mindim=1);
```

(vii) Solution of determining equations:

```
multipliers_sol := pdsolve(simplified_eqs[Solved]);
```

(viii) Computation of fluxes:

```
gem_get_CL_fluxes(multipliers_sol);
```

6.2 | Symmetries

A *symmetry group* of a differential equation maps each solution to another solution. Continuous (Lie) symmetries are therefore inherent to every non-trivial system. The foundational concepts were introduced by Sophus Lie in the late nineteenth century and have since evolved into a versatile tool for analysing ODEs and PDEs.

For an n-dimensional independent variable x and m-component dependent variable u, a one-parameter group of point transformations reads

$$(x')^i = f^i(x,u;\varepsilon), \qquad (u')^\nu = g^\nu(x,u;\varepsilon), \qquad i \in \{1,\dots,n\},\ \nu \in \{1,\dots,m\},$$

or, in infinitesimal form,

$$(x')^i = x^i + \varepsilon\,\xi^i(x,u) + O(\varepsilon^2), \qquad (u')^\nu = u^\nu + \varepsilon\,\eta^\nu(x,u) + O(\varepsilon^2).$$

The associated generator is

$$X = \xi^i(x,u)\,\frac{\partial}{\partial x^i} + \eta^\nu(x,u)\,\frac{\partial}{\partial u^\nu}.$$

Its K-th prolongation is obtained via the standard formulae involving total derivatives D_i; cf. (143). The group is a symmetry of (140) iff

$$X^{(K)}\,L^\sigma = 0 \quad \text{whenever} \quad L^\sigma = 0, \qquad \sigma \in \{1,\dots,N\}.$$

The `GeM` workflow for symmetries.

(i) Initialisation.

(ii) Declaration of variables and parameters.

(iii) Input of the PDE system.

(iv) Generation of determining equations:

```
det_eqs := gem_symm_det_eqs([...]);
```

(v) Selection of symmetry components:

```
sym_components := gem_symm_components();
```

(vi) Reduction of the system:

```
simplified_eqs := DEtools[rifsimp](det_eqs, sym_components, mindim=1);
```

(vii) Solution of determining equations:

```
symm_sol := pdsolve(simplified_eqs[Solved]);
```

(viii) Output of symmetries:

```
gem_output_symm(symm_sol);
```

Detailed examples for both the Boussinesq and the full cylindrical blood-flow models are provided below, together with the resulting conservation laws, symmetry generators, and the corresponding one-parameter transformation groups.

Conservation laws and symmetries for the Boussinesq system.
For the cylindrical Boussinesq system (68), (69) with constant radius r_0, a direct multiplier search yields the local conservation law

$$\left[\bar{u} + \frac{1}{2}\tilde{\beta}\,\gamma\,\eta_x + \frac{2}{3}\tilde{\alpha}\,\eta_{xt} - \frac{1}{18}r_0^2\,\bar{u}_{xx}\right]_t + \left[\frac{1}{2}\bar{u}^2 + \tilde{\beta}\,\eta + \frac{1}{2}\tilde{\beta}\,\gamma\,\eta_t - \frac{1}{9}r_0^2\,\bar{u}_{tx} + \frac{1}{3}\tilde{\alpha}\,\eta_{tt}\right]_x = 0. \quad (146)$$

Here $\tilde{\alpha} = \rho^{w}\,h/\rho$ and $\tilde{\beta}(x) = E\,h/(\rho\,r_0^2)$. The corresponding multipliers are

- $\lambda_1(t, x, \eta, \bar{u}, \dots) = 0$,
- $\lambda_2(t, x, \eta, \bar{u}, \dots) = 1$.

The infinitesimal symmetry generators are

$$X_1 = \partial_x, \qquad X_2 = \partial_t,$$

with associated point transformations

$$x' = x + \varepsilon_1, \quad t' = t, \quad \eta' = \eta, \quad \bar{u}' = \bar{u}; \qquad x' = x, \quad t' = t + \varepsilon_2, \quad \eta' = \eta, \quad \bar{u}' = \bar{u}.$$

These represent spatial and temporal translations.

Conservation laws and symmetries of the full cylindrical model (constant unperturbed radius).
For $r_0 = \text{const}$, the governing equations read

$$\frac{(r_0+\eta)^2}{2}\bar{u}_x - \frac{(r_0+\eta)^3}{12}\eta_x\,\bar{u}_{xx} - \frac{(r_0+\eta)^4}{48}\bar{u}_{xxx} + (r_0+\eta)\,\eta_t + (r_0+\eta)\,\eta_x\,\bar{u} = 0, \quad (147)$$

$$(r_0+\eta)\,\bar{u}_t + \frac{(r_0+\eta)^2}{6}\eta_t\,\bar{u}_{xx} + (r_0+\eta)\,\bar{u}\,\bar{u}_x + \frac{(r_0+\eta)^2}{6}\eta_x\,\bar{u}\,\bar{u}_{xx} + \left(\tilde{\alpha}\,\eta_{tt} + \tilde{\beta}\,\eta + \tilde{\beta}\,\gamma\,\eta_t\right)_x (r_0+\eta) + \frac{(r_0+\eta)^3}{12}\bar{u}_x\,\bar{u}_{xx} - \frac{1}{6}\frac{\partial}{\partial x}\left[(r_0+\eta)^3\left(\bar{u}_{xt} + \bar{u}\,\bar{u}_{xx} - \tfrac{1}{2}\bar{u}_x^2\right)\right] = 0. \quad (148)$$

In the multiplier ansatz adopted, *two* independent sets are obtained:

$$\{\lambda_1 = 1,\ \lambda_2 = 0\}, \qquad \{\lambda_3 = 0,\ \lambda_4 = \tfrac{1}{r_0+\eta}\}.$$

Each set generates a distinct local conservation law.

First conservation law: mass.

With $\lambda_1 = 1$ the characteristic form yields the balance of mass (equivalently, of the fluid cross-sectional area):

$$\left[\eta\, r_0 + \tfrac{1}{2}\,\eta^2\right]_t + \left[-\tfrac{1}{48}\,\eta^4\,\bar{u}_{xx} - \tfrac{1}{12}\,r_0\,\eta^3\,\bar{u}_{xx} - \tfrac{1}{8}\,r_0^2\,\eta^2\,\bar{u}_{xx} - \tfrac{1}{12}\,r_0^3\,\eta\,\bar{u}_{xx} - \tfrac{1}{48}\,r_0^4\,\bar{u}_{xx} + \tfrac{1}{2}\,\eta^2\,\bar{u} + r_0\,\eta\,\bar{u} + \tfrac{1}{2}\,r_0^2\,\bar{u}\right]_x = 0. \quad (149)$$

Second conservation law: momentum.

With $\lambda_4 = 1/(r_0+\eta)$ one obtains the balance of momentum:

$$\begin{aligned}\Big[&\bar{u} + \tfrac{1}{2}\,\tilde{\beta}\,\gamma\,\eta_x - \tfrac{1}{18}\,r_0^2\,\bar{u}_{xx} + \tfrac{2}{3}\,\tilde{\alpha}\,\eta_{xt} + \tfrac{5}{108}\,\bar{u}\,\eta_x^2 + \tfrac{1}{27}\,\bar{u}_{xx}\,\eta^2 + \tfrac{1}{36}\,r_0\,\bar{u}_{xx}\,\eta \\ &- \tfrac{5}{108}\,\eta\,\eta_x\,\bar{u}_x - \tfrac{5}{72}\,r_0\,\eta_x\,\bar{u}_x + \tfrac{5}{108}\,\eta\,\eta_{xx}\,\bar{u} + \tfrac{5}{72}\,r_0\,\eta_{xx}\,\bar{u}\Big]_t + \Big[-\tfrac{1}{6}\,\bar{u}\,\bar{u}_{xx}\,\eta^2 - \tfrac{1}{3}\,r_0\,\bar{u}\,\bar{u}_{xx}\,\eta - \tfrac{1}{6}\,r_0^2\,\bar{u}\,\bar{u}_{xx} \\ &+ \tfrac{1}{8}\,\bar{u}_x^2\,\eta^2 + \tfrac{1}{4}\,r_0\,\bar{u}_x^2\,\eta + \tfrac{1}{8}\,r_0^2\,\bar{u}_x^2 - \tfrac{11}{54}\,\eta^2\,\bar{u}_{tx} - \tfrac{13}{36}\,r_0\,\eta\,\bar{u}_{tx} - \tfrac{1}{9}\,r_0^2\,\bar{u}_{tx} \\ &+ \eta\,\tilde{\beta} + \tfrac{1}{2}\,\tilde{\beta}\,\gamma\,\eta_t + \tfrac{1}{3}\,\tilde{\alpha}\,\eta_{tt} - \tfrac{5}{108}\,\bar{u}\,\eta\,\eta_{tx} - \tfrac{5}{72}\,r_0\,\bar{u}\,\eta_{tx} + \tfrac{5}{54}\,\bar{u}_x\,\eta\,\eta_t + \tfrac{5}{36}\,r_0\,\bar{u}_x\,\eta_t \\ &- \tfrac{5}{108}\,\eta\,\eta_x\,\bar{u}_t - \tfrac{5}{72}\,r_0\,\eta_x\,\bar{u}_t + \tfrac{1}{2}\,\bar{u}^2\Big]_x = 0. \end{aligned} \quad (150)$$

Both conserved vectors (149), (150) are of third order in spatial derivatives. In the weakly nonlinear limit $\eta \ll r_0$ they reduce, respectively, to the classical *mass* and *momentum* balances for long waves in an elastic vessel: the conserved density $\eta\, r_0 + \frac{1}{2}\,\eta^2$ of (149) is the leading-order perturbation of the fluid cross-sectional area (hence a volume/mass measure), while the density of (150) reduces to the depth-averaged velocity $\bar{u}$ and thus represents the momentum of the flow.

Symmetry generators.

For the constant-radius system the complete Lie algebra of point symmetries is spanned by

$$X_1 = \partial_x, \qquad X_2 = \partial_t,$$

corresponding to spatial and temporal translations. The induced one-parameter groups act by

$$(x, t, \eta, \bar{u}) \mapsto (x + \varepsilon_1,\ t,\ \eta,\ \bar{u}), \qquad (x, t, \eta, \bar{u}) \mapsto (x,\ t + \varepsilon_2,\ \eta,\ \bar{u}).$$

Conservation laws and symmetries of the full cylindrical model (variable unperturbed radius)

For $r_0 = r_0(x)$ and $r^w = r_0(x) + \eta(x,t)$ the governing equations assume the form (62), (63). Only a single non-trivial multiplier survives, $\lambda_1 = 1$, $\lambda_2 = 0$, giving the conservation law

$$\left[\eta\, r_0 + \tfrac{1}{2}\,\eta^2\right]_t + \left[-\tfrac{1}{48}\,\eta^4\,\bar{u}_{xx} - \tfrac{1}{12}\,\eta^3\,r_0\,\bar{u}_{xx} - \tfrac{1}{8}\,\eta^2\,r_0^2\,\bar{u}_{xx} - \tfrac{1}{12}\,\eta\,r_0^3\,\bar{u}_{xx} - \tfrac{1}{48}\,r_0^4\,\bar{u}_{xx} + \tfrac{1}{2}\,\eta^2\,\bar{u} + r_0\,\eta\,\bar{u} + \tfrac{1}{2}\,r_0^2\,\bar{u}\right]_x = 0\,, \quad (151)$$

which reduces to (149) when r_0 is constant. The Lie algebra of point symmetries now depends on $r_0(x)$ and the parameters α, β, γ. The seven representative cases obtained with `GeM` are listed below:

Case 1. Generic data (α, β, γ arbitrary; $r_0(x)$ arbitrary): $X_1 = \partial_t$.
Case 2. $r_0'' = 0$ ($r_0 = r_{00} + r_{01}x$): $X_1 = \partial_t,\ X_2 = \partial_\eta - \dfrac{1}{r_{01}}\partial_x$.
Case 3. $\beta = 0$: $X_1 = \partial_t,\ X_2 = \bar{u}\,\partial_{\bar{u}} - t\,\partial_t$.
Case 4. $\beta = 0$ and $r_0'' = 0$: $X_1 = \partial_t,\ X_2 = \partial_\eta - \dfrac{1}{r_{01}}\partial_x,\ X_3 = \bar{u}\,\partial_{\bar{u}} - t\,\partial_t$.
Case 5. $r_0' = 0$ ($r_0 = r_{00}$): $X_1 = \partial_x,\ X_2 = \partial_t$.
Case 6. $\alpha = \gamma = r_0' = 0$: $X_1 = \partial_x,\ X_2 = \partial_t,\ X_3 = t\,\partial_x + \partial_{\bar{u}}$.
Case 7. $\alpha = \beta = r_0' = 0$: $X_1 = \partial_x,\ X_2 = \partial_t,\ X_3 = t\,\partial_x + \partial_{\bar{u}},\ X_4 = \bar{u}\,\partial_{\bar{u}} - t\,\partial_t$.

All multipliers, conserved vectors, and symmetry generators have been derived with the `GeM` package in Maple Cheviakov (2007). Complete worksheets and verification scripts are provided in accompanying codes.

On Galilean invariance.
It is worth emphasising that the cylindrical SGN and Boussinesq systems are, in general, *not* invariant under the Galilei boost generated by $X = t\,\partial_x + \partial_{\bar{u}}$, that is, the transformation $x \mapsto x + V\,t$, $\bar{u} \mapsto \bar{u} + V$. The symmetry classification above makes the obstruction precise: this generator is admitted *only* in Cases 6 and 7, namely when the vessel is uniform ($r_0' = 0$) and the wall inertia and viscoelasticity are switched off ($\alpha = \gamma = 0$). Equivalently, it is the wall-inertia term ($\alpha\,\eta_{tt}$), the viscoelastic relaxation (γ), and the geometric variation ($r_0(x)$) that break Galilean invariance. This is physically expected, and not a deficiency of the model: unlike the classical free-surface Serre–Green–Naghdi equations, the compliant wall endows the problem with a distinguished reference frame — the laboratory frame in which the vessel is at rest — so that boosts cannot be a symmetry once the wall responds dynamically. Restoring Galilean invariance through an asymptotically admissible change of variables would therefore amount to discarding precisely the fluid–structure coupling that the model is designed to capture.

6.3 | Hamiltonian structure, Noether's theorem and invariant solutions

The symmetry classification above, together with the variational and Hamiltonian formulations of Section 2.2.1, endows the model hierarchy with a rich geometric structure, which we now record. Throughout this subsection $r_0 = \text{const}$ and, unless stated otherwise, $\gamma = 0$.

Lemma 4 (Lie-algebra structure) *In the maximal-symmetry case $\tilde{\alpha} = \beta = r_0' = 0$ (Case 7 above), the generators $X_1 = \partial_x$, $X_2 = \partial_t$, $X_3 = t\,\partial_x + \partial_{\bar{u}}$, $X_4 = \bar{u}\,\partial_{\bar{u}} - t\,\partial_t$ span a four-dimensional solvable Lie algebra whose only non-vanishing commutators are*

$$[X_2, X_3] = X_1, \qquad [X_2, X_4] = -X_2, \qquad [X_3, X_4] = X_3. \tag{152}$$

In particular $\{X_1, X_2, X_3\}$ is the $(1+1)$-dimensional Galilei *algebra, X_3 generating the boost.*

Proof Direct evaluation of $[X_i, X_j] = X_i X_j - X_j X_i$ on a smooth test function gives (152), all other brackets vanishing (verified in `scripts/verify_lie_algebra.py`). The derived algebra $\{X_1, X_2, X_3\}$ is nilpotent, so the full algebra is solvable.

Lemma 5 (Symmetry reductions and invariant solutions) (i) *Invariance under $X_2 + c\,X_1 = \partial_t + c\,\partial_x$ forces $\eta, \bar{u}$ to depend only on $\zeta = x - ct$ and reduces the constant-radius system to the travelling-wave ODEs of Section 2.7.1, of which the solitary wave of Proposition 1 is the homoclinic orbit.* (ii) *When $\beta = 0$ the scaling generator $X_4 = \bar{u}\,\partial_{\bar{u}} - t\,\partial_t$ admits the self-similar invariant solution*

$$\bar{u}(x,t) = \frac{G(x)}{t}, \qquad \eta(x,t) = H(x), \tag{153}$$

where (G, H) solve the autonomous reduced system

$$\Big(\tfrac{\tilde{\alpha}\, r_0}{2} + \tfrac{r_0^2}{6}\Big) G'' + G\,G' - G = 0, \qquad -\tfrac{r_0^3}{48}\, G''' + \tfrac{r_0}{2}\, G' + \tfrac{1}{2}\, H\, G' + G\, H' = 0. \tag{154}$$

Proof (i) The invariant-surface condition for $X = \partial_t + c\,\partial_x$ is $\eta_t + c\,\eta_x = 0$ (and likewise for $\bar{u}$), whose general solution depends on $\zeta = x - ct$ only; substitution yields the travelling-wave system already integrated in Section 2.7.1. (ii) A functional basis of invariants of X_4 is $\{x, \eta, \bar{u}\,t\}$, giving (153). Inserting (153) into the constant-radius, $\beta = \gamma = 0$ Boussinesq system (88)–(89) (physical variables) and clearing the explicit powers of t leaves the t-independent ODEs (154); the exact cancellation of all explicitly t-dependent terms is verified in `scripts/verify_reductions.py`.

Proposition 3 (Noether correspondence) *Through the variational (*Luke*) formulation of Section 2.2.1, Noether's first theorem pairs each one-parameter variational symmetry with a conservation law: time translation ∂_t yields conservation of energy, space translation ∂_x yields conservation of momentum (150), and the particle-relabelling (gauge) symmetry yields the mass/volume law (149). The assignment $X \mapsto (\text{conserved density})$ is an anti-homomorphism from the symmetry algebra of Lemma 4 into the* Poisson *algebra of Proposition 4.*

Proof For a system in variational (equivalently, non-canonical Hamiltonian) form, Noether's theorem associates with every divergence symmetry a local conserved current, and conversely (Olver, 1993; Bluman et al., 2010; Benjamin and Olver, 1982). The translations ∂_t, ∂_x are variational symmetries whenever they are point symmetries (Cases 5–7), and the corresponding currents are, respectively, the energy density (Lemma 6) and the momentum density of (150); the constant multiplier $\Lambda = 1$ generating (149) is the relabelling current. That brackets of generators map to (minus) Poisson brackets of the associated Hamiltonians is the standard momentum-map property (Olver, 1993).

Lemma 6 (Energy balance and dissipation) *The KdV reduction (81) admits the energy functional $E_{\mathrm{K}} = \int_{\mathbb{R}} \frac{1}{2}\,\eta^2\,\mathrm{d}x$ and the BBM reduction (83) the functional $E_{\mathrm{B}} = \int_{\mathbb{R}} \frac{1}{2}\big(\eta^2 + \frac{(12\tilde{\alpha}+3r_0)\,r_0}{48}\,\eta_x^2\big)\mathrm{d}x$. Along solutions decaying at infinity, both satisfy*

$$\frac{\mathrm{d}E}{\mathrm{d}t} = -\,\frac{r_0\,\tilde{\beta}\,\gamma}{4} \int_{\mathbb{R}} \eta_x^2\,\mathrm{d}x \;\le\; 0. \tag{155}$$

Hence for $\gamma = 0$ the energy is exactly conserved—a third invariant beside mass (149) and momentum (150)—and for $\gamma > 0$ it is monotonically non-increasing, providing an a priori bound and identifying γ as the Kelvin–Voigt *dissipation rate.*

Proof Pairing (81) with η and integrating over $\mathbb{R}$, the advective, nonlinear and dispersive terms are exact x-derivatives and integrate to zero, leaving $\frac{\mathrm{d}}{\mathrm{d}t}\int \frac{1}{2}\eta^2 = -\frac{r_0\tilde{\beta}\gamma}{4}\int \eta_x^2$. For BBM the same conclusion follows on pairing the equation with η and using the regularising operator $(1 - c\,\partial_x^2)$, which converts the time-derivative term into $\frac{\mathrm{d}}{\mathrm{d}t}E_{\mathrm{B}}$. Both divergence identities are verified in `scripts/verify_energy.py`.

Proposition 4 (Poisson structure, Casimir and variational characterisation) *With $\gamma = 0$ the KdV reduction (81) is* Hamiltonian,

$$\eta_t = -\,\partial_x \frac{\delta \mathcal{H}}{\delta \eta}, \qquad \mathcal{H} = \int_{\mathbb{R}} \Big(\tfrac{a}{2}\,\eta^2 + \tfrac{b}{6}\,\eta^3 - \tfrac{c}{2}\,\eta_x^2 \Big) \mathrm{d}x, \tag{156}$$

with $a = \tilde{c}$, $b = 5\tilde{c}/2r_0$, $c = \tilde{c}(12\tilde{\alpha} + 3r_0)r_0/48$ and Poisson *operator $\mathcal{J} = -\partial_x$; the mass $M = \int_{\mathbb{R}} \eta \, \mathrm{d}x$ is its* Casimir, *$\mathcal{J}\,\delta M/\delta\eta = 0$. The BBM reduction is* Hamiltonian *with operator $\mathcal{J} = -(1 - c\,\partial_x^2)^{-1}\partial_x$ and Hamiltonian $\int_{\mathbb{R}} (\frac{a}{2}\eta^2 + \frac{b}{6}\eta^3)\,\mathrm{d}x$, the mass again a* Casimir. *Finally the solitary wave of Proposition 1 is a critical point of the constrained energy $\mathcal{H} - c_s P$, $P = \int_{\mathbb{R}} \frac{1}{2}\eta^2\,\mathrm{d}x$: the Euler–Lagrange equation $\delta(\mathcal{H} - c_s P)/\delta\eta = 0$ is its profile equation.*

Proof The variational derivative $\delta\mathcal{H}/\delta\eta = a\,\eta + \frac{b}{2}\eta^2 + c\,\eta_{xx}$ gives $-\partial_x\,\delta\mathcal{H}/\delta\eta = -(a\,\eta_x + b\,\eta\,\eta_x + c\,\eta_{xxx})$, which is (81) at $\gamma = 0$; since $\delta M/\delta\eta = 1$ and $\partial_x 1 = 0$, the mass is a Casimir. The BBM statement follows identically after applying $(1 - c\,\partial_x^2)$. Lastly $\delta(\mathcal{H} - c_s P)/\delta\eta = c\,\eta_{xx} + (a - c_s)\,\eta + \frac{b}{2}\eta^2 = 0$ coincides with the once-integrated travelling-wave equation of Proposition 1. All three identities are verified in `scripts/verify_poisson.py`.

Remark For $\gamma = 0$ the KdV reduction is, moreover, bi-Hamiltonian (Magri, 1978): it inherits the second Poisson structure and the recursion operator of the Korteweg–de Vries hierarchy, hence an infinite sequence of conservation laws and complete integrability. The viscoelastic term ($\gamma > 0$) breaks this structure, leaving the finite family of balances established above. The same integrable, defocusing character underlies the modulational stability established in Section 7.

7 | MODULATIONAL INSTABILITY ANALYSIS

The cubic Nonlinear Schrödinger (NLS) equation,

$$\mathrm{i}\,\psi_t + \psi_{xx} + |\psi|^2\psi = 0\,, \tag{157}$$

is the canonical envelope model for weakly non-linear, weakly dispersive wave packets Sulem and Sulem (1999); Zakharov and Shabat (1972). It emerges, *inter alia*, in non-linear optics, water-wave theory, Bose–Einstein condensates, and plasma physics, and admits both analytical solutions (in the integrable focusing case) and efficient numerical discretisations Agrawal (2013); Taha and Ablowitz (1984). The first systematic derivation of (157) as a modulation equation for narrow-band wave trains was given independently by Benney–Newell (1967) and Zakharov (1968). Subsequent refinements include rigorous error estimates for water waves (Sulem & Sulem 1992) and multi-scale expansions that connect the NLS to lower-order models such as the KdV and BBM equations; see the reviews by Peregrine (1983) and Hammack and Henderson (1993) for historical details.

7.1 | Plane waves and Modulational Instability (MI)

Equation (157) admits the spatially homogeneous solution

$$\psi_0(x,t) = A\,\exp\!\big(\mathrm{i}|A|^2 t\big), \qquad A \in \mathbb{C}\setminus\{0\}, \tag{158}$$

often called a *plane wave*. To test its spectral stability we set

$$\psi(x,t) = \left[A + \varepsilon\,\varphi(x,t)\right]\,\exp(\mathrm{i}|A|^2 t), \qquad 0 < \varepsilon \ll 1,$$

linearise in ε, and seek normal modes $\varphi \propto \exp(\sigma t + \mathrm{i}qx)$. A straightforward calculation gives the dispersion relation

$$\sigma^2 = q^2(2|A|^2 - q^2). \tag{159}$$

Thus long-wave perturbations with $0 < q^2 < 2|A|^2$ grow exponentially in time, a manifestation of the Benjamin–Feir (BF) or Modulational Instability (MI).

Theorem 5 (Benjamin–Feir criterion) *Let ψ_0 be the plane-wave solution (158) of the focusing NLS equation (157). Then ψ_0 is modulationally unstable: for every wavenumber q satisfying $0 < q^2 < 2|A|^2$, there exists a perturbation of the form $\varphi(x,t) = \left[\hat{a}\,\mathrm{e}^{\sigma t} + \hat{b}\,\mathrm{e}^{-\sigma t}\right]\mathrm{e}^{\mathrm{i}qx}$, with growth rate $\sigma(q) = q\sqrt{2|A|^2 - q^2} > 0$. Conversely, all perturbations with $q^2 \geq 2|A|^2$ remain neutral. Hence the homogeneous state (158) is unstable to side-band perturbations whose wavelengths exceed the critical value $\lambda_c = \pi/|A|$.*

Theorem 5 furnishes the standard Benjamin–Feir instability criterion and will serve as the reference point for the modulational stability of the KdV- and BBM-derived NLS envelopes obtained in the preceding sections.

7.2 | Modulational instability analysis in KdV wave packets

This section derives the cubic Nonlinear Schrödinger (NLS) equation from the Korteweg–de Vries–Burgers (KdVB) equation using a systematic multiple-scales expansion. The KdVB equation models wave propagation with dispersion, non-linearity, and dissipation, while the NLS equation governs the slow modulation of nearly monochromatic wave trains. The derivation clarifies the transformation process and the physical implications of each term. The KdVB equation is given by

$$\eta_t + \tilde{c}\eta_x + \frac{5\tilde{c}}{2r_0}\eta\eta_x + \frac{\tilde{c}(12\tilde{\alpha} + 3r_0)r_0}{48}\eta_{xxx} - \frac{r_0\tilde{\beta}\gamma}{4}\eta_{xx} = 0, \tag{160}$$

where $\eta(x,t)$ represents the wave amplitude, and $\tilde{c}$, r_0, $\tilde{\alpha}$, $\tilde{\beta}$, and γ are constants. For clarity, define the coefficients:

$$a = \tilde{c}, \qquad b = \frac{5\tilde{c}}{2r_0}, \qquad c = \frac{\tilde{c}(12\tilde{\alpha} + 3r_0)r_0}{48}, \qquad d = -\frac{r_0\tilde{\beta}\gamma}{4}. \tag{161}$$

Here, a relates to linear advection, b to non-linear steepening, c to dispersion, and d to dissipation.

| Multiple-Scales Expansion

To study the slowly varying dynamics of a wave train, we employ a multi-scale perturbation approach. We seek a solution of the form (*cf.* Dias and Bridges (2005)):

$$\eta(x,t) = \varepsilon\left(\eta_0 + \varepsilon\eta_1(x,X,t,T_1,T_2) + \varepsilon^2\eta_2(x,X,t,T_1,T_2) + \varepsilon^3\eta_3(x,X,t,T_1,T_2) + \dots\right), \tag{162}$$

where $\varepsilon \ll 1$ is a small parameter, and the slow scales are defined as:

$$X = \varepsilon x, \quad T_1 = \varepsilon t, \quad T_2 = \varepsilon^2 t.$$

Here, η_0 represents the leading-order wave train, and $\eta_1, \eta_2, \eta_3, \ldots$ are higher-order corrections that depend on both fast scales (x, t) and slow scales (X, T_1, T_2). We assume the leading-order solution η_0 is a slowly modulated plane wave:

$$\eta_0 = A(X, T_1, T_2) \exp(i\theta(x, t)) + \text{c.c.}, \tag{163}$$

where $A(X, T_1, T_2)$ is the complex amplitude modulating on the slow scales, $\theta(x, t) = Kx - \omega t$ is the phase with wave number K and frequency ω, and "c.c." denotes the complex conjugate. This form captures a rapidly oscillating wave with slowly varying amplitude. To account for the multiple scales, we expand the partial derivatives as follows:

$$\partial_t \to \partial_t + \varepsilon \partial_{T_1} + \varepsilon^2 \partial_{T_2}, \quad \partial_x \to \partial_x + \varepsilon \partial_X, \partial_x^3 \to \partial_x^3 + 3\varepsilon \partial_x^2 \partial_X + 3\varepsilon^2 \partial_x \partial_X^2 + \varepsilon^3 \partial_X^3.$$

These expansions reflect the contributions of derivatives across both fast and slow scales, enabling us to separate the dynamics at different orders of ε.

7.2.1 | Multi-Scale Analysis with Negligible Dissipation

In this section, we continue the multi-scale analysis of the KdVB equation, focusing on the case where the dissipative term coefficient satisfies $d = O(\varepsilon^3)$. This assumption simplifies the perturbation analysis, allowing us to derive the Nonlinear Schrödinger (NLS) equation for the amplitude of the wave train.

Substituting the ansatz (162) into the KdVB equation (160) and collecting terms at $O(\varepsilon)$, we obtain:

$$\left(\partial_t + a\partial_x + c\partial_x^3\right) \eta_0 = 0, \tag{164}$$

where $a = \tilde{c}$, $c = \dfrac{\tilde{c}(12\tilde{\alpha} + 3r_0) r_0}{48}$, and $\eta_0 = A(X, T_1, T_2) \exp(\mathrm{i}\theta) + \text{c.c.}$, with $\theta = kx - \omega t$. This yields the dispersion relation:

$$\omega = ak - ck^3. \tag{165}$$

The phase velocity is:

$$s = \frac{\omega}{k} = a - ck^2,$$

indicating dispersion since s depends on the wave number k. The group velocity, which governs the propagation of the wave packet's envelope, is:

$$c_g = \frac{d\omega}{dk} = a - 3ck^2.$$

At $O(\varepsilon^2)$, the equation becomes:

$$\left(\partial_t + a\partial_x + c\partial_x^3\right)\eta_1 = -\left(\partial_{T_1} + a\partial_X - 3ck^2\partial_X\right)\eta_0 - b\partial_x\left(\frac{1}{2}\eta_0^2\right), \tag{166}$$

where $b = \frac{5}{2}\frac{1}{r_0}\tilde{c}$. The right-hand side includes terms proportional to $\exp(\pm i\theta)$ and $\exp(\pm 2i\theta)$. The resonant terms proportional to $\exp(\pm i\theta)$ would lead to secular growth in η_1 and must be eliminated by imposing:

$$\left(\partial_{T_1} + c_g\partial_X\right)\eta_0 = 0 \implies A(X, T_1, T_2) = A(X - c_g T_1, T_2). \tag{167}$$

This indicates that the amplitude A propagates at the group velocity c_g. The remaining non-resonant term is:

$$\partial_x\left(\frac{1}{2}\eta_0^2\right) = ikA^2\exp(2i\theta) + \text{c.c.}.$$

We assume a solution for η_1 of the form:

$$\eta_1 = B_0(X, T_1, T_2) + \left[B_1(X, T_1, T_2)e^{i\theta} + \text{c.c.}\right] + \left[B_2(X, T_1, T_2)e^{2i\theta} + \text{c.c.}\right]. \tag{168}$$

Substituting into (166) and matching terms proportional to $\exp(2i\theta)$, we find:

$$B_2 = \frac{b}{6ck^2}A^2, \tag{169}$$

while B_0 and B_1 remain undetermined at this order.

At $O(\varepsilon^3)$, the equation is:

$$\left(\partial_t + a\partial_x + c\partial_x^3\right)\eta_2 = -\left(\partial_{T_2} + 3c\partial_x\partial_X^2\right)\eta_0 - \left(\partial_{T_1} + a\partial_X + 3c\partial_x^2\partial_X\right)\eta_1 - b\partial_x(\eta_0\eta_1) - b\eta_0\partial_X\eta_0. \tag{170}$$

To prevent secular growth, we eliminate resonant terms independent of θ and proportional to $\exp(\pm i\theta)$. The θ-independent terms give:

$$\left(\partial_{T_1} + a\partial_X\right)B_0 + b\partial_X|A|^2 = 0, \tag{171}$$

yielding:

$$B_0 = \phi(X - aT_1) - \frac{b|A|^2(X - c_g T_1)}{3ck^2},$$

where ϕ is an arbitrary function. The terms proportional to $\exp(i\theta)$ give:

$$i\partial_{T_2}A - 3k\partial_X^2 A + \left(\partial_{T_1} + (a - 3ck^2)\partial_X\right)B_1 + ikb\left(AB_0 + A^*B_2\right) = 0. \tag{172}$$

To ensure B_1 depends only on $X - c_g T_1$, we set:

$$\left(\partial_{T_1} + (a - 3ck^2)\partial_X\right)B_1 = 0,$$

since $a - 3ck^2 = c_g$. Substituting B_0 and B_2 into (172), we obtain:

$$\mathrm{i}\partial_{T_2} A - 3k\,\partial_X^2 A + \frac{b^2 k A|A|^2}{6ck^2} - \frac{kbA\phi}{3ck^2} = 0\,.$$

Assuming $\phi = 0$ (a common simplification for initial conditions), and defining $\zeta = X - c_g T_1$ and $\beta = \dfrac{b^2}{6ck}$, we rewrite the equation as the NLS equation:

$$\mathrm{i}\partial_{T_2} A - 3k\,\partial_\zeta^2 A + \beta A|A|^2 = 0. \tag{173}$$

7.2.2 | Multi-Scale Analysis with Significant Dissipation

In this section, we extend the multi-scale analysis of the KdVB equation to the case where the dissipative term coefficient satisfies $d = O(1)$. This introduces a significant dissipative effect, modifying the dispersion relation and resulting in a complex group velocity, ultimately leading to a modified Nonlinear Schrödinger (NLS) equation.

Substituting the ansatz (162) into the KdVB equation (160) and collecting terms at $O(\varepsilon)$, we obtain:

$$\left(\partial_t + a\,\partial_x + c\,\partial_x^3 + d\,\partial_x^2\right)\eta_0 = 0, \tag{174}$$

where $a = \tilde{c}$, $c = \frac{\tilde{c}(12\tilde{\alpha}+3r_0)r_0}{48}$, $d = -\frac{r_0}{4}\tilde{\beta}\gamma$, and $\eta_0 = A(X, T_1, T_2)\exp(i\theta) + \text{c.c.}$, with $\theta = kx - \omega t$. This yields the dispersion relation:

$$\omega = ak - ck^3 + \mathrm{i}dk^2. \tag{175}$$

The group velocity, accounting for dissipation, is:

$$c_g = \frac{d\omega}{dk} = a - 3ck^2 + 2\mathrm{i}dk. \tag{176}$$

The imaginary component in c_g reflects the dissipative effect on wave packet propagation.

At $O(\varepsilon^2)$, the equation is:

$$\left(\partial_t + a\,\partial_x + c\,\partial_x^3 + d\,\partial_x^2\right)\eta_1 = -\left(\partial_{T_1} + a\,\partial_X - 3ck^2\partial_X + 2\mathrm{i}dk\,\partial_X\right)\eta_0 - b\,\partial_x\left(\frac{1}{2}\eta_0^2\right), \tag{177}$$

where $b = \frac{5}{2}\frac{1}{r_0}\tilde{c}$. The right-hand side contains terms proportional to $\exp(\pm\mathrm{i}\theta)$ and $\exp(\pm 2\mathrm{i}\theta)$. To eliminate secular growth in η_1, we remove resonant terms proportional to $\exp(\pm\mathrm{i}\theta)$ by imposing:

$$\left(\partial_{T_1} + c_g\,\partial_X\right)\eta_0 = 0 \implies A(X, T_1, T_2) = A(X - c_g T_1, T_2). \tag{178}$$

This indicates that the amplitude A propagates at the complex group velocity c_g. The non-resonant term is:

$$\partial_x\left(\frac{1}{2}\eta_0^2\right) = \mathrm{i}kA^2\exp(2\mathrm{i}\theta) + \text{c.c.}.$$

We assume a solution for η_1 as in (168):

$$\eta_1 = B_0(X, T_1, T_2) + \left[B_1(X, T_1, T_2)\mathrm{e}^{\mathrm{i}\theta} + \text{c.c.}\right] + \left[B_2(X, T_1, T_2)\mathrm{e}^{2\mathrm{i}\theta} + \text{c.c.}\right].$$

Substituting into (177) and matching terms proportional to $\exp(2\mathrm{i}\theta)$, we find:

$$B_2 = \frac{b}{6ck^2 - 2\mathrm{i}dk}A^2, \tag{179}$$

while B_0 and B_1 remain undetermined at this order.

At $O(\varepsilon^3)$, the equation is:

$$\left(\partial_t + a\partial_x + c\partial_x^3 + d\partial_x^2\right)\eta_2 = -\left(\partial_{T_2} + 3c\partial_x\partial_X^2 + d\partial_X^2\right)\eta_0 - \left(\partial_{T_1} + a\partial_X + 3c\partial_x^2\partial_X + 2d\partial_x\partial_X\right)\eta_1 - b\partial_x(\eta_0\eta_1) - b\eta_0\partial_X\eta_0. \tag{180}$$

To prevent secular growth, we eliminate resonant terms independent of θ and proportional to $\exp(\pm\mathrm{i}\theta)$. The θ-independent terms give:

$$\left(\partial_{T_1} + a\partial_X\right)B_0 + b\partial_X|A|^2 = 0, \tag{181}$$

yielding:

$$B_0 = \phi(X - aT_1) - \frac{b|A|^2(X - c_gT_1)}{3ck^2 - 2\mathrm{i}dk},$$

where ϕ is an arbitrary function. The terms proportional to $\exp(\mathrm{i}\theta)$ give:

$$\left(\partial_{T_2} + 3\mathrm{i}k\partial_X^2 + d\partial_X^2\right)A + \left(\partial_{T_1} + (a - 3ck^2)\partial_X\right)B_1 + \mathrm{i}kb\left(AB_0 + A^*B_2\right) = 0. \tag{182}$$

Assuming B_1 depends only on $X - c_gT_1$, we set:

$$\left(\partial_{T_1} + (a - 3ck^2)\partial_X\right)B_1 = 0,$$

since $a - 3ck^2 = \mathrm{Re}(c_g)$. Substituting B_0 and B_2 into (182), and assuming $\phi = 0$, we obtain:

$$\partial_{T_2}A + 3\mathrm{i}k\partial_X^2A + d\partial_X^2A + ikb^2\left(\frac{1}{6ck^2 - 2\mathrm{i}dk} - \frac{1}{3ck^2 - 2\mathrm{i}dk}\right)|A|^2A = 0.$$

Defining $\zeta = X - c_gT_1$, $m = \frac{1}{6ck^2 - 2\mathrm{i}dk}$, and $n = \frac{1}{3ck^2 - 2\mathrm{i}dk}$, and rewriting in standard form, we obtain the modified NLS equation:

$$\mathrm{i}\partial_{T_2}A - 3k\partial_\zeta^2A + id\partial_\zeta^2A + ikb^2(m - n)|A|^2A = 0. \tag{183}$$

7.2.3 | Multi-scale Analysis with Intermediate Dissipation

In this section, we perform a multi-scale analysis of the KdVB equation for the case where the dissipative term coefficient is $d = O(\varepsilon^2)$, *i.e.*, $d = \varepsilon^2 \tilde{d}$, with $\tilde{d}$ a constant. This intermediate dissipation strength affects the dynamics at higher orders, leading to a modified Nonlinear Schrödinger (NLS) equation.

Substituting the ansatz (162) into the KdVB equation (160), the equations at $O(\varepsilon)$ and $O(\varepsilon^2)$ are identical to those in the first case ($d = O(\varepsilon^3)$), as the dissipative term $d\partial_x^2\eta$ only appears at $O(\varepsilon^3)$ or higher. Thus, we have:

- At $O(\varepsilon)$: The leading-order equation (164) and dispersion relation (165):

$$\left(\partial_t + a\partial_x + c\partial_x^3\right)\eta_0 = 0, \quad \omega = ak - ck^3,$$

  with group velocity $c_g = a - 3ck^2$.
- At $O(\varepsilon^2)$: The second-order equation (166), secular condition (167), and solution for B_2 (169):

$$A(X, T_1, T_2) = A(X - c_g T_1, T_2), \quad B_2 = \frac{b}{6ck^2}A^2.$$

The coefficients are $a = \tilde{c}$, $b = \frac{5}{2}\frac{1}{r_0}\tilde{c}$, and $c = \frac{\tilde{c}(12\tilde{\alpha} + 3r_0)r_0}{48}$.

At $O(\varepsilon^3)$, the dissipative term introduces a difference. The equation is:

$$\left(\partial_t + a\partial_x + c\partial_x^3\right)\eta_2 = -\left(\partial_{T_2} + 3c\partial_x\partial_X^2 + \tilde{d}\partial_x^2\right)\eta_0 - \left(\partial_{T_1} + a\partial_X + 3c\partial_x^2\partial_X\right)\eta_1 - b\partial_x(\eta_0\eta_1) - b\eta_0\partial_X\eta_0. \tag{184}$$

The θ-independent resonant terms yield the same equation as in the first case (171):

$$\left(\partial_{T_1} + a\partial_X\right)B_0 + b\partial_X|A|^2 = 0,$$

giving:

$$B_0 = \phi(X - aT_1) - \frac{b|A|^2(X - c_g T_1)}{3ck^2},$$

where ϕ is an arbitrary function. For simplicity, we set $\phi = 0$. The terms proportional to $\exp(i\theta)$ give:

$$\partial_{T_2}A + 3\mathrm{i}k\partial_X^2 A - k^2\tilde{d}A + \mathrm{i}kb\left(AB_0 + A^*B_2\right) = 0. \tag{185}$$

Substituting B_0 and B_2, and defining $\zeta = X - c_g T_1$, we obtain:

$$\partial_{T_2}A + 3\mathrm{i}k\partial_\zeta^2 A - k^2\tilde{d}A + \nu^\star A|A|^2 = 0,$$

where:

$$\nu^\star = \frac{-2\mathrm{i}kb^2 + \mathrm{i}kb}{ck^2} = \frac{\mathrm{i}kb(1 - 2b)}{ck^2}.$$

Rewriting in standard NLS form:

$$\mathrm{i}\partial_{T_2} A - 3k\,\partial_\zeta^2 A + \mathrm{i}k^2\tilde{d}A - \mathrm{i}\nu^\star A|A|^2 = 0. \tag{186}$$

7.2.4 | Instability analysis

Modulational instability is a fundamental phenomenon observed in nonlinear dispersive systems, such as surface waves on deep water Benjamin and Feir (1967); Yuen and Lake (1975, 1980); Remoissenet (1996) and light waves in optical fibres Tai et al. (1986). It describes the growth of small perturbations to a plane wave, leading to complex nonlinear dynamics. In one-dimensional systems, numerical simulations of the focusing cubic Nonlinear Schrödinger (NLS) equation on periodic domains reveal recurrent behaviour reminiscent of the Fermi–Pasta–Ulam (FPU) phenomenon Fermi et al. (1955); Yuen and Ferguson (1978); Hafizi (1981).

| Case 1: Negligible Dissipation ($d = O(\varepsilon^3)$)

For the Nonlinear Schrödinger (NLS) equation (173):

$$\mathrm{i}\partial_{T_2} A - 3k\,\partial_\zeta^2 A + \beta A|A|^2 = 0, \quad \beta = \frac{b^2}{6ck}, \quad \zeta = X - c_g T_1,$$

a plane wave solution is:

$$A_0(T_2) = a_0 \exp(\mathrm{i}\beta a_0^2 T_2), \tag{187}$$

where $a_0 \in \mathbb{R}^+$ is the amplitude. To study stability, we introduce a small perturbation:

$$A(\zeta, T_2) = A_0(T_2)[1 + a(\zeta, T_2)], \quad |a| \ll 1. \tag{188}$$

Substituting into (173) and linearizing, we obtain:

$$\mathrm{i}\partial_{T_2} a = 3k\,\partial_\zeta^2 a - \beta a_0^2 (a + a^*), \tag{189}$$

where a^* is the complex conjugate of a. Assuming $a = u(\zeta, T_2) + \mathrm{i}v(\zeta, T_2)$ with real u and v, we get:

$$\partial_{T_2}\begin{pmatrix} u \\ v \end{pmatrix} = \begin{pmatrix} 0 & 3k\,\partial_\zeta^2 \\ -3k\,\partial_\zeta^2 + 2\beta a_0^2 & 0 \end{pmatrix}\begin{pmatrix} u \\ v \end{pmatrix}. \tag{190}$$

Seeking solutions of the form $(u, v) \propto \exp(iq\zeta - i\lambda T_2)$, the dispersion relation is:

$$\lambda^2 = 9k^2q^4 + 6\beta k a_0^2 q^2. \tag{191}$$

If $\beta k > 0$, then for small $q^2 < -\frac{2\beta a_0^2}{3k}$, $\lambda^2 < 0$, indicating instability due to imaginary λ.

| Case 2: Significant Dissipation ($d = O(1)$)

For the modified NLS equation (183):

$$i\partial_{T_2} A - 3k\partial_\zeta^2 A + id\partial_\zeta^2 A + ikb^2(m-n)|A|^2 A = 0,$$

where $m = \frac{1}{6ck^2 - 2idk}$, $n = \frac{1}{3ck^2 - 2idk}$, a plane wave solution is:

$$A_0(T_2) = a_0 \exp(iRa_0^2 T_2), \tag{192}$$

where $a_0 \in \mathbb{R}^+$ and $R = kb^2(m-n) = kb^2\left(\frac{1}{6ck^2-2idk} - \frac{1}{3ck^2-2idk}\right)$. Perturbing as in (188), we obtain:

$$i\partial_{T_2} a = 3k\partial_\zeta^2 a - id\partial_\zeta^2 a - Ra_0^2(a + a^*). \tag{193}$$

Assuming $a = u + iv$, the system becomes:

$$\partial_{T_2}\begin{pmatrix} u \\ v \end{pmatrix} = \begin{pmatrix} -2Qa_0^2 - d\partial_\zeta^2 & 3k\partial_\zeta^2 \\ -3k\partial_\zeta^2 + 2Na_0^2 & -d\partial_\zeta^2 \end{pmatrix}\begin{pmatrix} u \\ v \end{pmatrix}, \tag{194}$$

where:

$$Q = \mathrm{Re}(R) = \frac{-2b^2k^2d(9c^2k^4 + 4d^2k^2) + 2b^2dk^2(36c^2k^4 + 4d^2k^2)}{(36c^2k^4 + 4d^2k^2)(9c^2k^4 + 4d^2k^2)},$$

$$N = \mathrm{Re}(R) = Q.$$

Seeking solutions $\propto \exp(iq\zeta - i\lambda T_2)$, the dispersion relation is:

$$\lambda^2 = (9k^2 + d^2)q^4 + (6Na_0^2 k - 2dQa_0^2)q^2. \tag{195}$$

The stability depends on the sign and magnitude of $6Nk - 2dQ$, with instability possible for certain q^2 when the coefficient of q^2 is negative.

| Summary of Multi-Scale Analysis and Modulational Instability

The multi-scale analysis of the KdV–Burgers equation (160) for a slowly modulated wave train $\eta_0 = A\exp(i\theta) +$ c.c. reveals how dissipation affects the governing Nonlinear Schrödinger (NLS) equations and the MI of plane wave solutions. The results vary with the dissipation coefficient d:

Negligible Dissipation ($d = O(\varepsilon^3)$): The NLS equation (173) is purely dispersive with a cubic nonlinearity. The MI analysis (191) shows that plane waves are unstable for wave numbers $q^2 < -\frac{2\beta a_0^2}{3k}$ when $\beta k > 0$, driven by the interplay of dispersion and nonlinearity.

Significant Dissipation ($d = O(1)$): The modified NLS equation (183) includes a dissipative term $id\partial_\zeta^2 A$, altering the dynamics. The instability condition (195) depends on the coefficient $6Nk - 2dQ$, indicating that dissipation modifies the range of unstable wave numbers, potentially stabilising or destabilising the wave depending on system parameters.

Intermediate Dissipation ($d = O(\varepsilon^2)$, $d = \varepsilon^2\tilde{d}$): The NLS equation (186) introduces a linear dissipative term $ik^2\tilde{d}A$. While MI was not analysed here, the presence of dissipation suggests a modified instability threshold, requiring further perturbation analysis.

These findings underscore the complex interplay between nonlinearity, dispersion, and dissipation in nonlinear wave dynamics. The derived NLS equations and instability conditions provide critical insights into the stability of wave trains in dispersive media, with applications to physical systems like water waves and optical fibres.

7.3 | Modulational instability analysis in BBM wave packets

The Benjamin–Bona–Mahony (BBM) equation models the unidirectional propagation of long waves in dispersive systems, offering an alternative to the Korteweg–de Vries (KdV) equation with improved high-frequency behavior. In normalized variables, the viscous BBM equation is:

$$\partial_t\eta + \tilde{c}\partial_x\eta + \frac{5}{2}\frac{1}{r_0}\tilde{c}\eta\partial_x\eta - \frac{(12\tilde{\alpha}+3r_0)r_0}{48}\partial_x^2\partial_t\eta - \frac{r_0}{4}\gamma\tilde{\beta}\partial_x^2\eta = 0, \tag{196}$$

where $\eta(x,t)$ is the wave field, $\tilde{c}$ is a characteristic speed, r_0, $\tilde{\alpha}$, $\tilde{\beta}$, and γ are system parameters, and subscripts denote partial derivatives.

To study nonlinear wave packets, we assume a slowly modulated wave train solution of the form:

$$\eta(x,t) = \varepsilon A(\xi,\tau)\exp(i\theta) + \text{c.c.}, \tag{197}$$

where $A(\xi,\tau)$ is the complex amplitude, $\theta = kx - \omega t$ with wave number k and frequency ω, and "c.c." denotes the complex conjugate. The slow scales are:

$$\xi = \varepsilon(x - pt), \qquad \tau = \varepsilon^2 t, \tag{198}$$

where $\varepsilon \ll 1$ is a small parameter and p is the group velocity to be determined. We expand the solution as an asymptotic series:

$$\eta(x,t) = \varepsilon\eta_1 + \varepsilon^2\eta_2 + \varepsilon^3\eta_3 + \dots. \tag{199}$$

Substituting (199) into (196), applying the slow scales (198), and collecting terms at $O(\varepsilon)$, we obtain:

$$\left(\partial_t + a\partial_x - c\partial_t\partial_x^2 + d\partial_x^2\right)\eta_1 = 0, \tag{200}$$

where $a = \tilde{c}$, $b = \frac{5}{2}\frac{1}{r_0}\tilde{c}$, $c = \frac{(12\tilde{\alpha}+3r_0)r_0}{48}$, and $d = -\frac{r_0}{4}\gamma\tilde{\beta}$. For $\eta_1 = A(\xi,\tau)\exp(i\theta) + \text{c.c.}$, this yields the dispersion

relation:

$$\omega = \frac{ak}{1+ck^2} + \mathrm{i}\frac{dk^2}{1+ck^2}. \tag{201}$$

The group velocity is:

$$p = \frac{d\omega}{dk} = \frac{a(1-ck^2)+2\mathrm{i}dk}{(1+ck^2)^2}.$$

At $O(\varepsilon^2)$, the equation is:

$$\left(\partial_t + a\partial_x - c\partial_t\partial_x^2 + d\partial_x^2\right)\eta_2 = \left(p\partial_\xi - a\partial_\xi + 2c\partial_t\partial_x\partial_\xi - cp\partial_x^2\partial_\xi - 2d\partial_x\partial_\xi\right)\eta_1 - b\partial_x(\eta_1\eta_1). \tag{202}$$

The right-hand side includes terms proportional to $\exp(\pm\mathrm{i}\theta)$ and $\exp(\pm 2\mathrm{i}\theta)$. To eliminate secular growth in η_2, resonant terms proportional to $\exp(\pm\mathrm{i}\theta)$ are set to zero, confirming the group velocity p above. The non-resonant terms give:

$$\left(\partial_t + a\partial_x - c\partial_t\partial_x^2 + d\partial_x^2\right)\eta_2 = -bikA^2\exp(2\mathrm{i}\theta) + \text{c.c.}. \tag{203}$$

Assuming a solution:

$$\eta_2 = B(\xi,\tau)\exp(2\mathrm{i}\theta) + \text{c.c.} + C(\xi,\tau), \tag{204}$$

we find:

$$B(\xi,\tau) = \sigma A^2, \quad \sigma = \frac{b(1+ck^2)}{6ack^2 + 4\mathrm{i}cdk^3 - 2\mathrm{i}dk}. \tag{205}$$

At $O(\varepsilon^3)$, the equation is:

$$\begin{aligned}\left(\partial_t + a\partial_x - c\partial_t\partial_x^2 + d\partial_x^2\right)\eta_3 = &-\left(\partial_\tau - c\partial_\tau\partial_\xi^2 + 2cp\partial_x\partial_\xi^2 - c\partial_x^2\partial_\tau + d\partial_\xi^2\right)\eta_1 \\ &-\left(-p\partial_\xi + a\partial_\xi - 2c\partial_t\partial_x\partial_\xi + cp\partial_x^2\partial_\xi + 2d\partial_x\partial_\xi\right)\eta_2 - b\partial_x(\eta_1\eta_2).\end{aligned} \tag{206}$$

Substituting (197) and (204), we eliminate resonant terms. The θ-independent terms give:

$$(p-a)\partial_\xi C + b\partial_\xi|A|^2 = 0, \tag{207}$$

yielding:

$$C(\xi,\tau) = \nu|A|^2, \qquad \nu = \frac{b}{p-a} = \frac{b(1+ck^2)^2}{-3ack^2 + 2\mathrm{i}dk - ac^2k^4}.$$

The $\exp(i\theta)$ resonant terms yield:

$$\left(-\partial_\tau - \mathrm{i}\omega c\partial_\xi^2 - 2\mathrm{i}kpc\partial_\xi^2 - cik^2\partial_\tau - d\partial_\xi^2\right)A - bik(AC + A^*B) = 0. \tag{208}$$

Substituting B and C, we obtain the NLS equation:

$$\partial_\tau A + \mathrm{i}M\partial_\xi^2 A + \mathrm{i}N A|A|^2 = 0, \tag{209}$$

where:

$$M = \frac{\omega c + 2kpc - \mathrm{i}d}{1 + ck^2}, \qquad N = \frac{bk(\nu + \sigma)}{1 + ck^2}.$$

The derivation of the NLS equation (209) from the viscous BBM equation provides a framework for understanding the modulation of wave packets in dispersive systems with viscosity. The coefficients M and N encapsulate the effects of dispersion, dissipation, and nonlinearity, offering insights into wave stability and dynamics in applications such as water waves.

7.3.1 | Instability Analysis

Modulational Instability (MI) is a key phenomenon in nonlinear wave systems, where small perturbations to a plane wave can grow exponentially, leading to complex dynamics. Here, we analyze the stability of plane wave solutions to the Nonlinear Schrödinger (NLS) equation (209) derived from the viscous Benjamin–Bona–Mahony (BBM) equation:

$$\partial_\tau A + \mathrm{i}M\partial_\xi^2 A + \mathrm{i}N A|A|^2 = 0,$$

where $M = \dfrac{\omega c + 2kpc - \mathrm{i}d}{1 + ck^2}$, $N = \dfrac{bk(\nu + \sigma)}{1 + ck^2}$, and the coefficients are defined as in the previous derivation. The NLS equation (209) admits a plane wave solution of the form:

$$A_0(\tau) = a_0 \exp(-\mathrm{i}N a_0^2 \tau), \tag{210}$$

where $a_0 \in \mathbb{R}^+$ is the constant amplitude. To study its stability, we introduce a small perturbation:

$$A(\xi, \tau) = a_0 e^{-\mathrm{i}N a_0^2 \tau}[1 + a(\xi, \tau)], \qquad |a| \ll 1. \tag{211}$$

Substituting (211) into (209) and linearizing, we obtain:

$$\mathrm{i}\partial_\tau a - M\partial_\xi^2 a - N a_0^2 (a + a^*) = 0, \tag{212}$$

where a^* is the complex conjugate of a. Assuming $a = u(\xi, \tau) + \mathrm{i}v(\xi, \tau)$ with real u and v, we rewrite (212) as a system:

$$\partial_\tau \begin{pmatrix} u \\ v \end{pmatrix} = \begin{pmatrix} 0 & M\partial_\xi^2 \\ M\partial_\xi^2 - 2N a_0^2 & 0 \end{pmatrix} \begin{pmatrix} u \\ v \end{pmatrix}. \tag{213}$$

Seeking solutions of the form $(u, v) \propto \exp(iq\xi - i\lambda\tau)$, we substitute into (213) to obtain the dispersion relation:

$$\lambda^2 = M^2 q^4 - 2NM a_0^2 q^2. \tag{214}$$

The stability of the plane wave depends on the sign of λ^2:

- If $\lambda^2 > 0$, perturbations oscillate, indicating stability.
- If $\lambda^2 < 0$, perturbations grow exponentially, indicating MI.

Instability occurs when $MN > 0$ and $q^2 < \dfrac{2Na_0^2}{M}$, as the quadratic term dominates for small q. The interplay between M (dispersion/dissipation) and N (nonlinearity) governs the instability threshold.

The dispersion relation (214) highlights the role of dispersion, dissipation, and nonlinearity in the stability of wave packets in the viscous BBM system. The coefficients M and N, derived from the BBM parameters, determine the range of wave numbers q for which instability occurs. This analysis provides insights into the dynamics of long waves in dispersive media, with applications to water waves and other nonlinear systems.

7.4 | Comparison of the KdV and BBM packets: modulational stability

Both reductions produce a cubic envelope equation of the canonical form $\mathrm{i}A_T + P\,A_{\xi\xi} + Q\,A|A|^2 = 0$, for which the Lighthill criterion states that modulational instability occurs *iff* $PQ > 0$ (focusing). For the dissipation-free ($\gamma = 0$) reductions,

$$\text{KdV:}\quad P = -3k,\quad Q = \beta = \frac{b^2}{6\,c\,k} \;\Rightarrow\; PQ = -\frac{b^2}{2c} < 0; \qquad \text{BBM:}\quad P = -M,\quad Q = -N \;\Rightarrow\; PQ = MN, \tag{215}$$

with b, c the KdV coefficients (161) ($c > 0$) and M, N the BBM coefficients defined below (209). The KdV product is negative for *every* carrier wavenumber; evaluating the BBM coefficients for the physiological data of Figure 4 gives $MN < 0$ throughout the long-wave range as well. Hence the following.

Proposition 5 (Modulational stability of the physiological pulse train) *For $\gamma = 0$ and the physiological parameters of Figure 4, the cylindrical KdV and BBM envelope equations are both of* defocusing *type ($PQ < 0$) for all carrier wavenumbers in the long-wave range $k\,r_0 \lesssim 1$. Consequently a slowly modulated arterial pulse train is modulationally stable and develops no* Benjamin–Feir *sidebands.*

Proof For KdV, (215) gives $PQ = -b^2/(2c) < 0$ identically since $c > 0$. For BBM with $\gamma = 0$ the coefficients reduce to $M = (\omega c + 2kpc)/(1 + ck^2)$ and $N = bk(\nu + \sigma)/(1 + ck^2)$ with ω, p, σ, ν as derived above; substituting the physiological values and scanning the carrier over $k \in (0, 100]\ \mathrm{m}^{-1}$ (i.e. $kr_0 \le 1$) yields $MN < 0$ throughout, as computed in `scripts/mi_growth.py`, which also generates Figure 8.

8 | CONCLUSIONS AND DISCUSSION

In this work, we have conducted a comprehensive investigation of pulsatile flow within deformable vessels featuring viscoelastic walls. Grounded in the assumption of axisymmetric flow and considering an ideal fluid for simplicity, we have developed a coherent hierarchy of asymptotic models that capture the complex dynamics governing these intricate systems. Our analysis centered on implementing model order reduction under the long-wave assumption of axi-symmetric Euler equations, which led to the derivation of novel asymptotic model equations offering a detailed

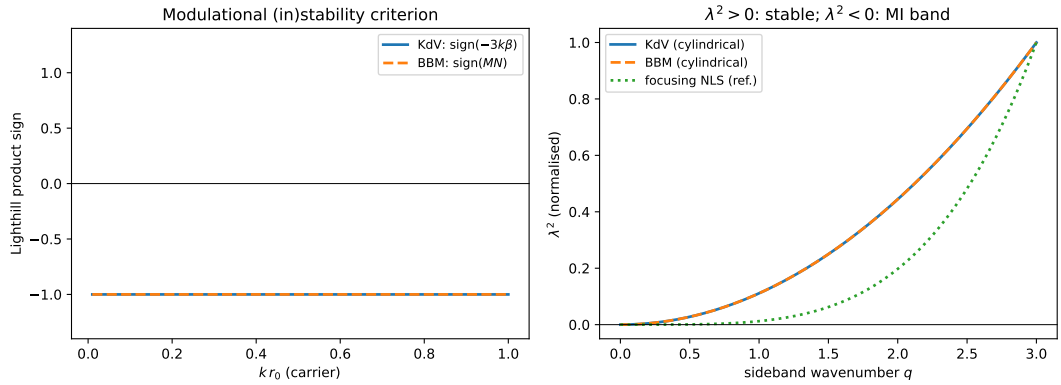


FIGURE 8 *Modulational (in)stability of the cylindrical reductions for the physiological parameters of Figure 4 ($\gamma = 0$).* Left: *the sign of the* Lighthill *product ($-3k\beta$ for KdV, MN for BBM) is negative across the long-wave range, so both envelopes are defocusing.* Right: *$\lambda^2(q)$ remains positive for the KdV and BBM reductions (empty instability band), in contrast with a focusing reference NLS (dotted), whose λ^2 becomes negative. Produced by* `scripts/mi_growth.py`.

portrayal of long-crested pulse propagation in deformable tubes with cylindrical symmetry. This conceptual framework builds upon foundations introduced in Mitsotakis et al. (2018).

Acknowledging the impact of viscous stresses in bio-fluids, we extended our study to incorporate these effects, traditionally addressed through hyperbolic systems of equations. The constraints posed by cylindrical symmetry prompted us to propose various weakly dispersive extensions, enhancing the accuracy of fluid dynamics representation in deformable vessels. Our research began with an in-depth scrutiny of the fully nonlinear regime, resulting in the formulation of a Serre–Green–Naghdi (SGN) system in cylindrical geometry. This system generalises the planar SGN equations to realistic vascular settings and serves as the parent model from which all subsequent approximations descend.

To enhance tractability, we presented a derivation of the weakly nonlinear counterpart of our SGN system. Additionally, we introduced one-dimensional reductions exemplified by Korteweg–de Vries (KdV) and Benjamin–Bona–Mahony (BBM) type equations, further simplifying the system while preserving essential dynamics.

Our analytical exploration provided valuable insights into the fundamental characteristics of these newly formulated systems. We systematically examined symmetries, conservation laws, and Whitham solitons, offering a comprehensive perspective on their behavior. Linear properties, such as dispersion relations, were meticulously discussed in relation to the base model. Furthermore, through multi-scale analysis, we derived cubic Nonlinear Schrödinger (NLS) equations and investigated MI, shedding light on the emergence of Stokes waves from both KdV and BBM equations.

The methodologies and insights developed in this work contribute significantly to our understanding of pulsatile flow within deformable vessels and represent a valuable addition to the broader fields of fluid dynamics and mathematical modeling. As we conclude, we anticipate that our findings will stimulate further investigations and contribute to the ongoing dialogue in this dynamic field, positioning this research at the forefront of discussions on complex fluid systems.

9 | PERSPECTIVES

This work provides a foundational understanding of the mathematical and numerical modeling of nonlinear wave propagation in fluid systems. However, several aspects remain open to further exploration and offer opportunities for extending the research presented here. Building upon the results of this study, we identify multiple promising directions for future research that would expand the scope of our analysis.

A natural extension of this work involves the study of solitary wave solutions for the Serre–Green–Naghdi (SGN) system in cylindrical coordinates. A comprehensive numerical investigation of this system would provide valuable insights into the stability, shape, and velocity of such solitary waves. Moreover, understanding how these waves interact in cylindrical settings would significantly enhance our knowledge of wave dynamics in confined geometries.

The reduced 0D system derived in this article offers a simplified framework for analyzing complex fluid dynamics. A detailed numerical study of this system would enable validation of the theoretical results and exploration of its dynamical features. Such an analysis could uncover critical aspects of its solution space, including fixed points, oscillatory behavior, and transitions between regimes, helping identify practical scenarios where this reduced model can serve as a reliable approximation for more complex systems.

Further advancement in this field would benefit from deriving and studying the Nonlinear Schrödinger (NLS) equations for the SGN system in cylindrical coordinates and for the Boussinesq system. This investigation would include examining modulation instability phenomena in these systems, shedding light on the stability and evolution of wave trains in cylindrical geometries.

Additionally, finding and characterizing Stokes waves in the SGN and Boussinesq systems in cylindrical coordinates represents an essential step toward a complete understanding of these models. This analysis would complement the solitary wave solutions and provide further insights into the dynamics of nonlinear waves in such configurations.

These perspectives aim to extend the theoretical and numerical framework developed in this study, contributing to a deeper understanding of nonlinear wave propagation in fluid dynamics. The proposed extensions not only advance the theoretical and numerical understanding of the systems studied in this work but also hold significant implications for biomedical applications. The investigation of nonlinear wave phenomena in cylindrical geometries is particularly relevant for modeling blood flow in arteries, where such configurations naturally occur. By addressing these perspectives, future research can provide deeper insights into the mechanics of pulse propagation in blood vessels, bridging the gap between mathematical modeling and physiological applications, and offering new tools for understanding and predicting hemodynamic behaviors in real-world scenarios.

Code availability

All scripts associated with this study are freely available. These comprise the Maple (Maplesoft™) worksheets used for the conservation-law and symmetry computations, the Python (SymPy/NumPy/Matplotlib) scripts that symbolically verify the analytical results—the asymptotic consistency of the model hierarchy, the energy balances, the dispersion-relation expansion, the Lie-algebra and reduction computations, the Hamiltonian/Poisson structure, and the hodograph linearisation—and generate the modulational-instability figure, together with the complete LaTeX source files of this article. They are openly available at the following URL:

https://github.com/dutykh/SerreCyl/

Acknowledgements

DD and REC would like to acknowledge the invaluable support of Prof. Stéphane Gerbi (University Savoie Mont Blanc, Chambéry, France), without whom this work would have appeared much later, or perhaps not at all. DD, DM, and REC would like to thank Dr. Hervé Le Meur (Université de Picardie Jules Verne, Amiens, France) for very helpful and enlightening discussions on the topics of nonlinear and dispersive waves.

Conflict of interest

The authors declare that there are no conflicts of interest regarding the publication of this article.

Supporting Information

Further details are available in the PhD manuscript of Rim El Cheikh, which serves as supplementary material to this article.

Data availability statement

No experimental data were used in the preparation of this study. However, any other data or materials related to this research are available from the corresponding author upon reasonable request by email.

Acronyms

BBM Benjamin–Bona–Mahony. 1, 3, 4, 24, 25, 27, 30, 32, 33, 39, 41, 42, 49–51, 59–63, 65, 66
BF Benjamin–Feir. 50, 51

FPU Fermi–Pasta–Ulam. 56

IBVP Initial Boundary Value Problem. 6

KdV Korteweg–de Vries. 1, 3, 4, 24–27, 29, 30, 32, 33, 36, 37, 39, 40, 42, 49–51, 58, 59, 62, 63
KdVB Korteweg–de Vries–Burgers. 51, 52, 54, 55

MI Modulational Instability. 1, 3, 4, 50, 58, 61, 63

NLS Nonlinear Schrödinger. 50–58, 60–63

PDE Partial Differential Equation. 21

SGN Serre–Green–Naghdi. 1, 3, 18, 19, 21, 22, 30, 48, 63, 64

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A | A WINDKESSEL MODEL FOR PULSATILE FLOW IN VISCOELASTIC VESSELS BASED ON SERRE-TYPE APPROXIMATIONS

In the late nineteenth century, Otto Frank developed the *Windkessel* model, drawing an analogy between the cardiovascular system and a closed hydraulic circuit. In his description, the heart functions as a pump connected to a chamber that is almost entirely filled with water except for a small pocket of air; compression of the water forces air out of the chamber in a manner reminiscent of ventricular ejection. Windkessel representations are therefore routinely employed to quantify the load placed upon the heart during a cardiac cycle Catanho et al. (2012).

When modelling the cardiac cycle, the Windkessel approach accounts for *arterial compliance*, *peripheral resistance*, and *inertia*. For a hydraulic system, it may be regarded as the counterpart of Poiseuille's law, comparing blood motion through arteries to viscous flow in pipes.

Windkessel and other lumped-parameter models are often derived from electrical-circuit analogies in which current represents volumetric blood flow rate and voltage represents arterial pressure Avolio (1980); Segers et al. (1998); Milišić and Quarteroni (2004). In this picture, electrical resistances mimic viscous losses, capacitors represent vascular compliance (volume storage), and inductors reproduce the inertial effects of the blood column.

In the electrical analogy, pressure plays the role of voltage, whereas flow rate corresponds to current. A two-element Windkessel model incorporates arterial compliance and total peripheral resistance. Here the compliance of the systemic arteries is represented by a capacitor C $\left[\mathrm{cm}^3\,\mathrm{mmHg}^{-1}\right]$, and the peripheral vascular resistance is represented by a resistor R $\left[\mathrm{mmHg\,s\,cm}^{-3}\right]$. The inflow rate $Q(t)$ $[\mathrm{cm}^3\,\mathrm{s}^{-1}]$ corresponds to an electric current, whereas the outlet pressure $P(t)$ $[\mathrm{mmHg}]$ corresponds to a voltage source.

In what follows, we obtain a zero-dimensional (0D) ordinary differential system by spatially averaging the viscous BBM-type equation,

$$\eta_t + \tilde{c}\,\eta_x + \frac{5}{2}\,\frac{\tilde{c}}{r_0}\,\eta\,\eta_x - \frac{(12\tilde{\alpha} + 3r_0)\,r_0}{48}\,\eta_{xxt} - \frac{r_0}{4}\,\gamma\,\tilde{\beta}\,\eta_{xx} = 0. \tag{216}$$

Assuming $\eta_x = -\eta_t + O(\varepsilon, \delta^2)$ and substituting this relation into (216) we obtain

$$\eta_t + \tilde{c}\,\eta_x + \frac{5}{2}\,\frac{\tilde{c}}{r_0}\,\eta\,\eta_x + \frac{(12\tilde{\alpha} + 3r_0)\,r_0}{48}\,\eta_{xtt} - \frac{r_0}{4}\,\gamma\,\tilde{\beta}\,\eta_{tt} = 0. \tag{217}$$

Multiplication by 48 yields

$$48\,\eta_t + 48\,\tilde{c}\,\eta_x + \frac{120\,\tilde{c}}{r_0}\,\eta\,\eta_x + (12\tilde{\alpha} + 3r_0)\,r_0\,\eta_{xtt} - 12\,r_0\,\gamma\,\tilde{\beta}\,\eta_{tt} = 0. \tag{218}$$

Introducing

$$A' = 48, \qquad B' = (12\tilde{\alpha} + 3r_0)\,r_0, \qquad C' = -12\,r_0\,\gamma\,\tilde{\beta},$$

equation (218) becomes

$$A'\,\eta_t + 48\,\tilde{c}\,\eta_x + \frac{120\,\tilde{c}}{r_0}\,\eta\,\eta_x + B'\,\eta_{xtt} + C'\,\eta_{tt} = 0. \tag{219}$$

Integrating over $x \in [a, b]$ gives

$$A' \int_a^b \eta_t \, \mathrm{d}x + 48\,\tilde{c} \int_a^b \eta_x \, \mathrm{d}x + \frac{120\,\tilde{c}}{r_0} \int_a^b \eta\,\eta_x \, \mathrm{d}x + B' \int_a^b \eta_{xtt} \, \mathrm{d}x + C' \int_a^b \eta_{tt} \, \mathrm{d}x = 0. \tag{220}$$

Defining

$$\hat{\eta}(t) = \frac{1}{b-a} \int_a^b \eta(x,t) \, \mathrm{d}x, \qquad \eta_1(t) = \eta(a,t), \qquad \eta_2(t) = \eta(b,t),$$

and setting $A = A'(b-a)$, $B = B'(b-a)$ and $C = C'(b-a)$, we obtain the lumped relation

$$A\,\dot{\hat{\eta}} + B\,\ddot{\eta}_2 - B\,\ddot{\eta}_1 + C\,\ddot{\hat{\eta}} + \frac{120\,\tilde{c}}{r_0}\left(\eta_2^2 - \eta_1^2\right) + 48\,\tilde{c}\,(\eta_2 - \eta_1) - B\,\ddot{\eta}_1 + \frac{120\,\tilde{c}}{r_0}\left(\eta_2^2 - \eta_1^2\right) + 48\,\tilde{c}\,(\eta_2 - \eta_1) = 0. \tag{221}$$

By integrating the governing BBM-type equation over space, we have derived a consistent 0D system that provides a compact yet accurate description of the haemodynamics in a compliant arterial segment. Such lumped models, when embedded in larger cardiovascular networks, furnish an invaluable tool for predicting pressure-flow relations and assessing ventricular afterload in both physiological and pathological conditions.